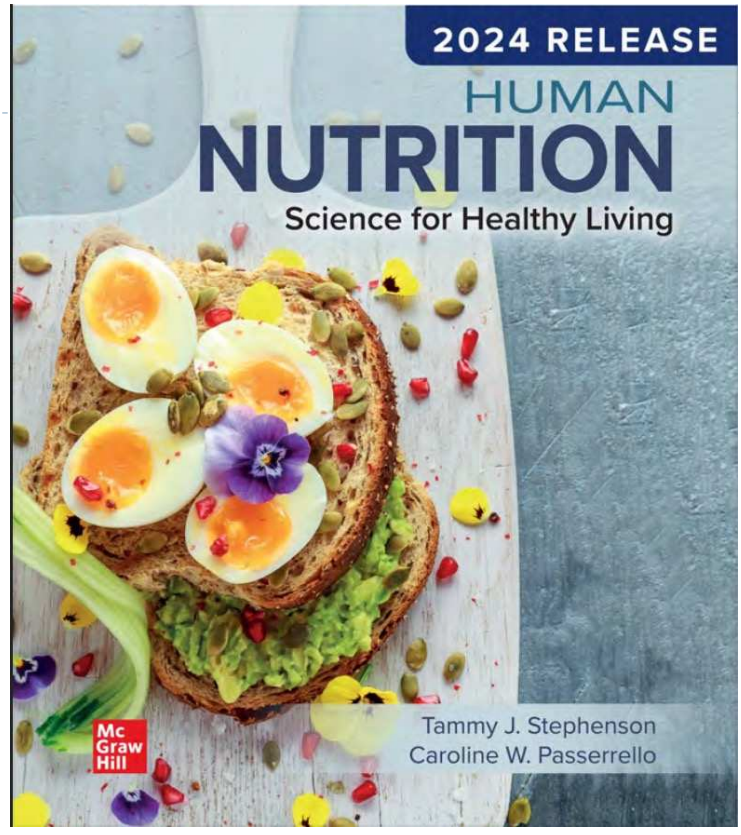
Principles of Nutrition

Dr. Marwan Jalambo
Assistant Professor (Nutrition)



CONTENTS

Chapter 1: Introduction to Nutrition

Chapter 2: Basis of a Healthy Diet

Chapter 3: Human Digestion, Absorption, and Transport

Chapter 4: Carbohydrates: Sugars, Starches, and Fiber

Chapter 5: Lipids: Triglycerides, Phospholipids, and Sterols

Chapter 6: Proteins: Amino Acids

Chapter 7: Metabolism

Chapter 8: Vitamin Overview and Fat-Soluble Vitamins

Chapter 9: Water-Soluble Vitamins

Chapter 10: Water and Major Minerals

Chapter 11: Trace Minerals

Chapter 12: Obesity, Energy Balance, and Weight Management

Chapter 13: Eating Disorders and Disordered Eating

Advice

Read everyday ….. Don't wait mood ….

Mood won't come …..

Welcome to the world of Nutrition



1

Introduction to Nutrition

Part 1

Introduction

Why Study Nutrition?

Food is a basic human need for survival.

Nutrients are necessary for the growth, maintenance, and repair of the body's cells.

While humans have instincts, environmental cues and food processing can alter the body's innate sense of its needs.

We must learn about the nutritional value of foods and the effects of our diet on health.

Motivation is key for change. Common motivators include:

- Losing or gaining weight
- Improving athletic performance
- Managing health concerns (e.g., high blood pressure)

What People Eat

Above Recommendations: Meat, eggs, nuts, and grains.

Below Recommendations: Vegetables, dairy products, and fruit.

Access to ultra-processed foods continues to increase.

This trend contributes to individuals being overweight yet undernourished.

Processed Foods & Public Health

Processing (refining, canning, freezing, cooking) can make food more nutritious, safer, and less perishable.

However, some processing removes naturally occurring nutrients.

It can also add sodium, sugar, and "solid" fat.

Public Health Concern: Many serious diseases, like obesity and type 2 diabetes, are associated with these dietary practices and have become more common.

People may live longer, healthier lives by improving their diet's nutritional quality.

The Nutrients

The Six Classes of Nutrients:

1. Carbohydrates
2. Lipids (fats and oils)
3. Proteins
4. Vitamins
5. Minerals
6. Water

Both food and the human body are made of these nutrients.

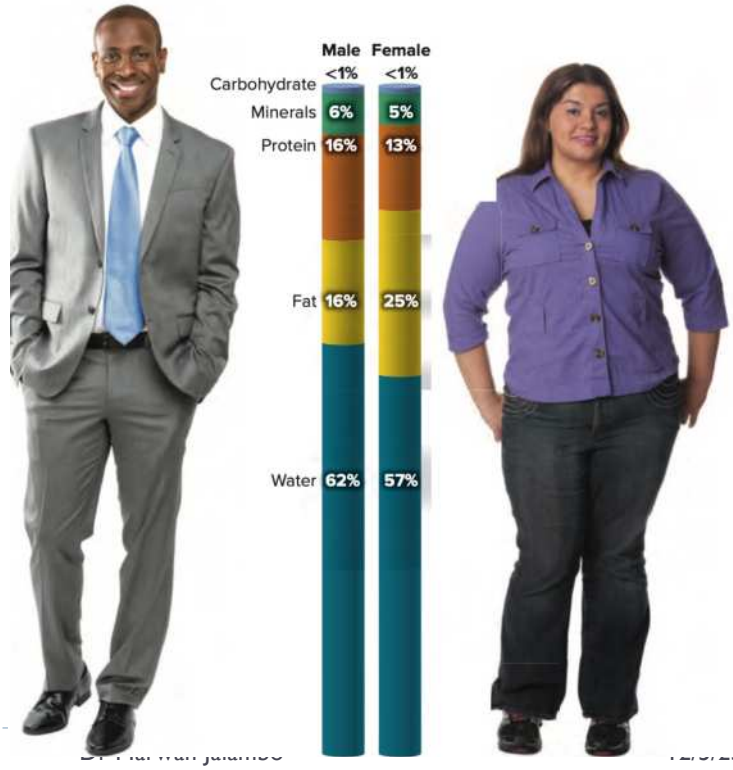
About 50-70% of the body's total weight is water.

Figure: Comparing Body Composition

Young females tend to have less water and protein, and more fat than young males.

Bodies with higher fat content tend to have less water.

Vitamins are in such small amounts they are not shown.



Nutrient Class	Major Functions
Carbohydrates	• Major source of energy (most forms) • Maintenance of normal blood glucose levels • Elimination of solid waste from gastrointestinal tract (fiber)
Lipids	• Major source of energy • Cellular development, physical growth and development • Regulation of body processes (certain hormones, for example) • Growth and development of the brain • Absorption of fat-soluble vitamins
Proteins	• Formation of structural components, such as muscle fibers, and functional components, such as enzymes • Cellular development, growth, and maintenance • Regulation of body processes (certain hormones, for example) • Transportation of substances within the blood • Energy (normally a minor source)

Vitamins	• Regulation of body processes • Immune function • Production and maintenance of cells • Protection against agents that can damage cellular components
Minerals	• Regulation of body processes, including fluid balance and energy metabolism • Formation of certain chemical messengers • Formation of structural and functional components of various substances and tissues • Cellular development, growth, and maintenance
Water	• Maintenance of fluid balance • Regulation of body temperature • Elimination of wastes • Transportation of substances • Participation in many chemical reactions

Metabolism & Elements

Metabolism: The total of all chemical processes that occur in living cells.

- Breaking down molecules (supplying energy)
- Building larger molecules
- Eliminating waste

Element: A substance that cannot be separated into simpler substances (e.g., carbon, oxygen).

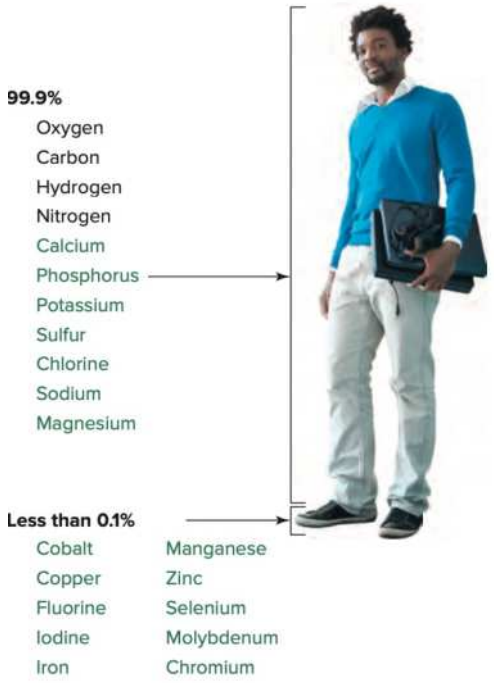
Nutrients are sources of elements the body needs.

Elements in Nutrients & the Body

Figure: Nutrient-related elements in the human body

99.9% of body weight: Oxygen, Carbon, Hydrogen, Nitrogen, Calcium, Phosphorus, Potassium, Sulfur, Chlorine, Sodium, Magnesium.

Less than 0.1%: Cobalt, Copper, Fluorine, Iodine, Iron, Manganese, Zinc, Selenium, Molybdenum, Chromium.



Organic vs. Inorganic Nutrients

Organic: Refers to compounds that contain carbon.

- Carbohydrates
- Lipids
- Proteins
- Vitamins

Inorganic: Refers to substances that do not contain carbon.

- Minerals
- Water

Essential Nutrients

Essential Nutrient: Must be supplied by food because the body does not synthesize it or make enough to meet its needs.

There are about 50 dietary essentials.

Water is the most essential nutrient.

Essential Nutrients

Criteria for an Essential Nutrient:

1. Deficiency Disease: If the nutrient is missing, a deficiency disease occurs.
2. Signs & Symptoms: This state causes abnormal physiological changes.
   - Signs: Observable or measurable (e.g., rash, bleeding gums).
   - Symptoms: Subjective complaints (e.g., dizziness, fatigue).
3. Correction: When the nutrient is added back, the abnormal changes are corrected.

Biological Role: Scientists identify the nutrient's specific roles in the body.

Example: Vitamin C & Scurvy



Scurvy is the vitamin C deficiency disease.

Mechanism: Vitamin C is needed to maintain collagen, a protein that holds cells together (e.g., in blood vessels and for wound healing).

Signs: Without vitamin C, blood vessels leak, causing gums to bleed (Fig.) and wounds to heal slowly.

Correction: Feeding vitamin C-rich foods reverses the signs and symptoms.

Conclusion: Vitamin C meets all criteria of an essential nutrient.

Conditionally Essential & List of Essentials

Conditionally Essential Nutrients: Nutrients that are normally nonessential but become essential under certain conditions (e.g., metabolic disorders, serious diseases, prenatal development).

Table: Essential Nutrients for Humans

CHO	Proteins (Amino Acids*)	Lipids	Vitamins	Minerals	Water
Glucose	The following amino acids are generally recognized as essential: Histidine Leucine Isoleucine Lysine Methionine Phenylalanine Threonine Tryptophan Valine	Fats that contain:Lin-oleic acid Alpha-linolenic acid	A Thiamin Riboflavin Niacin Pantothenic acid Biotin‖ Folate B-6 B-12 C D E K Choline	**Major minerals:** Calcium Chloride Magnesium Phosphorus Potassium Sodium Sulfur **Trace minerals:** Chromium Copper Iodine Iron Manganese Molybdenum Selenium Zinc	Water

Macronutrients vs. Micronutrients

Macronutrients: Nutrients the body needs in large amounts (grams).

1. Carbohydrates
2. Fats (Lipids)
3. Proteins

Micronutrients: Nutrients the body needs in very small amounts (milligrams or micrograms).

1. Vitamins
2. Minerals

Energy:

Macronutrients supply energy for cells.

Micronutrients do not supply energy.

Water does not provide energy and is not usually classified as a macronutrient.

Phytochemicals

Phytochemicals (phyto = plant): Substances in plants that are not nutrients but may have health benefits.

Beneficial Examples:

- Caffeine (stimulant)
- Beta-carotene (body can convert to vitamin A)

Not all are beneficial. Some are toxic or interfere with nutrient absorption:

- Nicotine (in tobacco)
- Ricin (in castor beans)
- Oxalic acid (in rhubarb leaves)

Food as Fuel

Most foods are sources of biological fuel, providing energy for cells to carry out activities.

Key Definitions:

calorie: The amount of heat necessary to raise the temperature of 1 g of water 1° Celsius (C).

kilocalorie (kcal) or Calorie (with a capital C): The amount of heat energy needed to raise the temperature of 1000 g (1 liter) of water 1° C.

This is the unit used to report energy in food.

Example: 1 small apple = 40,000 calories = 40 kcal = 40 Calories.

Measuring Food Energy: Direct Calorimetry

Bomb Calorimeter: A device used to measure the calorie content of food.

Process:

1. A specific amount of food is placed in the "bomb" chamber.
2. The chamber is surrounded by water.
3. An electrical spark ignites the food, causing it to burn completely.
4. The burning food releases heat, raising the water temperature.
5. The change in water temperature is measured to determine the calories.

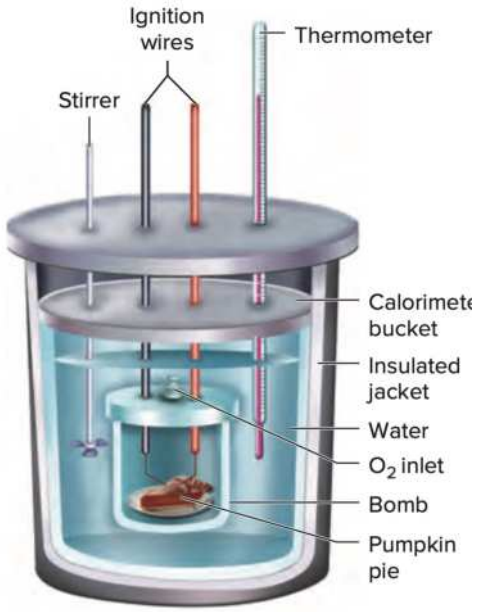


Figure: Bomb calorimeter

Calculating Food Energy

Energy is supplied by macronutrients and alcohol.

Alcohol is not a nutrient (body doesn't need it to survive) but is a source of energy.

Carbohydrate: 4 kcal/g

Protein: 4 kcal/g

Fat: 9 kcal/g

Alcohol: 7 kcal/g

Example: Ice Cream Cone (24g Carb, 8g Fat, 3g Protein)

STEP 1:

24 g Carb 4 kcal/g = 96 kcal

8 g Fat X 9 kcal = 72 kcal

3 g Protein X 4 kcal/g = 12 kcal

STEP 2:

Total = 96 + 72 + 12 = 180 kcal

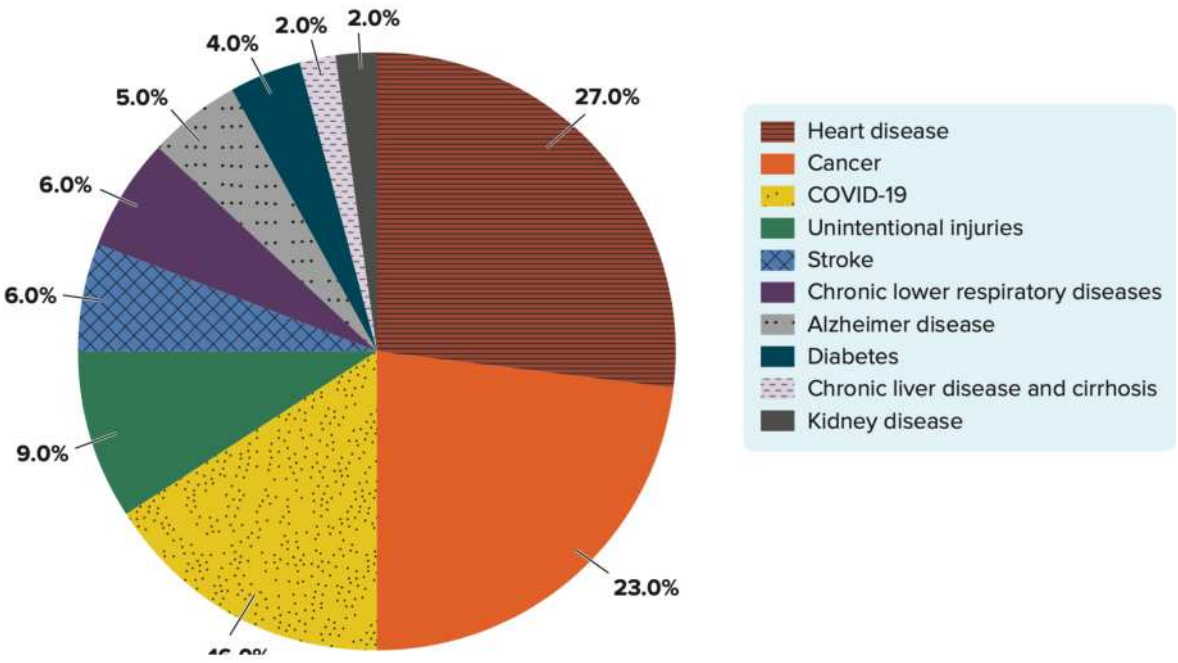


Does Diet Matter?

Poor food choices can have a significant impact on health.

Several of the leading causes of death are diet-related and lead to premature death.

Figure: Death Leading Causes (U.S., 2021)



Risk Factors & Lifestyle

Chronic Diseases: Long-term conditions that take years to develop and have complex causes (e.g., heart disease, diabetes, cancer).

Risk Factor: An attribute, characteristic, or exposure that increases a person's chances of developing a chronic disease.

- Genetic background / Family history
- Unsafe environmental conditions
- Psychological factors (e.g., excess stress)
- Social determinants of health (e.g., lack of access to food)
- Advanced age
- Unhealthy lifestyle

Lifestyle: A routine way of living, including dietary practices and exercise habits.

Healthy lifestyle choices may reduce the risk or delay the onset of chronic diseases.

Sedentary Time & National Health Goals

The Importance of Physical Activity

Being sedentary for >8 hours a day without physical activity has a mortality risk comparable to smoking.

60-70 minutes of moderate-intensity physical activity per day seem to eliminate the increased mortality risk associated with 8 hours of sitting

Healthy People 2030

A report issued by the U.S. DHHS every 10 years.

Includes national health promotion and disease prevention goals.

Provides science-based objectives to improve the health of all Americans.

"Nutrition and Healthy Eating" is a key topic area.

Factors That Influence Eating Habits

Food is more than just nutrients; it has personal meaning, evokes feelings, and conveys status and ethnicity.

What influences your eating practices?

Factors include:

- Biological & Physiological
- Cognitive & Psychological
- Environmental
- Expert Advice

Biological, Physiological & Sensory Factors

Biological & Physiological Factors:

- Age / Life Stage: Nutrient needs change during childhood, pregnancy, or older adult years.
  - Caregivers ("gatekeepers") choose food for young children.
  - Older adults may change diets due to chronic health conditions (e.g., high blood pressure, tooth loss).
- Internal Sensations: Hunger and thirst

Sensory Information:

- Humans rely on taste, texture, odor, and appearance of food to develop likes and dislikes.

Cognitive & Psychological Factors

Cognitive Factors: Learning processes.

- We learn acceptable foods from past experiences, cultural practices, and religious teachings

Psychological Factors:

- *Mood & Stress:* Food affects mood, and mood affects food choices (bidirectional link).
- *Positive Associations:* Using food as a reward; "comfort foods" evoking feelings of safety.
- *Negative Associations:* Associating a food with a time you became ill, even if the food wasn't the cause.

Environmental & Expert Advice Factors

Environmental Factors:

- Food cost
- Proximity of home to food stores
- Availability of food on campus or at work
- Food Marketing: Store layout, packaging, and media advertising are designed to stimulate purchases.

Expert Advice Factors:

- Registered Dietitian Nutritionist (RDN) / Registered Dietitian (RD): A college-trained health care professional and "food and nutrition expert".
- Qualifications: Requires a (master's) degree, extensive supervised practice, passing a national exam, and ongoing continuing education.
- The title "nutritionist" is not standardized and may be used by anyone, regardless of training.

Sustainability and Your Diet

Food System: All components related to production, processing, distribution, access, consumption, and waste recovery of food.

Sustainable Food System: Delivers food security and nutrition for all without compromising the economic, social, and environmental bases for future generations.

Nutrition involves considering how food is grown, processed, purchased, and what happens to the waste.

Summary

1: Nutrition is the scientific study of nutrients, which are needed for growth, maintenance, and repair. Poor diet is linked to chronic diseases.

2: The six classes of nutrients are carbs, lipids, proteins, vitamins, minerals, and water. About 50 nutrients are essential.

3: Energy in food is measured in kilocalories (kcal). Carb/Protein = 4 kcal/g, Fat = 9 kcal/g, Alcohol = 7 kcal/g.

Summary

4: Diet-related chronic diseases (heart disease, cancer) are leading causes of death. Lifestyle changes can improve health.

5: Food choices are influenced by biological, physiological, cognitive, psychological, environmental, and expert advice factors.



Principles of Nutrition

Dr. Marwan Jalambo
Assistant Professor (Nutrition)

1

Part 2:Nutrition Concepts

Key Nutrition Concepts

The **Six** Key Nutrition Concepts

1. Most foods are mixtures of nutrients.
2. Variety, moderation, and balance can help ensure a diet's nutritional adequacy.
3. Food is the best source of nutrients.
4. Foods and the nutrients they contain are one of many components of health.
5. Malnutrition includes overnutrition as well as undernutrition.
6. Nutrition is a dynamic science.

Concept 1: Most Foods Are Mixtures of Nutrients

Most foods contain a mix of nutrients; water is often the major component.

- **Example**: Fat-free milk is ~91% water, but an excellent source of protein, carbs, vitamins, and minerals.
- **Example**: A chocolate-almond bar is ~44% sugar, but 1/3 of its energy is from fat; it also has small amounts of protein, vitamins, and minerals.

Concept 1: Most Foods Are Mixtures of Nutrients

Nutrient-Dense: Describes a food that supplies more key beneficial nutrients (protein, fiber, vitamins, minerals) in relation to its total calories.

- It has little or no solid fats, added sugars, refined starches, or sodium.

Examples: Vegetables, low-fat milk, fruits, lean meats, whole-grain cereals.

Energy Density

Energy Density (or Calorie Density): The amount of energy (kcal) a food provides per given weight (grams) of the food (kcal/g).

Fat (9 kcal/g) supplies the most energy, so fatty foods tend to be energy-dense.

Water (0 kcal/g) provides no energy, so watery foods (fruits, vegetables) have low energy density.



Figure: Energy Density Comparison

Doughnut: ~100g = 400 kcal (Energy Density: 4.0)

Strawberries: ~100g = 32 kcal (Energy Density: 0.32)

You'd have to eat 100 strawberries to get the same energy as one doughnut.

Empty Calories & Nutrient/Energy Density

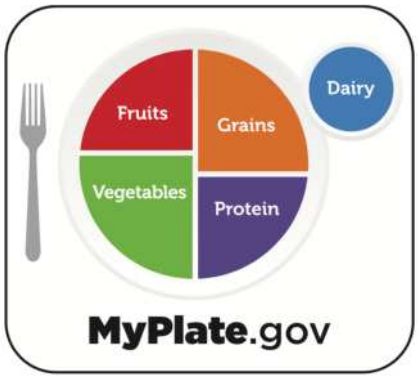
Empty Calories: Calories from solid fats and added sugars (SoFAS).

- Examples: Candy, snack chips, sugar-sweetened drinks, alcohol.
- These foods are generally not good sources of vitamins and minerals.
- Excess consumption can lead to an unbalanced diet.

Can a food be both energy-dense and nutrient-dense?

- Yes. Nuts, for example, are high in fat (energy-dense) but are also nutrient-dense, providing protein, vitamins, minerals, and fiber.

Concept 2: Variety, Moderation, and Balance



No naturally occurring food is "perfect" or contains all nutrients in the required amounts.

Variety: A diet that contains foods from each food group.

Moderation: Eating reasonable amounts of each food.

> Involves obtaining enough nutrients to meet needs while avoiding excessive amounts.

Balance: A diet containing a variety of nutrient-dense foods and limiting empty calories.

Figure: Beverage Comparison (8 fluid ounces [1 CUP])

Nutrient	Fat-Free Cow's Milk	Sugar-Sweetened Soft Drink
Energy	83 kcal	91 kcal
Protein	8.26 g	0.17 g
Calcium	299 mg	5 mg
Phosphorus	247 mg	25 mg
Riboflavin	0.446 mg	0 mg
Vitamin A	149 µg	0 µg



Concept 3: Food Is the Best Source of Nutrients

The most reliable and economical way to obtain nutrients and beneficial phytochemicals is to base the diet on "whole" and minimally processed foods.

Processing (especially refining) may remove healthful parts.

Some individuals do require supplements (e.g., older adults, pregnant females, those with certain medical conditions).

Dietary Supplement: A product containing a vitamin, mineral, herb, amino acid, or other dietary substance to supplement the diet.

Concept 3 (Cont.): Dietary Supplements & DSHEA

Dietary Supplement Health and Education Act of 1994 (DSHEA):

- This legislation allows manufacturers to classify nutrient supplements and herbal products as "foods," not drugs.
- Consequence: Dietary supplements do not undergo the rigorous testing for safety and effectiveness that medications must[243].

FDA Regulation:

- The FDA does not regulate supplements as strictly as drugs.
- The FDA alerts consumers after a product is found to present a significant risk.

Quality Seals:

- Nongovernmental labs (e.g., U.S. Pharmacopeia) can test product quality.
- A seal of approval assures the product was manufactured properly but does not guarantee it is safe or effective.

Concept 3 (Cont.): Range of Safe Intake

More is not always better.

Physiological Dose: The amount of a nutrient that is within the range of safe intake and enables optimal body function.

Megadose: An amount that greatly exceeds the recommended amount, most common with supplements.

- **Can behave like drugs and cause toxic side effects.**

Figure: Nutrient Intake Continuum

- **Deficient**: Not enough; leads to deficiency disease.
- **Marginal**: Just enough to function, but not enough to handle stress.
- **Safe**: Optimal range for body function.
- **Toxic**: Too much; leads to toxic side effects.



Concept 4: Foods and the nutrients they contain are one of many components of health

Nutrients cure deficiency diseases (like scurvy), but they do not "cure" other ailments.

Diet is only one aspect that influences a person's health.

Functional Foods: Foods (often manufactured) that contain nutrient and/or phytochemical ingredients that may provide health benefits.

> **Example**: Calcium-fortified orange juice to promote bone health.

Medical Nutrition Therapies: Nutritionally modified ("special") diets planned by RDNs for people with chronic health conditions (e.g., diabetes, food allergies).

Concept 5: Malnutrition Includes Overnutrition as Well as Undernutrition

Malnutrition (mal = bad): A state of health that occurs when the body is improperly nourished.

Malnutrition includes both:

> Undernutrition: Lack of energy or nutrients.
>
> Overnutrition: Long-term excess consumption of energy or nutrients.

Chronic overnutrition characterized by excessive energy intake results in obesity.

It is possible to be overweight and undernourished at the same time.

At-Risk Groups: Those with low incomes, food insecurity, severe eating disorders, substance use disorders, or serious medical conditions.

Concept 6: Nutrition Is a Dynamic Science

Researchers are constantly exploring the complex relationships between diet and health.

As a result, nutrition information and dietary recommendations constantly change.

Dietary recommendations undergo revision as new scientific evidence becomes available and is accepted by experts.

Summary

The six key concepts include: foods are mixtures; variety, moderation & balance; food is the best source; food is one part of health; malnutrition includes over- and undernutrition; and nutrition is a dynamic science.


Thank you!

Principles of Nutrition

Dr. Marwan Jalambo
Assistant Professor (Nutrition)

2

PART 1: Basis of Healthy Diet

What Is a Nutrient Requirement?

Nutrient Requirement: The smallest amount of a nutrient that maintains a defined level of nutritional health.

This amount typically prevents a deficiency disease or saturates cells with the nutrient.

Influencing Factors: An individual's nutrient needs are influenced by:

- Lifestyle
- Age
- Sex
- General health status
- Use of medications

Nutrient Storage: Many nutrients are stored in the liver, body fat, muscles, and bones. When intake is low or needs increase (like during illness), the body withdraws from storage.

Optimal storage levels can delay or prevent deficiency symptoms.

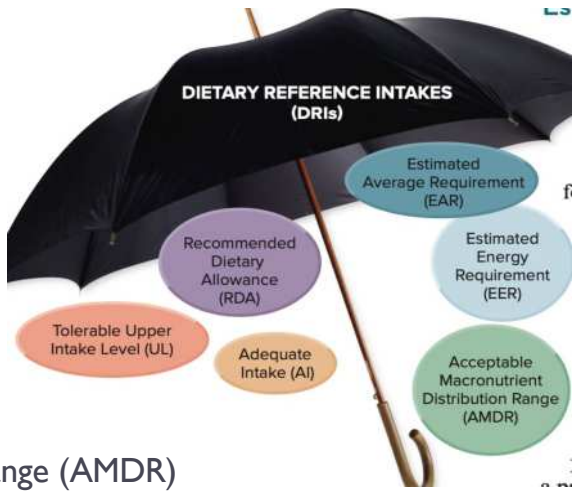
Dietary Reference Intakes (DRIs)

DRIs (Dietary Reference Intakes): A set of energy and nutrient intake standards used by nutrition experts as references for making dietary recommendations.

Developed by: Food and Nutrition Board of the National Academy of Medicine.

The 6 DRI Standards:

1. Estimated Average Requirement (EAR)
2. Estimated Energy Requirement (EER)
3. Recommended Dietary Allowance (RDA)
4. Adequate Intake (AI)
5. Tolerable Upper Intake Level (UL)
6. Acceptable Macronutrient Distribution Range (AMDR)



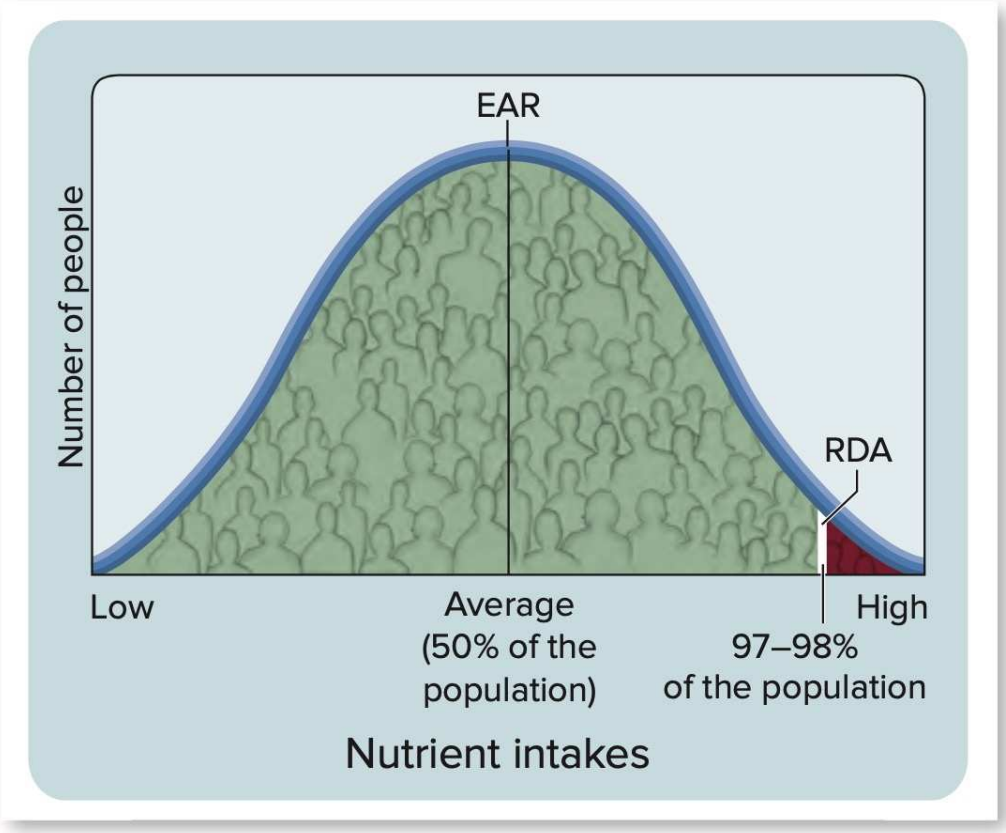
Estimated Average Requirement (EAR)

EAR: The amount of a nutrient that meets the needs of **50%** of healthy people in a specific life-stage/sex group.

How it's established:

The Food and Nutrition Board (FNB) identifies a **physiological marker**, a substance in the body that reflects proper functioning and can be measured.

Example: The amount of vitamin C in certain white blood cells. When these cells are saturated, it indicates an optimal supply.



Estimated Energy Requirement (EER)

EER: The average daily energy intake (calories) that meets the needs of a healthy person who is **maintaining their weight**.

Takes into account:

1. Physical activity level
2. Height
3. Weight
4. Sex
5. Age

Key Feature: The EER does **not** include a "margin of safety." It may be higher or lower than an individual's actual needs.

Recommended Dietary Allowances (RDAs)

RDA: Standards for recommended daily intakes of several nutrients.

Meets the needs of: Nearly all healthy individuals (**97-98%**) in a particular life-stage/sex group.

How it's established:

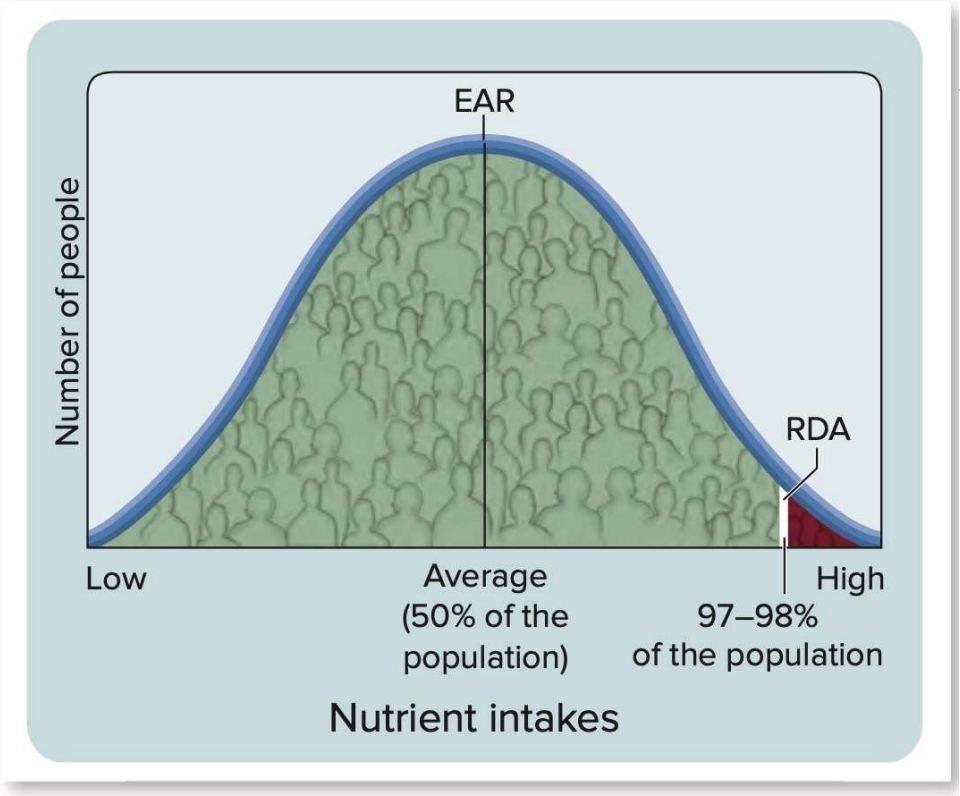
RDA = EAR + a "margin of safety"

The margin of safety accounts for individual variations and helps maintain body stores.

Example: Adult male Vit. C EAR = 75 mg

Adult male Vit. C RDA = 90 mg

(The margin of safety is 15 mg)



Adequate Intakes (AI)

AI: Set for nutrients when there is **not enough scientific information** available to determine an EAR (and therefore an RDA).

How it's established:

Scientists record the diets of a group of healthy people and estimate their average daily intake of the nutrient.

If this group shows no signs of deficiency, this average intake value is used as the AI.

Dietitians use AI along with RDAs as nutrient intake goals.

Nutrients Assigned Adequate Intakes (AI)

Vitamins	Minerals	Carbo-hydrates	Lipids	Water
Pantothenic acid	Sodium	Fiber	Total fat (infants)	Water
Biotin	Chloride		Linoleic acid	
Vitamin K	Potassium		Alpha-linolenic acid	
Choline	Fluoride			
	Manganese			
	Chromium			

Tolerable Upper Intake Level (UL)

UL: The **highest** average daily amount of a nutrient that is unlikely to harm most people when consumed daily.

Toxicity: The risk of a toxicity disorder increases when a person regularly consumes amounts *greater* than the UL.

Example:

The UL for Vitamin C is 2000 mg/day for adults.

It is difficult to exceed this from food (would take 24 oranges).

It is *easy* to exceed this by taking dietary supplements.

DRI: Acceptable Macronutrient Distribution Ranges (AMDRs)

AMDRs: Ranges of carbohydrate, fat, and protein intakes (as a percentage of total calories).

Purpose:

Provide adequate amounts of vitamins and minerals.

May reduce the risk of diet-related chronic diseases (e.g., heart disease, type 2 diabetes).

Help determine if a diet has a healthful *balance* of macronutrients.

AMDRs for Adults

Carbohydrate: 45-65%

Fat: 20-35%

Protein: 10-35%

TABLE: Summary of Dietary Reference Intakes

Standard	Definition	Example
(EAR)	Amount of a nutrient that meets the needs of 50% of healthy people who are in a particular life-stage and sex group	**Vitamin C** 75 mg/day for males and 60 mg/day for females ages 19 through 50 years
(RDA)	Amount of a nutrient that meets the needs of nearly all healthy individuals in a particular life- stage and sex group	**Vitamin C** 90 mg/day for males and 75 mg/day for females ages 19 through 50 years
(AI)	Amount of a nutrient that is considered to be adequate based on the population's typical intakes, but there is not enough scientific information available to determine an RDA for the nutrient at this time	**Vitamin C** 40 mg/day for infants from birth through 6 months of age
(UL)	Highest average amount of a nutrient that is unlikely to harm most people when the amount is consumed daily	**Vitamin C** 2000 mg/day for adults

Applying Nutrient Standards

How DRIs are Used:

Evaluate the nutritional adequacy of a population's diet.

Plan nutritious diets for groups.

Assess an individual's diet (using RDAs and AIs as targets).

Risk Assessment:

Deficiency Risk: A person's diet is consistently **below the EAR**.

Toxicity Risk: A person's intake is consistently **above the UL** (most common with high-dose supplements).

Commercial Applications:

Developing infant formulas.

Creating special formulas for people who cannot eat regular foods (e.g., tube feeding).

Major Food Groups

Dietary Guidance System (Food Guide): Translates complex DRIs into practical, food-based dietary recommendations for consumers.

Food Groups: Foods are classified according to their natural origins and key nutrients.

USDA Major Food Groups:

Grains

Dairy

Fruits

Vegetables

Protein-rich foods

(Oils are also often included)

Importance of Variety: Nutritional content varies widely *within* a food group.



Grains Group

Includes: Products made from wheat, rice, oats, corn, barley, etc. (e.g., pasta, bread, tortillas).

Primary Macronutrients: Carbohydrate (starch) and protein.

Enrichment vs. Fortification:

Enrichment: Replaces *some* nutrients lost during refining. Specific amounts of iron and B vitamins (thiamin, riboflavin, niacin, folic acid) are added back to refined grains.

Fortification: The addition of nutrients to *any* food to boost its nutritional value. (e.g., calcium added to orange juice, Vitamins A & D added to milk) .

1 regular slice bread
1 cup ready-to-eat cereal (flakes)
1⁄2 English muffin
1⁄2 cup cooked rice, pasta, or cereal such as oatmeal
3 cups popped popcorn
5 small whole-wheat crackers

Grains (continued)

Whole Grains: Contain the *entire* grain kernel—the bran, germ, and endosperm.

Examples: Whole wheat, oatmeal, whole-grain cornmeal, brown rice.

Benefits: Compared to refined grains, whole grains naturally contain more fiber and micronutrients that are not replaced during enrichment.

Dairy Foods

Includes: Cow's milk and products made from it *that retain their calcium content*, such as yogurt and hard cheeses.

Excellent Sources of: Protein, phosphorus, and riboflavin (a B vitamin).

Fortification: Most cow's milk in the U.S. is fortified with vitamins A and D.

Substitutes: Plant-based "milks" (soy, almond) can substitute *if* they are fortified with calcium and other micronutrients.

What's NOT included: Cream cheese, cream, and butter. They are dairy products but have little/no calcium and are high in solid fat.

Dairy Food Equivalents

1 cup low-fat cow's milk or calcium- fortified almond or soy milk alternative 1 cup plain yogurt, frozen yogurt, or pudding (made with cow's milk)
1/2 cup evaporated cow's milk
2 cups cottage cheese
1/2 cup ricotta cheese
1/2 oz natural cheese such as Swiss or cheddar
1/2 cup shredded cheese 2 oz processed cheese

Protein Foods

Includes: Beef, pork, fish, poultry, beans, eggs, nuts, tofu, and seeds.

Rich Sources of: Micronutrients, especially iron, zinc, and B vitamins.

Bioavailability: The body absorbs minerals (like iron and zinc) more easily from animal foods than from plant foods.

Fat Categories:

Very Lean: White meat of turkey.

Lean: Ground beef (<=15% fat), tuna.

High-Fat: Pork sausage, bacon, hot dogs, regular cheeses.

Protein Foods

oz cooked lean beef, pork, ham, poultry (without skin), tempeh (a soybean product), or fish or shellfish
1 egg
1 Tbsp peanut or almond butter 1/2 oz nuts (e.g., 12 almonds or 24 pistachios) or seeds
1/4 cup cooked beans, cooked dried peas, roasted soybeans, baked or refried beans, or tofu 2 Tbsp hummus

Fruits

Includes: Fresh, dried, frozen, canned fruit, and 100% fruit juice.

Good Sources of: Phytochemicals, potassium, vitamin C, and folate.

Fiber: Whole or cut-up fruit is often a good source of fiber.

Recommendation: The majority of choices should be whole or cut-up fruits, *not* juice. Fruit is more nutrient-dense

(more fiber, no added sugars).

Fruit Equivalents

1 small apple

1 medium grapefruit, pear

1 large banana, orange, peach

8 large strawberries

1 cup diced melon, grapes, pineapple 1 cup applesauce or 100% fruit juice

1/2 cup dried fruit (e.g., raisins, apricots, prunes)

Vegetables

Includes: Fresh, cooked, canned, frozen, dried vegetables, and 100% vegetable juice.

Subcategories: Dark green, orange, and starchy.

Rich Sources of: Potassium, vitamin A, vitamin C, folate, fiber, and phytochemicals.

Dried Beans & Peas: Can be counted in *either* the vegetable group OR the protein foods group (but not both at the same time).

Vegetable Equivalents

1 cup raw or cooked vegetables

1 cup vegetable juice

2 cups leafy greens (raw)

1 large sweet potato, ear of corn, tomato

Oils & Solid Fats

Oils:

Definition: Fats that are liquid at room temperature (e.g., canola, corn, olive oils).

Includes: Spreadable foods like mayonnaise, tub margarine, salad dressing.

Sources of: Vitamin E and essential fatty acids.

USDA Status: Not a food group, but consumption is recommended.

Solid Fats:

Definition: Fats that are fairly hard at room temperature (e.g., beef fat, butter, stick margarine, lard).

Includes: Coconut oil (liquid but chemically a solid fat), cream, and cream cheese.

Health Risk: High consumption is associated with increased risk of cardiovascular disease (CVD).

2

PART 2: Dietary Guidelines

Dietary Guidelines

Dietary Guidelines for Americans: A set of recommended eating and physical activity practices for *healthy* people.

Purpose: Incorporates scientific knowledge to promote health and prevent chronic disease.

Published by: U.S. Department of Health and Human Services (USDHHS) and USDA.

Revised: About every 5 years.

2020-2025 Guidelines: The first edition to include recommendations for all life stages, from infancy to older adults.

Used for: Developing national nutrition policies, educational materials, and menu planning.

Developing & Applying the Guidelines

Development Process:

A **Dietary Guidelines Advisory Committee (DGAC)**, comprised of medical and nutrition experts, is appointed.

The DGAC reviews all scientific evidence and prepares a *Scientific Report*.

The report is made available for public comment.

The USDHHS and USDA (government agencies) create the final *Dietary Guidelines* (they may not use all DGAC recommendations).

Core Principle: A healthy eating pattern includes foods from all 5 food groups and oils, but **limits** saturated fats, trans fats, added sugars, and sodium.

Applying the *2020–2025 Dietary Guidelines for Americans* to Usual Food Choices

If One Usually Eats	Consider Replacing With
White bread and rolls	Whole-wheat bread and rolls
Sugary ready-to-eat cereals	Overnight oats with berries, bananas, peaches, or other fruit you choose to add
Cheeseburger, French fries, and a regular (sugar-sweetened) soft drink	Roasted chicken or turkey sandwich, carrots and peppers with hummus, and water flavored with fresh fruit
Potato salad or cole slaw	Leafy greens or three-bean salad
Doughnuts, chips, or salty snack foods	Small bran muffin or whole-wheat bagel topped with peanut butter or almond butter, unsalted nuts, and dried fruit
Regular soft drinks	Water (if desired, add fresh fruit for flavor)
Boiled vegetables	Raw or steamed vegetables (often retain more nutrients than boiled)
Canned vegetables	Low-sodium varieties or frozen vegetables (retain more nutrients during processing)
Breaded and fried meat, fish, or poultry	Broiled or roasted meat, fish, or poultry
Fatty meats such as barbecued ribs, sausage, and hot dogs	Chicken, turkey, or fish; lean meats such as ground round
Whole or 2% cow's milk, cottage cheese with 4% fat, or yogurt made from whole cow's milk	1% milk, low-fat cottage cheese (1% fat), or low-fat yogurt
Regular ice cream	Low-sugar frozen yogurt or "lite ice cream"
Cream cheese	Low-fat cottage cheese (mashed) or reduced-fat cream cheese
Creamy salad dressings or dips made with mayonnaise or sour cream	Oil and vinegar dressing, reduced-fat salad dressings, or dips made from low-fat sour cream or plain yogurt
Chocolate chip or cream-filled cookies	Fruit-filled bars, oatmeal cookies, or fresh fruit
Salt added to season foods	Herbs, spices, or lemon juice

Food Guides: A Brief History

1943: First USDA guide based on RDAs (7 food categories).

Mid-1950s: Simplified to 4 food groups (Milk, Meat, Fruit & Vegetable, Bread & Cereal).

1984: The Food Wheel (5 groups).

1992: Food Guide Pyramid

Incorporated knowledge about health benefits and risks.

Layered format, with grains at the base.

2005: MyPyramid Plan

Online, interactive food guidance system

The Food Guide Pyramid



MyPyramid



MyPlate (2011 - Present)

In 2011, the USDA replaced MyPyramid with **MyPlate**.

It's an interactive dietary and menu-planning guidance system.

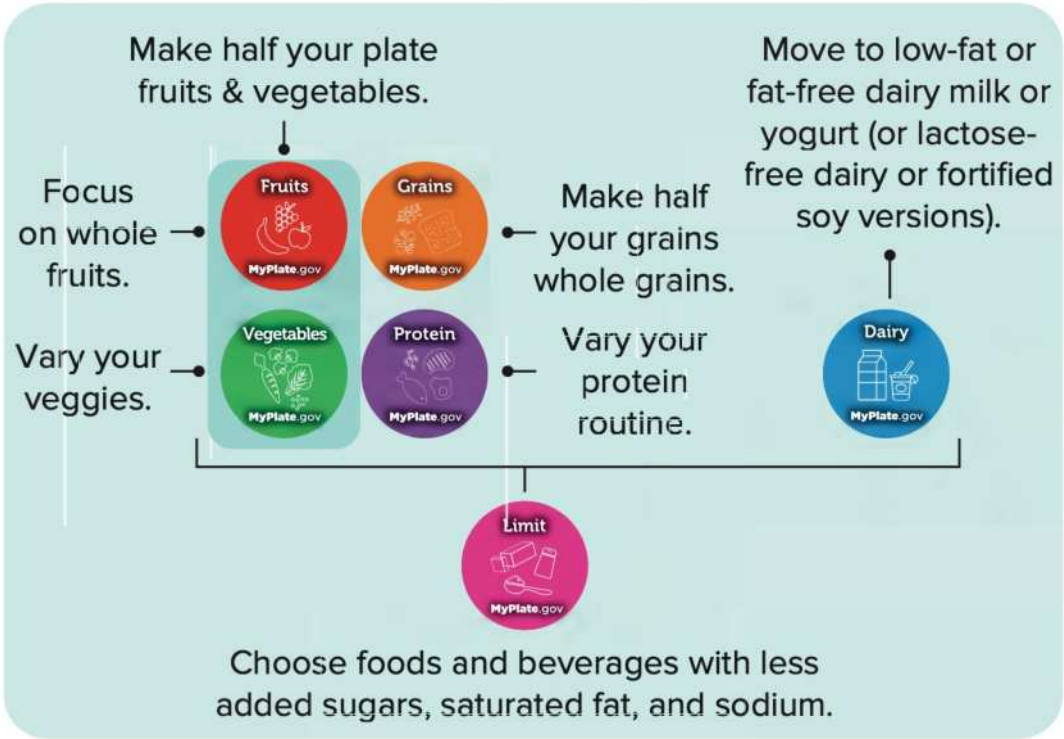
Focuses on 5 food groups: Fruits, Vegetables, Protein, Grains, and Dairy.

The website (https://www.myplate.gov/) offers:

- "MyPlate Plan" for personalized daily needs.
- Information on each food group.
- "Start Simple with MyPlate" App.

Guidelines for physical activity.

MyPlate Plan



What Are Empty Calories?

Definition: Energy (calories) supplied by **solid fats** and **added sugars**.

Alcohol-containing beverages are also a source of empty calories.

Major Sources of Empty Calories in American Diets:

Cakes, cookies, pastries, doughnuts (added sugars & solid fat)

Sugar-sweetened drinks (sports drinks, energy drinks, soda) (added sugars)

Cheese and pizza (solid fat)

Ice cream (added sugars & solid fat)

Fatty meats (sausage, hot dogs, ribs) (solid fat).

In a healthy plan: After eating all recommended nutrient-dense foods, a small number of discretionary "empty calories" may remain (e.g., 240 kcal in a 2000-kcal plan).

Individualizing MyPlate

The MyPlate website can create a personalized daily food plan.

It is based on an estimate of your daily energy needs and indicates how much to eat from each food group.

Personalization is based on:

- Age
- Sex
- Height
- Weight
- Physical activity level

MyPlate: Recommendations for Average, Healthy 20-Year-Old Adults

MyPlate Guidelines (Daily)	Females	Males
Kilocalories (kcal)	1600-2400	2600-3200
1. Fruits	1.5-2 cups	2-2.5 cups
2. Vegetables	2.5-3 cups	3.5-4 cups
3. Grains	5-8 oz (150-240 g)	9-10 oz (270-300 g)
4. Protein foods	5.0-6.5 oz (150-190 g)	6.5-7 oz (190-210 g)
5. Dairy	3 cups	3 cups

MyPlate Tips: Mixed Dishes & Portion Sizes

Classifying Mixed Dishes (e.g., Pizza, Casseroles):

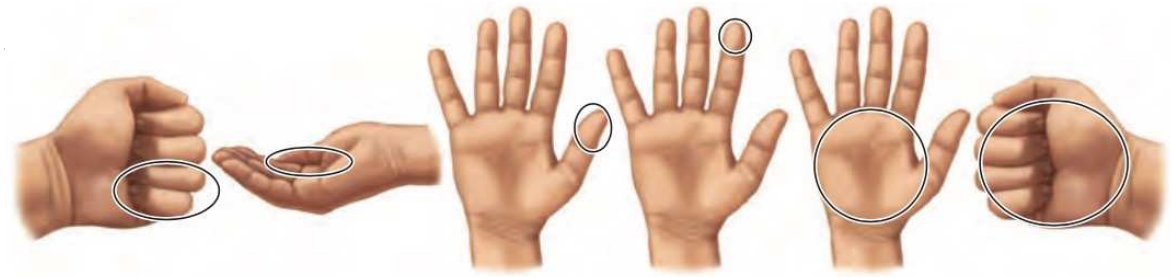
Determine all the ingredients (e.g., Pizza = crust, sauce, cheese).

Classify each ingredient into its appropriate food group (Crust=Grains, Sauce=Vegetable, Cheese=Dairy).

Estimate the number of cups or ounces for each ingredient.

Estimating Portion Sizes:

Instead of measuring cups, you can use familiar objects or your hand to judge amounts.



½ cup	1 oz	1 Tbsp	1 tsp	3 oz	1 cup
Pasta, rice, or potatoes	Salad dressing or cheese	Peanut butter or vegetable oil	Margarine or butter	Cooked meat or chicken	Large apple or cereal
					

Sustainability and Your Diet

Sustainable Diets: Diets with low environmental impacts that contribute to food/nutrition security and a healthy life for present and future generations.

Characteristics of a Sustainable, Healthy Diet:

- Is diverse and includes a wide variety of foods.
- Is balanced (energy intake vs. energy needs).
- Is based on minimally processed whole grains, legumes, fruits, and vegetables.
- **Limits meat intake**.
- Includes moderate amounts of dairy or alternatives.
- Includes unsalted seeds/nuts and small quantities of fish.
- Limits foods high in certain fats, added sugars, and/or salt.
- Focuses on **water** as the preferred beverage.

Other Menu-Planning Tools

Choose Your Foods:

A tool for estimating the energy, protein, carbohydrate, and fat contents of foods.

Commonly used for managing **diabetes** and for **weight management**.

It categorizes foods into 3 main groups: Carbohydrates, Meat/Meat Substitutes, and Fats.

It uses **exchange lists**, where foods on the same list (e.g., the fruit list) provide about the same amount of macronutrients and calories and can be "exchanged".

Carb Counting:

Another popular menu-planning tool, especially for people with diabetes.

The Role of Culture in Nutrition

Culture: All the knowledge and values shared by a group.

The Iceberg Model of Culture:

- **Surface Culture (Seen):** Observable actions like language, dress, food choices, holidays.
- **Deep Culture (Unseen):** Perceptions, attitudes, beliefs, values, religion, work ethic.



The Role of Culture in Nutrition

Key Terms:

- **Cultural Competence:** The capacity to work effectively in cross-cultural situations.
- **Cultural Humility:** An *ongoing process* of self-reflection, being aware of one's own biases, and managing power imbalances.
- **Cultural Competemility:** Viewing cultural competence through the lens of cultural humility.

Constructs of Cultural Competence (ASKED)

The mnemonic "A.S.K.E.D." helps remember the 5 constructs of cultural competence.

Awareness: Being aware of your *own* cultural identities and potential biases.

Skill: Knowing how to conduct a medical/food history and physical assessment in a culturally sensitive manner.

Knowledge: Understanding culturally specific disease rates and the role of social determinants of health.

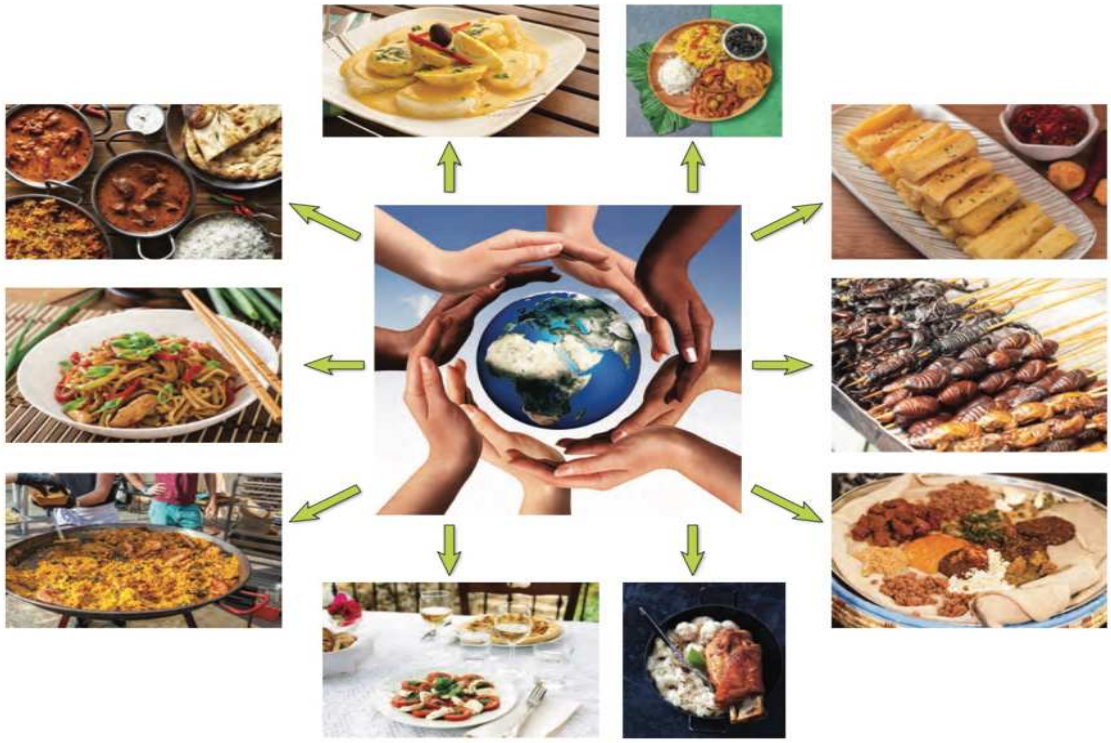
Encounters: Spending time with people from different cultures and suspending judgment during interactions.

Desire: Truly caring about recognizing and discussing cultural differences (engaging in the process because you *want* to).

Cultural Competemility Summary

Cultural awareness	Cultural skill	Cultural knowledge	Cultural encounters	Cultural desire
Being sensitive to similarities and differences between one's own culture and the cultures of others.	Conducting a medical history, food history, and a nutrition-focused physi-cal assessment in a culturally sensitive manner.	Understandin g culturally specific disease prevalence rates and the role of social determinants of health in dis- ease risk.	Suspending judgment in one's interactions with cultur-ally different others.	Exhibiting cultural knowledge and skills because one truly cares about recog-nizing and dis-cussing cultural differences.

Global Influences on Food Choices



Food and Dietary Supplement Labels

Regulated by: The Food and Drug Administration (FDA).

What's Required on a Label?

Product's name.

Manufacturer's name and address.

Amount of product in the package.

List of ingredients (in descending order by weight).

Country of Origin:

Must be declared for fresh/frozen fruits & vegetables, chicken, fish, shellfish, and certain nuts.

Daily Values (DVs)

DVs (Daily Values): A simplified, practical set of nutrient standards developed *for labeling purposes*.

Based on: A standard **2,000 kcal/day** diet.

How to Use DVs:

Goal: Obtain *at least* 100% of the DV for fiber, vitamins, and minerals.

Limit: Consume *less than* 100% of the DV for total fat, saturated fat, added sugars, and sodium.

The 5/20 Rule of Thumb:

5% DV or less of a nutrient = a **low** source.

20% DV or more of a nutrient = a **high** source.

The Nutrition Facts Panel

The FDA-required format for displaying energy and nutrient content.

Serving Size:

Must be listed in household units (e.g., 2/3 cup) AND grams (e.g., 55g).

Serving sizes are standardized among similar foods (e.g., all ice cream brands must use 2/3 cup).



What's on the NEW Nutrition Facts Panel?

Required Nutrients:

- Total Fat, Saturated Fat, Trans Fat
- Cholesterol, Sodium
- Total Carbohydrate, Dietary Fiber
- Total Sugars, **Includes Added Sugars** (NEW)
- Protein
- **Vitamin D** (NEW)
- **Potassium** (NEW)
- Calcium
- Iron

What's on the NEW Nutrition Facts Panel?

What's Removed?

- "Calories from fat" is no longer required.
- Vitamins A and C are no longer required (but D and Potassium are).

Exemptions: Fresh fruits/vegetables, fish, shellfish, and meats are not required to have labels (though many provide this info voluntarily).

Label Claims: 1) Health Claims

Health Claim: An FDA-permitted claim on a label that describes the relationship between a food and a *reduced risk* of a disease or health-related condition.

NOT Allowed: Cannot claim to "diagnose, cure, mitigation, or treatment of disease".

> *Example (Not Allowed):* "Prevents heart disease".

Allowed: Must be worded carefully.

> *Example (Allowed):* "Diets low in saturated fat and cholesterol *may reduce the risk* of heart disease".

Restrictions: A food cannot make a health claim if it contains high levels of "bad" nutrients (e.g., >4g saturated fat per serving). This is why whole milk (high in sat. fat) cannot make a claim about calcium and osteoporosis.

Label Claims: 2) Structure/Function

Structure/Function Claim:

- Describes the role a nutrient plays in maintaining a structure (bone) or a normal bodily function (digestion).
- **Does not** require FDA approval, but cannot mention a disease.
- *Example*: "Calcium builds strong bones" (Allowed).
- *Example*: "Fiber maintains bowel regularity" (Allowed).
- *Example*: "Promotes low blood pressure" (Not allowed, implies treating a disease).

Label Claims: 3) Nutrient Content

Nutrient Content Claim:

- Describes the *level* of a nutrient in a product using terms like
- "Free,"
- "High,"
- "Low,"
- "Reduced," or
- "More".

These terms are legally defined by the FDA.

Nutrient Content Claim Definitions (1)

Sugar	**Sugar Free:** The product provides less than 0.5 g sugar per serving. **Reduced Sugar:** The food contains at least 25% less sugar per serving than the reference food.
Calorie	**Calorie Free:** The food provides fewer than 5 kcal per serving. **Reduced or Fewer Calories:** The food contains at least 25% fewer kcal per serving than the reference food.
Fat	**Fat Free:** The food provides less than 0.5 g fat per serving. **Low Fat:** The food contains 3 g or less fat per serving. two percent milk is not "low fat" because it has more than 3 g fat per serving. the term *reduced fat* can be used to describe 2% milk. **Reduced or Less Fat:** the food supplies at least 25% less fat per serving than the reference food.

Nutrient Content Claim Definitions (2)

Fiber	**High Fiber:** The food contains 5 g or more fiber per serving. foods that include high-fiber claims on the label must also meet the definition for low fat. **Good Source Of Fiber:** The food supplies 2.5–4.9 g fiber per serving.
Meat and Poultry Products Regulated by the USDA	**Extra Lean:** The food provides less than 5 g fat, 2 g saturated fat, and 95 mg cholesterol per serving. **Lean:** The food contains less than 10 g fat, 4.5 g saturated fat, and 95 mg cholesterol

Dietary Supplement Labels

Required Panel: Must have a **"Supplement Facts"** panel.

Must identify: The product as a "Dietary Supplement" (or similar, e.g., "Herbal supplement").

Panel Lists: Serving size, amount per serving, and %DV (if a DV has been established).

Claims:

Supplements are **not** intended to "treat, diagnose, cure, or alleviate" diseases.

They **are allowed** to use Structure/Function claims (e.g., "Iron is necessary for healthy red blood cell formation").

Supplement Labels & The FDA Disclaimer

FDA Disclaimer: If a supplement label makes a structure/function claim that the FDA has *not* evaluated, it **must** include this disclaimer:

"This statement has not been evaluated by the Food and Drug Administration. This product is not intended to diagnose, treat, cure, or prevent any disease."

Manufacturer Responsibility: Manufacturers must notify the FDA of their claims but are not required to provide evidence *before* marketing. They are responsible for ensuring their claims are truthful and not misleading.

Organic Foods

Organically Produced Foods: Produced *without* the use of:

- Antibiotics or hormones
- Synthetic fertilizers and pesticides
- Genetic improvements (GMOs)
- Ionizing radiation

Popularity: In 2022, organic food sales were $61.7 billion, making up 6% of total U.S. food sales.

Organic Foods

Are they more nutritious?

- The science is mixed. Organic crops may have higher antioxidants; conventional crops may have higher protein.
- Nutritionally relevant differences are "increasingly accepted," but the ultimate impact on human health remains uncertain.
- Organic foods *have been shown* to contain fewer pesticides and fewer antibiotic-resistant bacteria.

Organic vs. Conventional Farming

Organic	Conventional
Synthetic fertilizers are not allowed.	Limited restrictions on fertilizers
Sewage sludge products are not allowed.	Sludge products may be used on some fields.
Restrictions on use of raw manure on fields used for food crops.	Few restrictions on raw manure use for edible crop fields.
Conventional pesticides are not allowed; natural pest management practices are encouraged.	Any government-approved pesticide may be used according to label instructions. Natural pest management practices may also be used.
Genetically modified organisms (GMOs) are not allowed.	Government-approved GMOs are permitted.

Organic vs. Conventional Farming

Organic	Conventional
Feeding livestock mammal and poultry by-products and manure is not allowed.	Certain mammal and poultry by-products are allowed in livestock feed.
Use of growth hormones and antibiotics in livestock production is not allowed.	Government-approved hormone and antibiotic treatments are permitted.
Food irradiation (a food preservation method) is not allowed.	Food irradiation may be used.
Detailed recordkeeping and site inspections by regulators are required.	Some records are required, but no on-site checks by regulators are necessary.

Labeling Organic Foods

- **USDA Organic Symbol:** This symbol indicates the product meets the USDA's strict standards for organic food.
- **Organic Labeling Categories:**



"100% Organic"

- *Requirement:* 100% organic ingredients.
- *Seal:* May use USDA seal.

"Organic"

- *Requirement:* Contains at least 95% organic ingredients.
- *Seal:* May use USDA seal.

"Made with organic ingredients"

- *Requirement:* Contains 70-95% organic ingredients.
- *Seal:* **Cannot** use USDA seal.



PRINCIPLE OF NUTRITION

Dr. Marwan Jalambo
Assistant Professor (Nutrition)

3

PART 1: Human Digestion, Absorption, & Transport

Overview of the Digestive System

Body cells need nutrients, not food.

Primary Roles of the Digestive System:

1. Break down food into smaller components (nutrients).
2. Absorb nutrients.
3. Eliminate solid waste products.

Key Terms:

1. **Digestion:** The process of breaking down large food molecules into usable nutrients.
2. **Absorption:** The process of nutrient uptake from the gastrointestinal (GI) tract.

Components of the Digestive System

Gastrointestinal (GI) Tract: A hollow, muscular tube (~16 feet long).

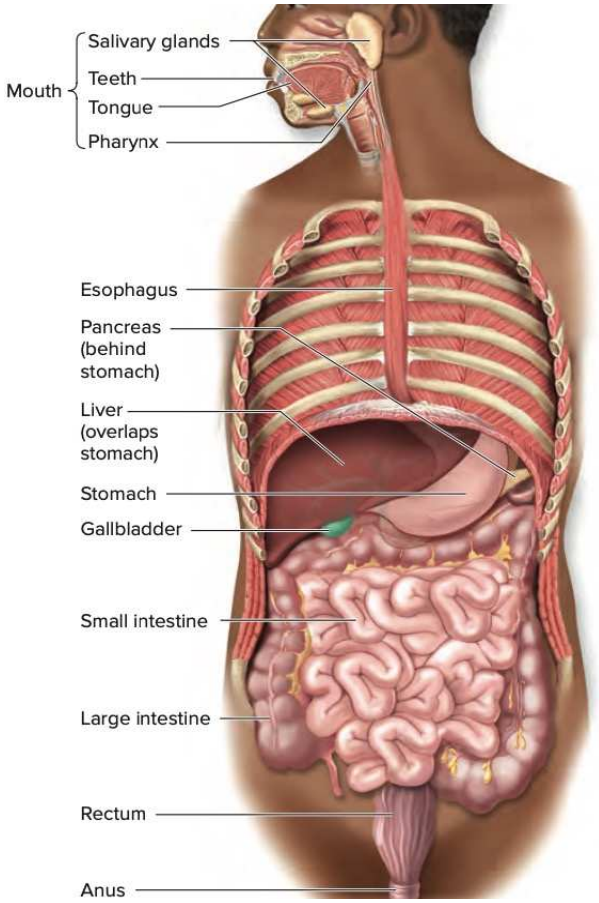
Primary Components (The GI Tract):

1. Mouth
2. Esophagus
3. Stomach
4. Small intestine
5. Large intestine (colon)
6. Rectum
7. Anus

Accessory Organs (Assist the GI Tract):

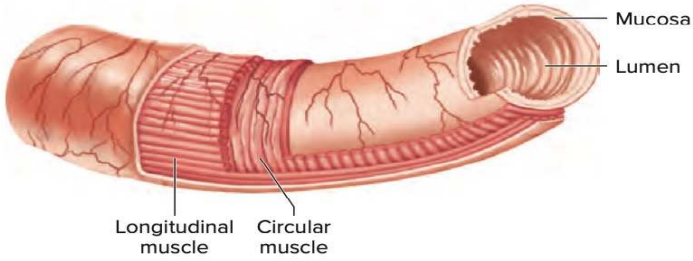
1. Teeth
2. Tongue
3. Salivary glands
4. Liver
5. Gallbladder
6. Pancreas

Figure: The Digestive System



Layers of the GI Tract Wall

Mucosa: The innermost layer, surrounds the **lumen** (the hollow space food flows through). Contains cells that secrete mucus.



Muscular Layer: Has two types of tissue:

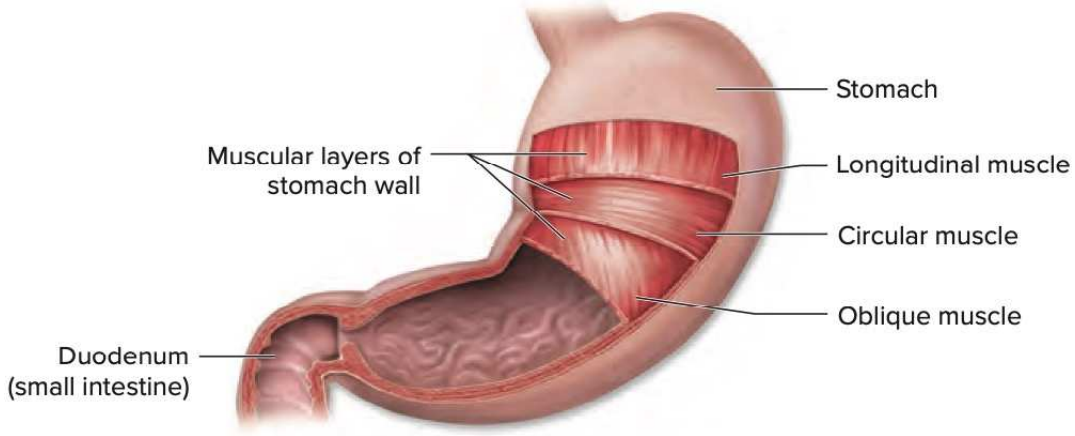
- **Circular muscles**
- **Longitudinal muscles**

These muscles relax and contract to mix and move material (motility).

Muscle Layers of the Stomach

The stomach wall is the **thickest and strongest** of the GI tract.

It has a **third muscle layer:** the **diagonal (oblique) muscle.**



Sphincters: Controlling the Flow

Sphincters: Thickened regions of circular muscle that function like valves.

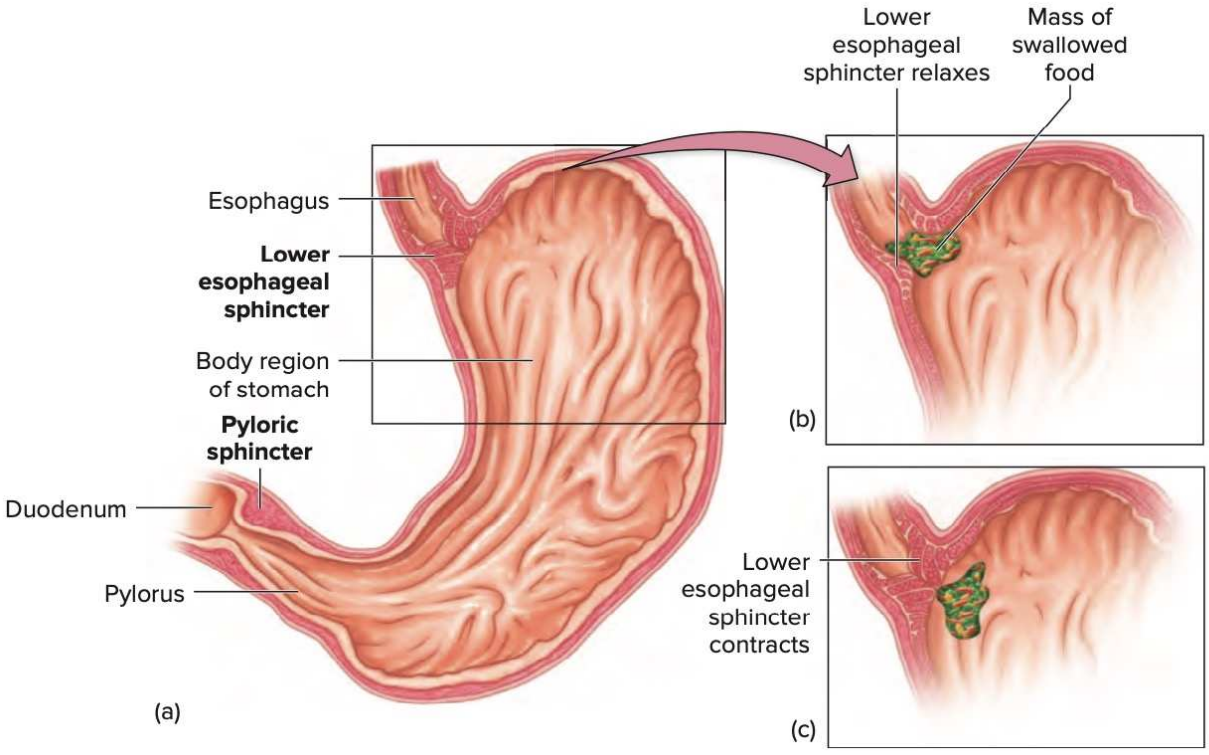
Function: They control the flow of contents at various points in the GI tract.

Relax: The passageway opens.

Contract: The passageway closes.

Importance: Essential for controlling the rates of digestion and absorption.

Examples: Lower esophageal sphincter and pyloric sphincter.



Mechanical vs. Chemical Digestion

Mechanical Digestion: The **physical breakdown** of food.

Example: Chewing a cracker into smaller pieces.

Chemical Digestion:

The breakdown of large nutrient molecules into smaller components.

Primarily driven by stomach acid and **enzymes**.

Enzyme: A protein that allows chemical reactions to occur at a rapid rate.

Lipases: Break down fats.

Proteases: Digest proteins.

Note: Water, cholesterol, minerals, and most vitamins are absorbed intact (no digestion needed).

The Mouth (Oral Cavity)

Digestion starts in the mouth.

Mechanical Digestion (Mastication):

- ✓ Chewing prepares solid foods for swallowing.
- ✓ It increases the food's surface area, giving enzymes greater access.

Tongue: Helps direct mashed-up food to the **pharynx** (throat).

Pharynx: Connects the nasal cavity with the top of the esophagus.

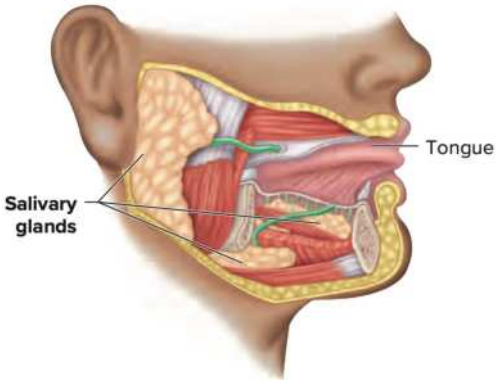
Salivary Glands

Salivary Glands: Secrete **saliva** as you chew.

Saliva: A watery fluid containing mucus and enzymes.

Function: Moistens and lubricates food to make it easier to swallow.

Production: A healthy person produces 4-6 cups (960-1440 ml) daily.



Bolus: The mass of food that has been chewed and mixed with saliva.

Enzymes in Saliva

Lysozyme: An enzyme that can destroy some bacteria in food or the mouth.

Salivary Amylase: Initiates chemical digestion by beginning the breakdown of **starch**.

Lingual Lipase: An enzyme that breaks down fat, but it is **inactive** until it reaches the stomach's acidic environment.

Note: Only a small amount of digestion occurs in the mouth because food spends little time there.

Taste and Smell

Taste Buds: Specialized sensory cells on the tongue that distinguish six basic tastes.

The Six Basic Tastes:

1. **Sweet:** (Carbohydrates)
2. **Sour:** (Acids)
3. **Salty:** (Sodium/minerals)
4. **Bitter:** (Often indicates poisons)
5. **Savory (Umami):** (Amino acids/proteins)
6. **Fat (Oleogustation):**

Taste and Smell

Smell (Olfaction): Airborne chemicals from food stimulate the olfactory region in the nasal passages.

Flavor: The brain combines information from taste buds and olfactory bulbs to identify flavors.

Disorders of Taste and Smell

Anosmia: Complete inability to detect odors.

Hypogeusia: Diminished ability to taste.

Ageusia: Total loss of the sense of taste.

Causes: Upper respiratory or middle ear infections, radiation, medications, head injury, poor dental health, or degenerative nerve diseases (e.g., Alzheimer's, Parkinson's).

COVID-19: Loss of taste and smell is a common symptom.

Impact: Can negatively affect appetite, weight, and quality of life; may lead to overuse of sugar and sodium.

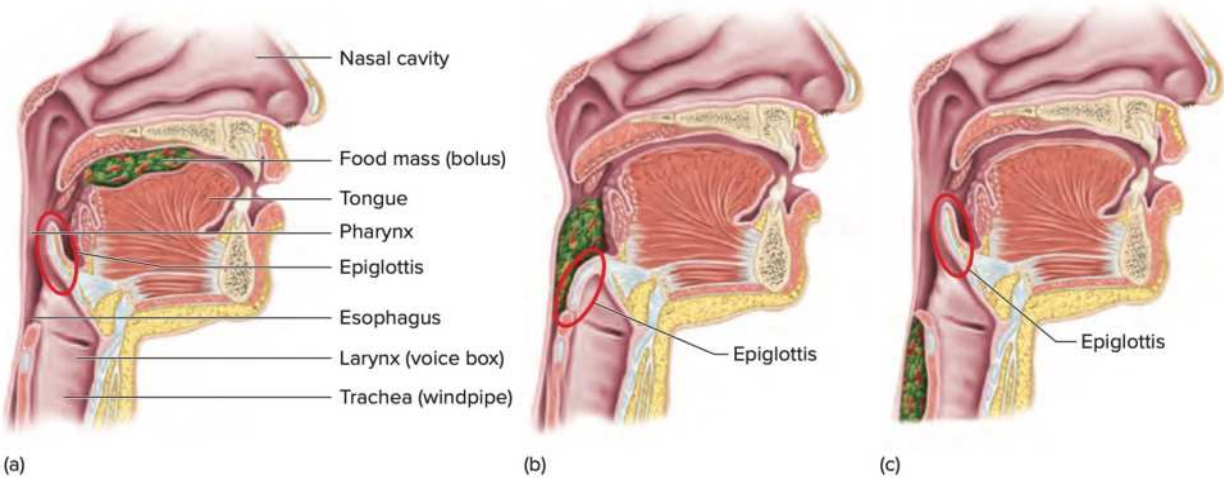
The Esophagus

Esophagus: A muscular tube, about 10 inches long, that extends from the pharynx to the stomach.

Primary Function: To transfer a bolus into the stomach.

Epiglottis: A flap of tissue that prevents food from entering the **larynx** (voice box) and **trachea** (windpipe).

Swallowing: When you swallow, breathing automatically stops, and the epiglottis covers the larynx to prevent choking.



a) While chewing, the epiglottis does not cover the larynx.

b) When swallowing, the epiglottis shifts to prevent the bolus from entering the larynx and trachea.

c) After swallowing, the epiglottis returns to its usual position and allows air to enter the larynx and trachea.

Peristalsis and Esophageal Sphincters

Peristalsis: Waves of muscular contractions that help move material through the intestine.

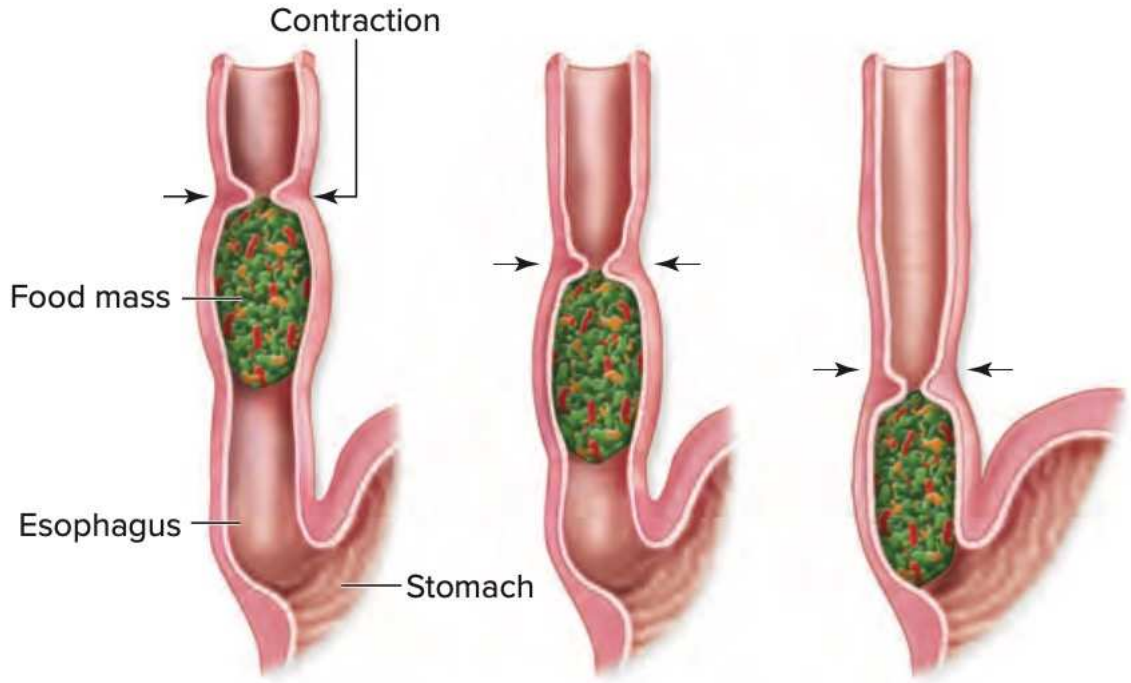
- It is an **involuntary** response.

Upper Esophageal Sphincter (UES): A section of the pharynx that relaxes to allow the bolus to enter the esophagus.

Lower Esophageal Sphincter (LES):

- Also known as the gastroesophageal sphincter.
- Located at the bottom of the esophagus.
- Normally **closed** to prevent the backflow of acidic stomach contents.

Heartburn (Acid Reflux): Occurs when the LES fails to close properly.



The Stomach

Stomach: A muscular sac that stores and mixes food.

Capacity: Can stretch to hold about 4 to 6 cups (960-1440 ml) of semiliquid material during a typical meal.

Gastric Glands: Tiny glands in the upper stomach lined with specialized cells.

Gastric Juice: Secreted by gastric glands; contains mucus, hydrochloric acid (HCl), intrinsic factor, and digestive enzymes.

Gastric Gland Cells and Secretions

Mucous Cells: Secrete **mucin** (which forms mucus).

Parietal Cells: Secrete:

1. Hydrogen and chloride ions (which form **Hydrochloric Acid, HCl**).
2. **Intrinsic Factor** (necessary for vitamin B-12 absorption).

Chief Cells: Secrete:

1. **Pepsinogen** (an inactive enzyme, or zymogen).
2. **Gastric Lipase**.

G Cells: Secrete the hormone **Gastrin**.

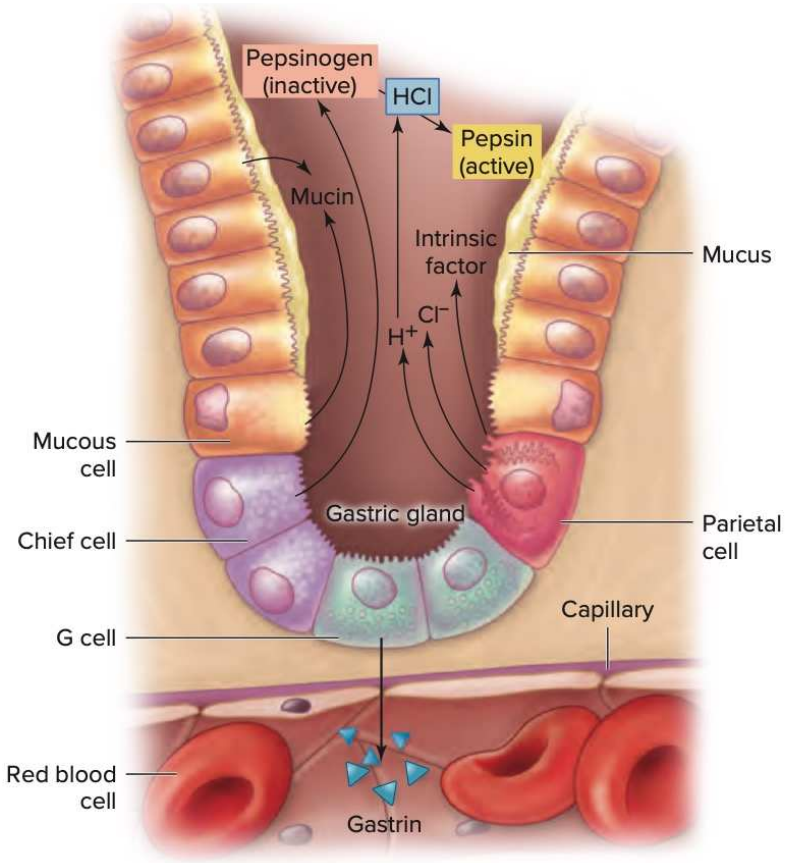


TABLE: Gastric Gland Cells

Gastric Gland Cells	Secretion	Major Functions
Mucous	Mucus	Protects stomach mucosa from HCl and enzymes
Parietal	Components that form HCl (hydrogen ions [H+] and chloride ions [Cl−])	Activates pepsinogen, destroys many food-borne microbes
	Intrinsic factor	Facilitates the absorption of vitamin B-12
Chief	Pepsinogen	Digests protein after conversion to pepsin
	Gastric lipase	Digests small amounts of fat
G cell	Gastrin	Stimulates stomach motility and gastric gland activity

Digestion in the Stomach

Chyme: The semiliquid mass formed when food mixes with gastric juice.

Pyloric Sphincter: Regulates the flow of chyme from the stomach into the small intestine.

HCl (Acid):

- Gives chyme a very acidic pH of about 2.0.
- **Stops** the action of salivary amylase (starch digestion).
- **Activates** lingual lipase (fat digestion, ~10%).

Protein Digestion:

- HCl converts inactive **pepsinogen** into active **pepsin**.
- Pepsin begins the enzymatic digestion of proteins.

Stomach Protection and Emptying

Stomach Protection:

- A thick layer of **mucus** produced by mucous cells protects the stomach wall from being damaged by its own acid and enzymes.
- **Gastric Ulcer:** A sore that forms if this mucus layer breaks down.

Stomach Emptying:

- Takes about **4 hours** following a meal.
- *Faster:* Watery meals (e.g., soup).
- *Slower:* Fatty and high-fiber meals (this increases the sense of fullness).

Dumping Syndrome: A disorder where chyme flows too rapidly from the stomach into the small intestine, often after stomach surgery.

The Small Intestine

The primary site for nutrient digestion and absorption.

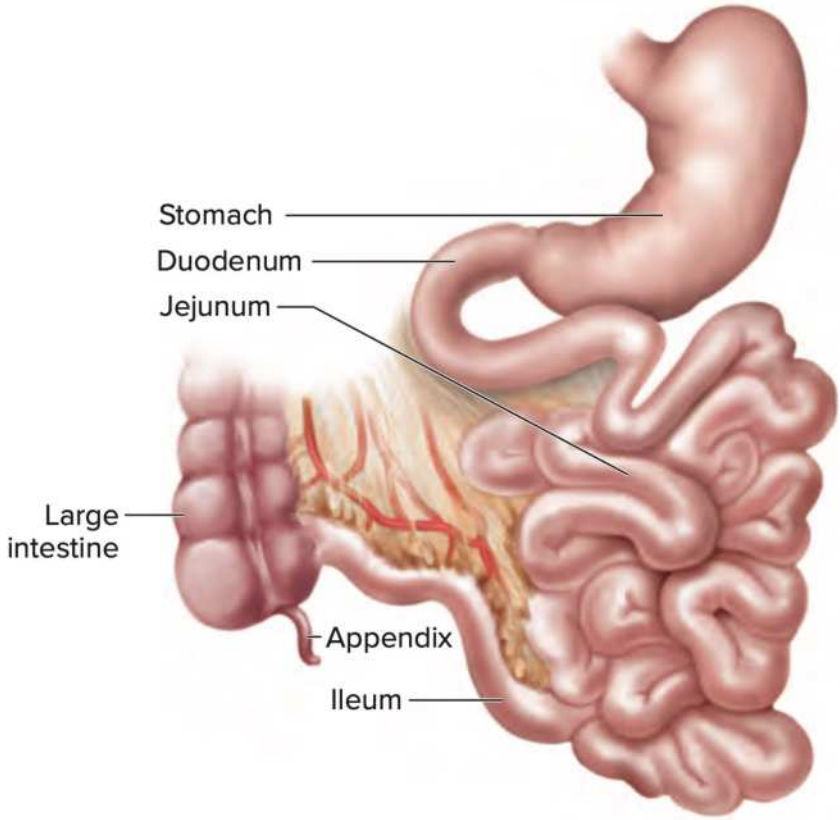
It is the longest component of the GI tract.

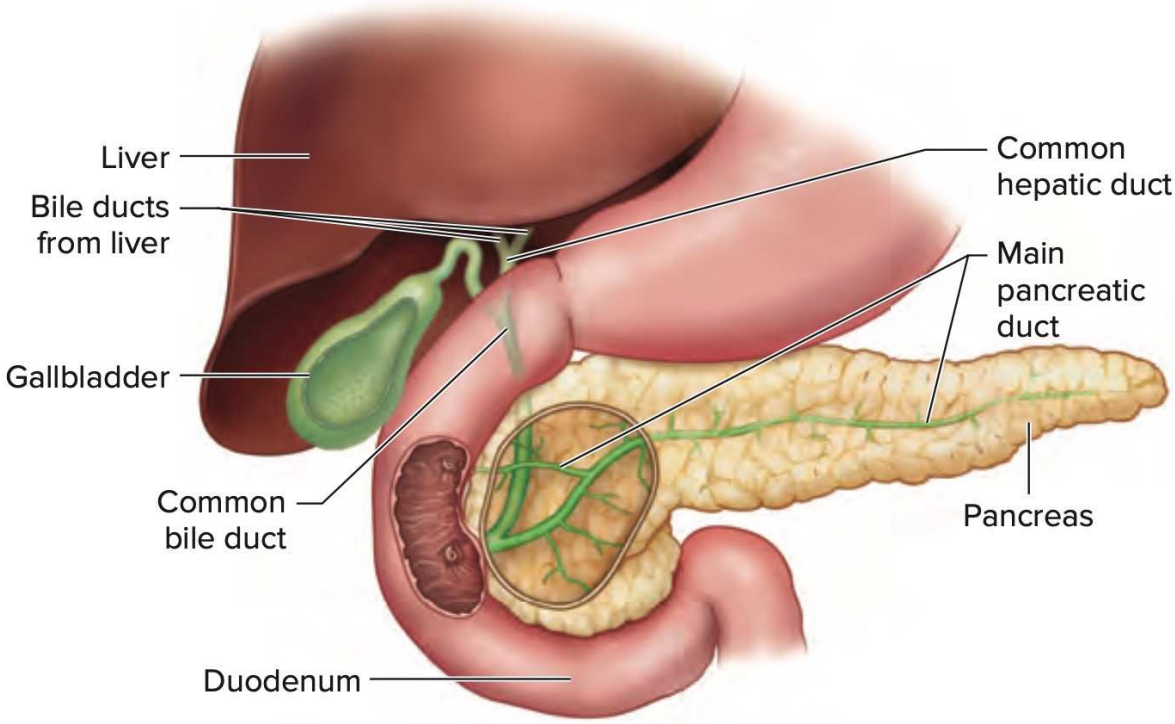
Three Sections:

1. **Duodenum:** First section (~10 inches).
2. **Jejunum:** Middle section (3 - 5.5 feet).
3. **Ileum:** Last section (5 - 9 feet).

Note: Most digestion and absorption occurs in the **duodenum and upper jejunum**.

Ileocecal Sphincter: Controls the rate of emptying from the ileum into the large intestine.


Stomach
Duodenum
Jejunum
Large intestine
Appendix
Ileum


Liver
Bile ducts from liver
Gallbladder
Common bile duct
Duodenum
Common hepatic duct
Main pancreatic duct
Pancreas

Accessory Organs: Liver, Gallbladder, Pancreas

Liver:

- Processes and stores many nutrients.
- Synthesizes cholesterol and uses it to make **bile**.
- *Bile:* A greenish-yellow fluid needed for fat digestion and absorption.

Gallbladder: Concentrates and **stores bile** until needed.

Pancreas:

- Produces and secretes many digestive **enzymes** for carbohydrates, protein, and fat.
- Secretes **bicarbonate ions (HCO_3^-)** to neutralize the highly acidic chyme from the stomach.

Table: Major Pancreatic and Intestinal Enzymes

Enzyme	Source	Action
Pancreatic amylase	Pancreas	Digests starch
Pancreatic lipase	Pancreas	Digests fat
Trypsin, chymotrypsin	Pancreas	Partially digests proteins
Carboxypeptidase	Pancreas	Breaks down certain partially digested proteins
Nuclease	Pancreas	Digests DNA and RNA into nucleotides
Peptidase	Small intestine	Breaks down certain partially digested proteins
Sucrase, maltase, lactase	Small intestine	Breaks down certain sugars into simpler sugars
Intestinal lipase	Small intestine	Digests fat

Hormones of the Small Intestine

Secretin:

- *Released from:* Duodenum and jejunum (in response to acidic chyme).
- *Action:* Stimulates the pancreas and liver to release a bicarbonate-rich solution to neutralize chyme.

Cholecystokinin (CCK):

- *Released from:* Small intestine (in response to fat and partially digested proteins).
- *Key Actions:*
  1. Stimulates the gallbladder to contract and release **bile**.
  2. Triggers the pancreas to release enzyme-rich juice.
  3. Reduces gastric secretions and motility (slows down the stomach).

Table: Actions of Key Digestive Hormones

Hormone	Primarily Released From	Secreted in Response to	Key Actions
Gastrin	Stomach	Food entering the stomach	Triggers parietal cells (HCl) & chief cells (pepsinogen); Stimulates stomach & small-intestinal motility
Secretin	Small intestine	Acidic chyme entering duodenum/jejunum	Stimulates release of bicarbonate-rich solution from liver & pancreas
Cholecystokinin (CCK)	Small intestine	Fat & protein breakdown products entering small intestine	Stimulates release of bile from gallbladder; Stimulates release of pancreatic enzymes; Decreases stomach secretions & slows motility

Villi: Increasing Surface Area

The mucosa of the small intestine is covered by **villi** (tiny, fingerlike projections).

Surface Area: These folds and villi create an enormous surface area (~200 square meters, the size of a tennis court) for efficient nutrient absorption.

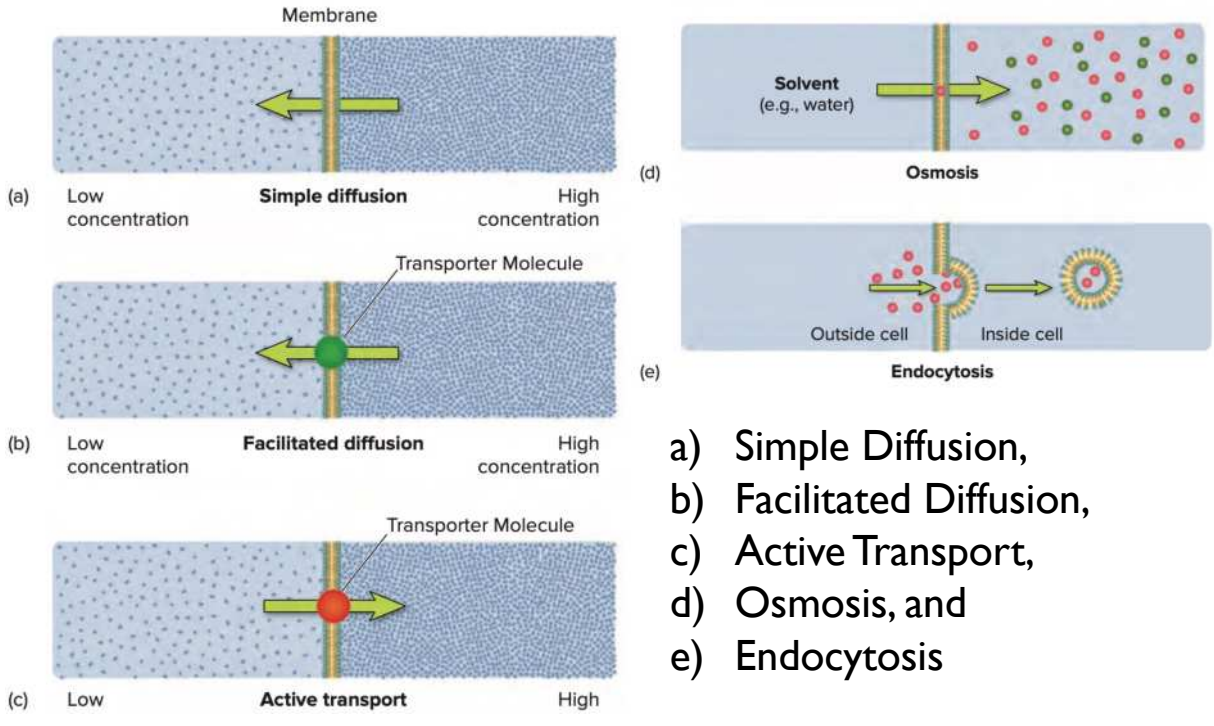
Enterocytes: The absorptive cells that form the outer layer of a villus.

Microvilli: Thousands of hairlike projections on each enterocyte, forming the **"brush border"**.

> Contain "brush border enzymes" that help digest protein and carbs.

Cell Turnover: Enterocytes live for only 3-6 days and are constantly replaced.

Nutrient Absorption Mechanisms



a) Simple Diffusion,
b) Facilitated Diffusion,
c) Active Transport,
d) Osmosis, and
e) Endocytosis

Nutrient Absorption Mechanisms

1. **Simple Diffusion:** Nutrients move down a concentration gradient (high to low). No energy required. (e.g., lipids, some vitamins) .
2. **Facilitated Diffusion:** Nutrients move down a gradient but require a special transport protein. No energy required. (e.g., fructose) .
3. **Active Transport:** Nutrients move *against* a concentration gradient (low to high). **Requires energy** and a transport protein. (e.g., glucose, amino acids) .
4. **Osmosis:** The movement of water across the cell membrane.
5. **Endocytosis:** The cell membrane surrounds and "swallows" large substances. (e.g., how infants absorb whole proteins from milk) .

Nutrient Transport & Bioavailability

Transport from Enterocytes:

1. **To Blood (Capillary Network):** Water-soluble nutrients and certain fats. They travel to the liver via the **hepatic portal vein**.
2. **To Lymph (Lacteal):** Most lipids and fat-soluble vitamins. They enter the lymphatic system.

Bioavailability: The amount of a nutrient that can be effectively digested, absorbed, and used by the body.

Factors Affecting Bioavailability:

1. *Dietary:* Chemical form of the nutrient, food processing, other foods eaten.
2. *Physiological:* Individual's digestion, nutrient status, age, sex, health.

The Large Intestine

Anatomy: About 5 feet long, with a larger diameter than the small intestine.

Path of Food Residue: Ileocecal sphincter → Cecum → Ascending colon → Transverse colon → Descending colon → Sigmoid colon → Rectum → Anus.

Functions of the Large Intestine

Absorption: Very little absorption occurs (no villi).

- Mainly absorbs **water** and **some minerals**.

Secretion: The primary secretion is **mucus** from goblet cells.

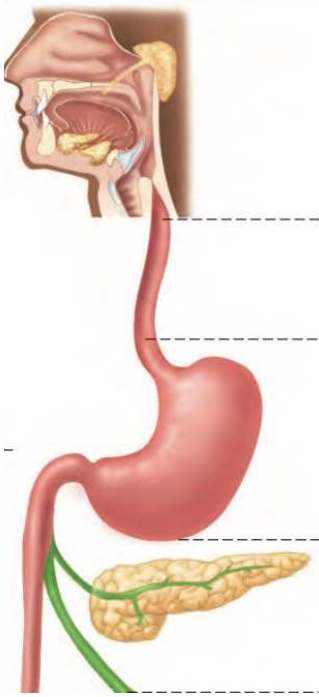
Feces (Stool): The semisolid material formed from remaining food residue.

- Composition: ~75% water. The rest is bacteria and undigested dietary fiber.

Defecation (Elimination):

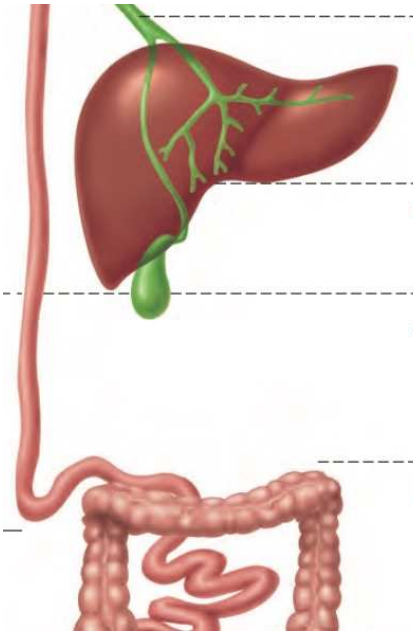
- Feces move to the **internal anal sphincter** (involuntary control), which relaxes.
- This creates the urge to defecate.
- The **external anal sphincter** is under voluntary control.

Summary of Digestion



Organ	Secretions	Digestive Functions
Mouth and salivary glands	Saliva and water	Moisten food
	Mucus	Lubrication
	Amylase	Starch-digesting enzyme
Teeth		Mastication
Esophagus		Moves food to stomach by peristaltic waves
	Mucus	Lubricates food
Stomach		Stores, mixes, dissolves, and continues digestion; regulates emptying of chyme into small intestine
	HCl	Kills microbes; activates pepsinogen to pepsin
	Pepsin	Digests proteins
	Mucus	Lubricates and protects stomach wall
Pancreas		Secretes enzymes and bicarbonate
	Enzymes	Digest carbohydrates, fats, proteins, and nucleic acids
	Bicarbonate	Neutralizes HCl entering small intestine from stomach

Summary of Digestion



Liver		Receives water-soluble products of digestion from small intestine
		Synthesizes and secretes bile
	Bile salts	Aid fat digestion
Gallbladder		Stores and concentrates bile between meals
		Releases bile during meals
Small intestine		Digests and absorbs most substances; mixes and propels contents
	Enzymes	Food digestion
	Water	Maintains fluidity of intestinal contents
	Mucus	Lubricates inner walls
Large intestine		Stores and concentrates undigested material; absorbs some minerals and water; eliminates wastes
	Mucus	Lubricates mucosa



3

PART 2: Common Digestive Tract Disorders

Microbes in the Digestive Tract

A healthy large intestine is home to **vast numbers of bacteria**.

Gut Microbiota: The ecosystem of bacteria, fungi, and viruses in the intestine.

Unique Composition: Varies from person to person based on diet, genetics, age, environment, and medication use.

Symbiotic Relationship: Microbes and their human host mutually benefit.

Actions of Intestinal Bacteria

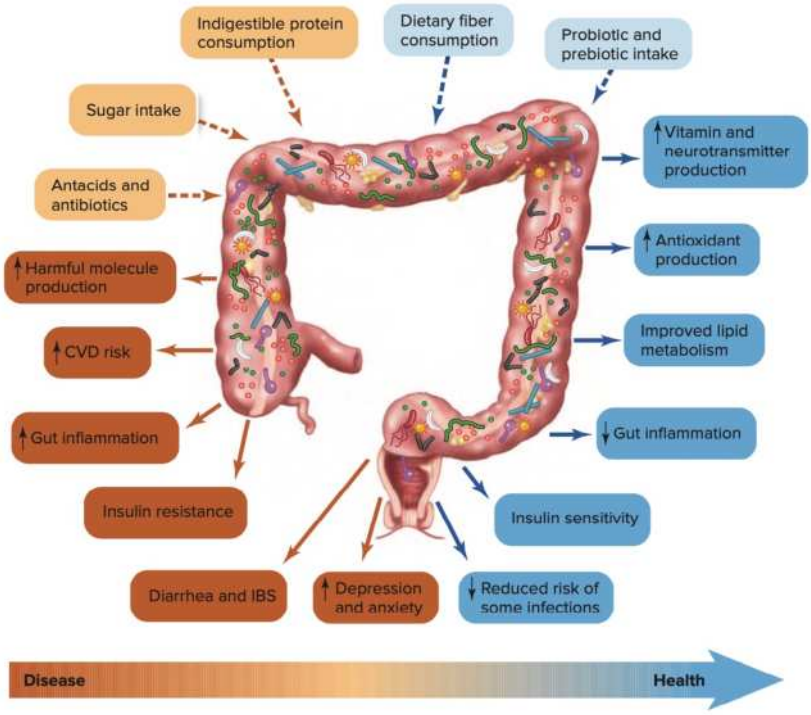
Benefits to Host:

- Help digest foods indigestible to humans (e.g., fiber).
- Produce important molecules (e.g., some vitamins, serotonin, GABA).
- Defend the body by warding off pathogens.
- Help develop the immune system and control inflammation.

Dysbiosis:

- An **imbalance** where harmful bacteria override beneficial ones.
- *Causes:* Antibiotic use, high-stress, diet high in processed foods.
- *Linked to:* Heart disease, obesity, liver diseases, anxiety, and depression.

Figure: Gut Microbiota in Health and Disease



Probiotics, Prebiotics & Fermented Foods

Probiotics:

- **Definition:** Live, beneficial microbes (primarily bacteria) that promote human health.
- **Sources:** Supplements and fermented foods like yogurt and sauerkraut.
- **Examples:** *Lactobacillus* and *Bifidobacterium*.

Prebiotics:

- **Definition:** Non-digestible substances (typically fiber) that promote the growth of beneficial bacteria.
- **Sources:** Fruits, most vegetables, and some grains (like oats).

Fermented Foods:

- Made through the controlled activity of live microbes.
- ***Examples***: Yogurt, cheese, kimchi, miso, sourdough bread.
- May be a better source of probiotics than capsules.

Fecal Microbiota Transplantation (FMT)

FMT (Stool Transplantation): The introduction of fecal material from a healthy donor into a recipient's large intestine.

Use: Highly effective for treating **Clostridium difficile (C. diff)** infection.

What is C. diff?

- A common but potentially harmful bacterium.
- Antibiotic use can kill "good" gut bacteria, allowing *C. diff* to survive, multiply, and secrete toxins that destroy intestinal tissue.

Method: Can be done via enema or "stool capsules".

Common Digestive Tract Disorders

Gastroesophageal Reflux (GER):

- Commonly called **heartburn** or **acid reflux**.
- A burning sensation caused by acidic contents from the stomach passing into the esophagus.

Gastroesophageal Reflux Disease (GERD):

- **Chronic, frequent GER**.
- *Risks of Untreated GERD:* Irritates the esophagus lining, leading to esophageal ulcers, bleeding, and a higher risk of **esophageal cancer**.

Factors Worsening GER: Excess body fat, alcohol, coffee, overeating, chocolate, greasy/spicy foods, tomatoes, smoking.

Recommendations to Reduce Heartburn

1. Avoid lying down within 3 hours after eating a meal.
2. Do not overeat at mealtimes.
3. Avoid smoking cigarettes or breathing secondhand smoke.
4. Elevate the head of the bed (6-8 inches).
5. Do not wear tight belts or clothes with tight waistbands.
6. Learn to recognize and limit foods that cause heartburn (e.g., spicy/greasy foods, peppermint, caffeine, alcohol, chocolate).
7. If obesity is contributing, losing weight may be beneficial.

Vomiting

Definition: The body's strategy for expelling noxious substances from the stomach.

Function: Prevents harmful substances from entering the small intestine.

Causes: Toxins, alcohol, blocked intestines, motion sickness, pregnancy (morning sickness), migraines.

Major Risk: Repeated vomiting can deplete body water, leading to **dehydration**.

When to see a doctor:

- Lasts > 1 day (adults) or > few hours (children).
- Signs of dehydration (dry lips, decreased urination).
- **Blood in vomit** (looks like "coffee grounds").
- Accompanied by fever, severe abdominal pain, weakness, or confusion.

Gastric (Peptic) Ulcer

Definition: A sore that occurs in the lining of the stomach or the upper small intestine.

Symptom: Deep, dull upper abdominal pain, often ~2 hours after eating.

Primary Cause: Infection with **Helicobacter pylori (H. pylori)** bacteria, which makes the stomach lining susceptible to acid damage.

Other Risk Factors:

Regular use of **NSAIDs** (aspirin, ibuprofen, naproxen).

Smoking cigarettes.

Heavy consumption of alcohol.

Treatment: Antibiotics (to kill *H. pylori*) and medications that reduce stomach acid production.

Factors That Increase the Risk of Gastric Ulcers

1	Nonsteroidal anti-inflammatory drugs (NSAIDs)
2	Smoking
3	Alcohol consumption
4	Infection with *H. pylori*
5	Genetics
6	Excess stomach acid production

Constipation

Definition: Having fewer than three bowel movements per week, or stools that are difficult to eliminate.

Causes: Lack of dietary fiber, low water intake, lack of physical activity, medications (e.g., opiates), or changes to routine.

Hemorrhoids:

Clusters of small veins in the anal canal.

Chronic constipation and straining can cause them to become inflamed and swollen, leading to pain, itching, and bleeding.

Treatment: Adding more fiber-rich foods, water, and fermented foods/probiotics to the diet.

Diarrhea

Definition: Frequent bowel movements with loose or watery stools.

Cause: Occurs when the balance of water absorption/secretion in the large intestine is abnormal.

Most cases are the body's attempt to **flush out bacterial or viral infections**.

Major Risk: Can quickly deplete body fluid, causing **dehydration** and loss of minerals (sodium and potassium).

Treatment: Avoid solid foods; drink replacement fluids that contain minerals.

Note: Prompt treatment is crucial for infants and older adults.

Irritable Bowel Syndrome (IBS)

Definition: A condition characterized by intestinal cramps and abnormal bowel function (diarrhea, constipation, or alternating episodes of both).

Cause: Unknown. Associated with intestinal infections, eating certain foods, and emotional stress.

IBS vs. IBD:

IBS does **NOT** result from chronic inflammation.

IBS does **NOT** appear to increase the risk of colorectal cancer.

Treatment:

Elimination diets (e.g., avoiding gluten, certain carbohydrates, artificial sweeteners).

Stress management strategies.

Eating fermented foods/probiotics may be helpful.

Dietary Changes to Treat IBS Symptoms

1. **Avoid or eliminate foods containing gluten, such as wheat, rye, and barley products.**
2. **Reduce consumption of or eliminate foods containing certain carbohydrates that are more difficult to digest, such as dairy products, wheat products, and some fruits and vegetables.**
3. **Reduce or eliminate consumption of foods and beverages that contain large amounts of artificial sweeteners.**

Inflammatory Bowel Disease (IBD)

Definition: Characterized by **chronic inflammation** of the GI tract.

Nature: An **autoimmune disorder**.

Symptoms: Diarrhea, rectal bleeding, abdominal cramps, fever, and unintentional weight loss.

Phases: Patients experience "flares" (symptom periods) followed by "remissions" (healthy periods).

Causes: Unknown, but an environmental trigger (e.g., infection, stress, diet, smoking) may cause the immune system to function abnormally.

Treatment: May include **Exclusive Enteral Nutrition (EEN)**—a nutrient-rich formula instead of food for 6-8 weeks.

Types of IBD: Crohn's vs. Ulcerative Colitis

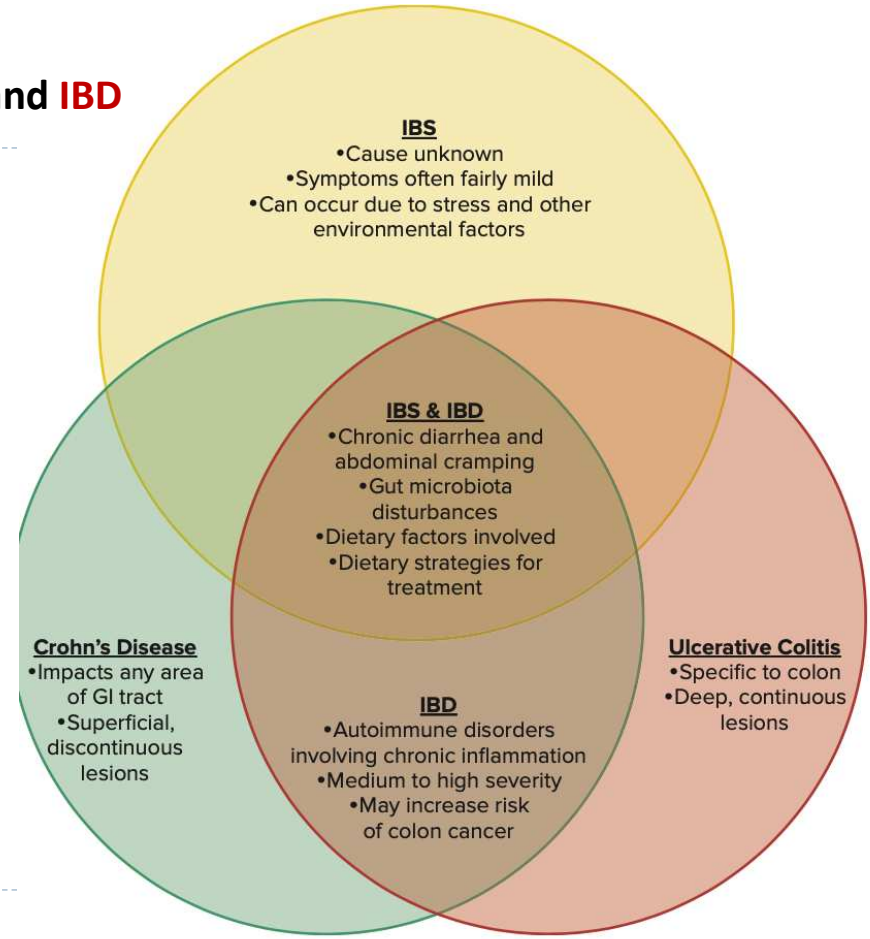
Crohn's Disease:

- Causes **patches** of inflammation.
- Can affect **any part** of the GI tract (mouth to anus).
- Can affect all layers of the intestinal wall.
- Can result in intestinal abscesses, blockages, and **fistulas** (abnormal openings).

Ulcerative Colitis (UC):

- Develops ulcers in a **continuous pattern**.
- Affects **only the colon and rectum**.
- Both types increase the risk of colon cancer.

Comparison of IBS and IBD



Diverticulosis and Diverticulitis

Diverticula: Small pouches that form in the inner lining of the large intestine, pushing through weak regions.

Diverticulosis: The condition of *having* these pouches. It is common and most people have no symptoms.

Diverticulitis: Occurs if bacteria or feces become trapped in the pouches, causing **painful inflammation and infection**.

Complications: Can damage blood vessels, causing rectal bleeding.

Treatment: May require antibiotics and, in some cases, surgery.


Thank you!

PRINCIPLE OF NUTRITION

Dr. Marwan Jalambo
Assistant Professor (Nutrition)

4

LIPIDS (FATS)

Part 1: Triglycerides, Phospholipids, & Sterols

What Are Lipids?

- **Lipids** are a class of nutrients that **do not dissolve in water**.
- This includes triglycerides, phospholipids, and sterols.
- This property is key to understanding their function and transport in the body, which is mostly water.

Example: Oil (lipid) floats on top of vinegar (water).

Major Functions of Lipids

- ✓ Serve as structural components of cell membranes.
- ✓ Store energy (as subcutaneous fat).
- ✓ Insulate against cold temperatures.
- ✓ Protect against minor bruising and cushion major organs.
- ✓ Serve as important signaling molecules.
- ✓ Enhance the flavor, texture, and aroma of food.

TABLE: Major Functions of Lipids in the Body

The body uses fats and other lipids to:
Absorb fat-soluble vitamins and phytochemicals
Cushion the body against bumps and blows
Form and maintain cell membranes
Participate in cell signaling
Insulate the body against cold temperatures
Produce steroid hormones
Provide and store energy (triglycerides)

Fatty Acids

Fatty Acid Structure

- A **fatty acid** is a hydrocarbon chain (carbon and hydrogen atoms).
- **Omega (methyl) end:** A carbon with three hydrogen atoms (−CH3).
- **Carboxylic acid end:** (−COOH).
- **Chain Lengths:**
  - **Short-chain:** 2 to 4 carbons
  - **Medium-chain:** 6 to 12 carbons
  - **Long-chain:** 14 to 24 carbons

Fatty Acid Saturation

Saturated Fatty Acid (SFA):

- Each carbon in the chain is "saturated" (filled) with hydrogen atoms.
- Contains **only single bonds** between carbons.

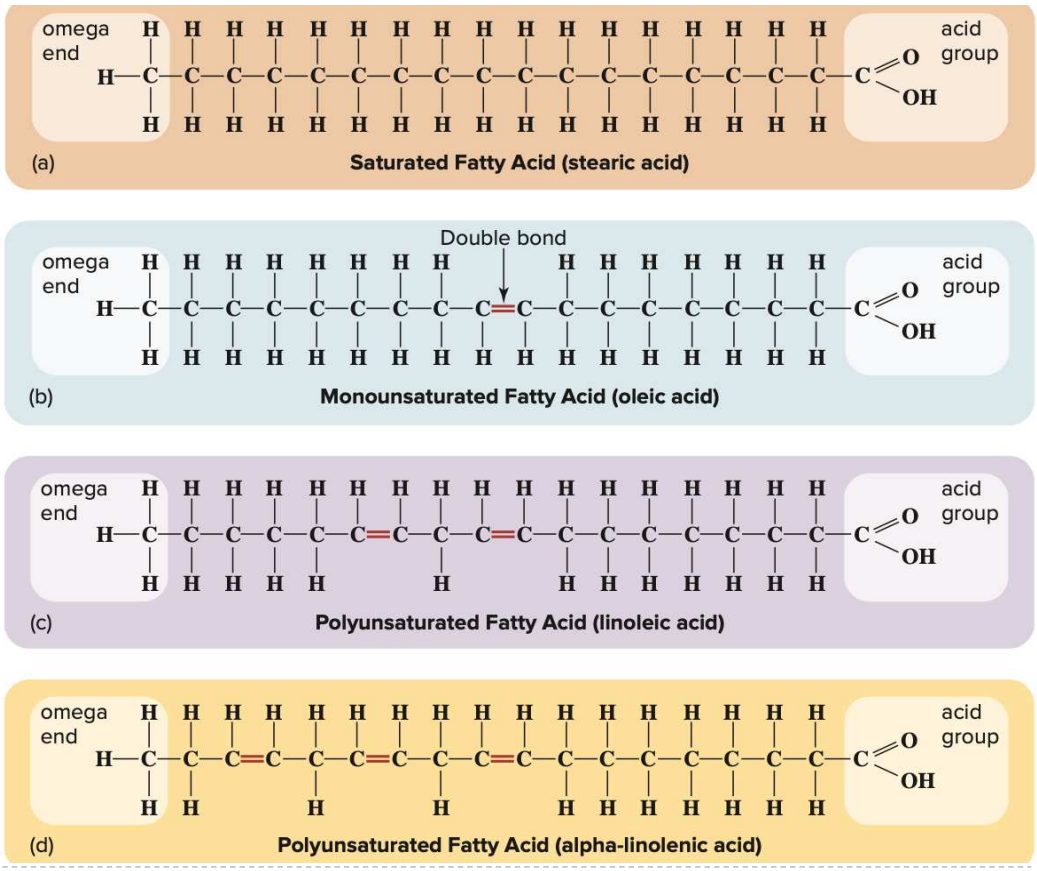
Unsaturated Fatty Acid:

- Is missing hydrogen atoms.
- Contains **at least one double bond** between carbons.

Types of Unsaturated Fats:

- **Monounsaturated (MUFA):** Has *one* double bond.
- **Polyunsaturated (PUFA):** Has *two or more* double bonds.

FIGURE: Fatty acids



Omega Fatty Acids

PUFAs are classified by the location of the first double bond from the **omega (methyl) end**.

Omega-3 Fatty Acid:

- First double bond is at the **third** carbon.
- Example: **Alpha-linolenic acid**.

Omega-6 Fatty Acid:

- First double bond is at the **sixth** carbon.
- Example: **Linoleic acid**.

Essential Fatty Acids

The body **cannot synthesize** these; they must be supplied by the diet.

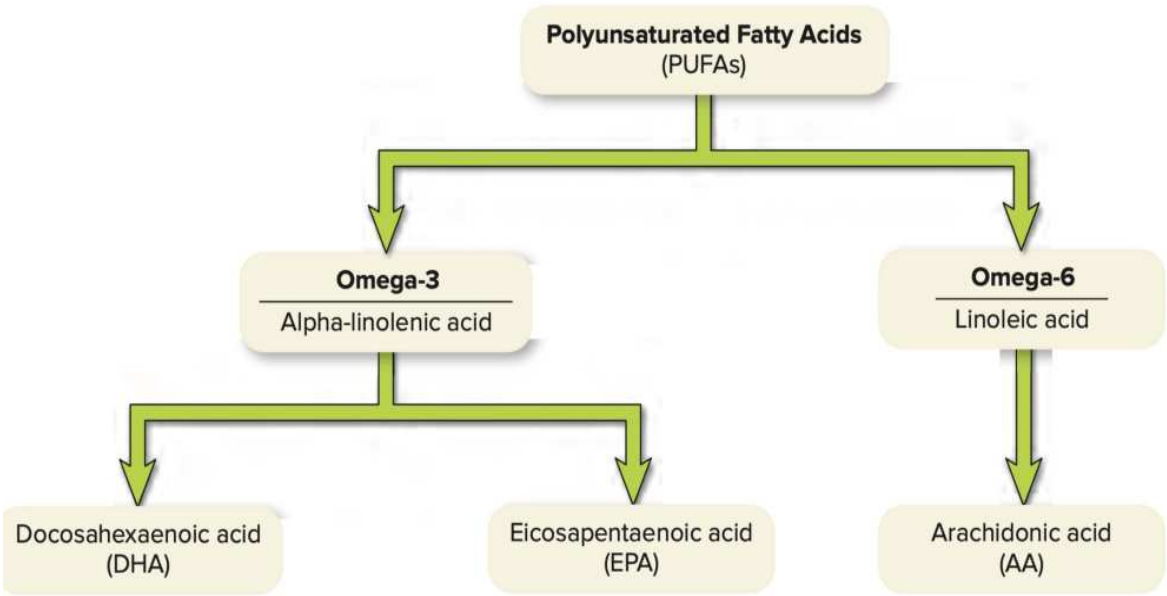
1. **Linoleic acid** (an omega-6 fatty acid)
2. **Alpha-linolenic acid** (an omega-3 fatty acid)

Adequate Intake (AI) for adults:

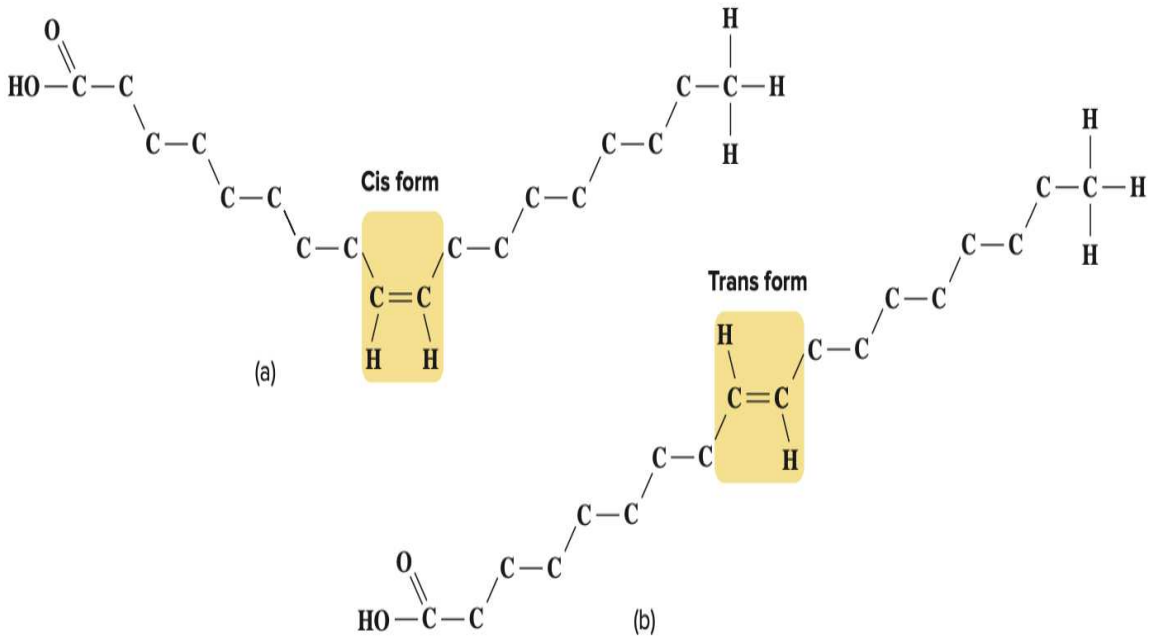
1. **Alpha-linolenic acid: 1.6** g/d (males), **1.1** g/d (females).
2. **Linoleic acid: 17**g/day (males), **12**g/day (females, 19-50).

Deficiency Signs: Scaly skin, hair loss, poor wound healing, and improper growth in infants.

Figure: Essential Fatty Acid Conversion



Trans Fats vs. Cis Fats



Trans Fats vs. Cis Fats

Cis Fatty Acid:

- Most naturally occurring unsaturated fats.
- Hydrogen atoms on the **same side** of the double bond.
- This creates a "kink" or bend in the chain.

Trans Fatty Acid:

- Hydrogen atoms on **opposite sides** of the double bond.
- This makes the chain **straighter**, similar to a saturated fatty acid.
- Fats with high trans fat are more solid at room temperature.

Trans Fats in Foods

Small amounts of natural trans fat (e.g., conjugated linoleic acid (CLA)) are in milk and meat.

Most (artificial) trans fats are created through **partial hydrogenation**.

- This process adds hydrogen to liquid vegetable oil.
- The resulting **Partially Hydrogenated Oils (PHOs)** are more solid and stable, extending shelf life and preventing rancidity.

Health Risk: PHOs raise unhealthy cholesterol, increasing heart disease risk.

FDA Ban: The FDA banned PHOs from processed foods as of 2020.

Triglycerides

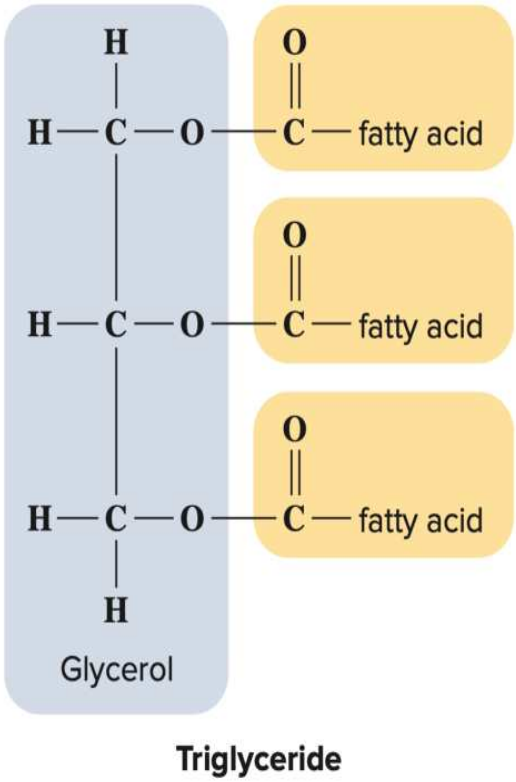
Triglyceride Structure

Triglycerides comprise about **95% of lipids** in food and in the human body.

Structure: Composed of 3 fatty acids bonded to a **glycerol** "backbone".

Related Structures:

- **Monoglyceride:** Glycerol + 1 fatty acid.
- **Diglyceride:** Glycerol + 2 fatty acids.



Triglyceride

Triglycerides in Foods

Most foods contain a **mixture** of SFA, MUFA, and PUFA.

Animal Foods:

- Generally contain **more saturated fatty acids** (SFAs).
- Examples: Beef fat, butter fat.

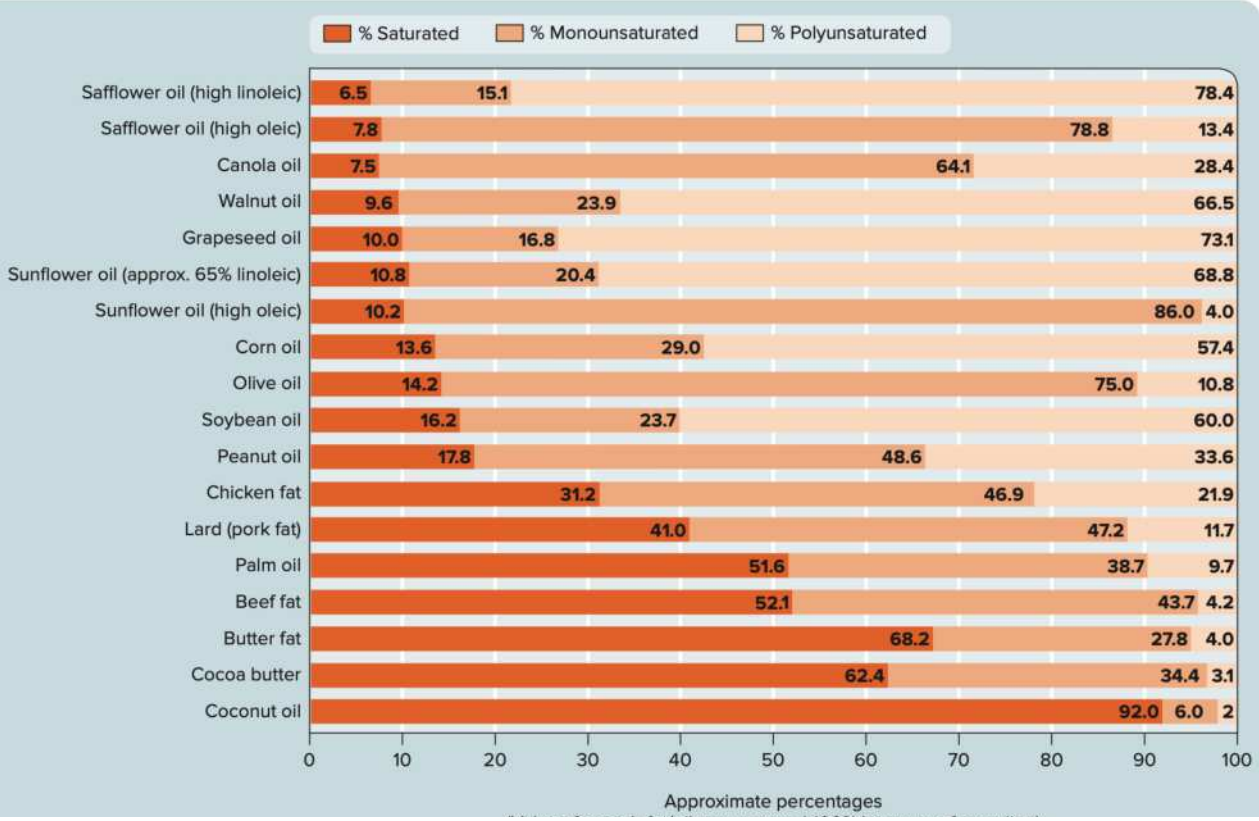
Plant Foods:

- Generally contain **more unsaturated fatty acids**.
- Examples: Olive oil (rich in MUFAs), Corn oil (rich in PUFAs).

❖ Exceptions (Tropical Oils):

- **Coconut oil** and **palm oil** are plant-based but are very high in saturated fat.

Fats and oils



Phospholipids

Structure: Similar to a triglyceride, but one fatty acid is replaced by a chemical group containing **phosphorus**.

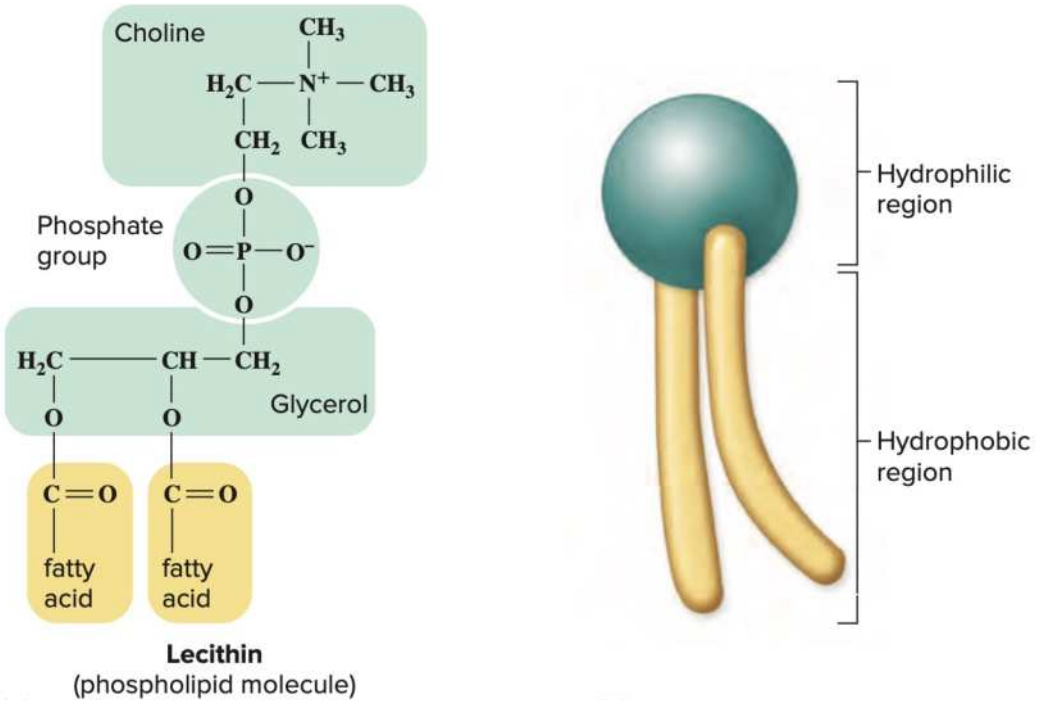
Dual Nature:

- **Hydrophilic (water-loving):** The phosphate "head".
- **Hydrophobic (water-fearing):** The fatty acid "tails".

Function: Emulsifier

- This dual nature allows phospholipids to act as emulsifiers, substances that keep water and oil mixed together.

Figure Of Lecithin



Lecithin
(phospholipid molecule)

Phospholipids in the Body and Food

In the Body:

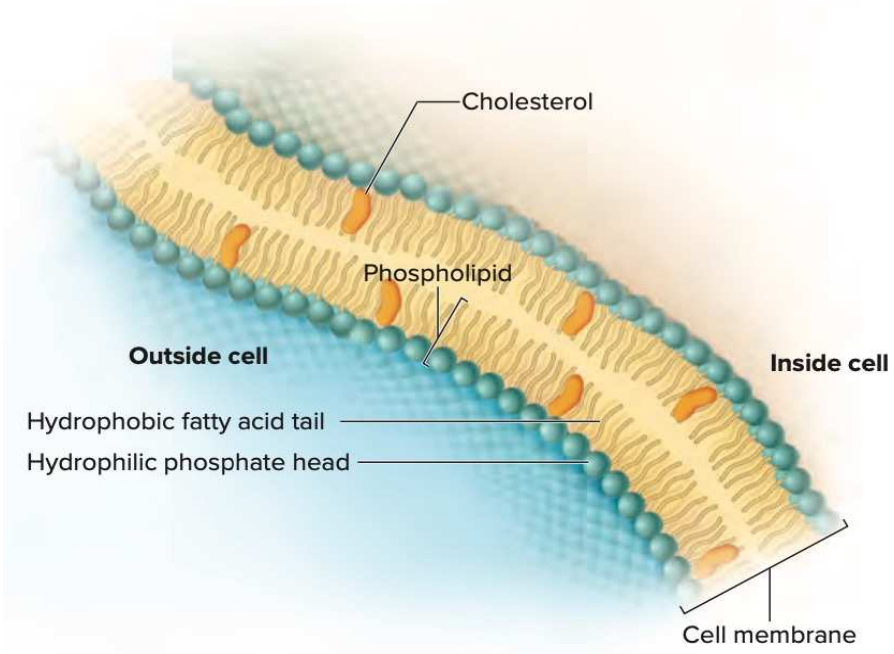
- Major structural component of all **cell membranes**.
- Essential for the normal functioning of **nerve cells**.

In Food:

- **Lecithin** is the most abundant phospholipid in food.
- Rich sources: **Egg yolks**, liver, wheat germ, peanut butter, soybeans.

Lecithin contains **choline**, a vitamin-like compound used to make the neurotransmitter *acetylcholine*

Phospholipids in cell membranes



Cholesterol and Other Sterols

Cholesterol

Sterols are lipids with a more complex chemical structure (rings).

Cholesterol is the most well-known sterol.

It is **essential to health**.

Functions:

- Integral component of every **cell membrane**.
- Used to synthesize: **Steroid hormones** (e.g., testosterone, estrogen) , **Vitamin D** , **Bile**

Cholesterol is **not** metabolized for energy.

Cholesterol Sources

Dietary Sources:

- Cholesterol is found **ONLY in animal-derived foods**.
- Plants do not contain cholesterol.

Body Synthesis:

- About **two-thirds** of total body cholesterol is made by the **liver**.
- About **one-third** comes from food sources.

TABLE Approximate Cholesterol Content of Some Foods/Serving

Food	Serving Size	Cholesterol (mg)
Liver	3oz (85 gm)	234
Egg	1 large	186
Egg yolk	1	186
Sardines	3oz (85 gm)	121
Single-patty cheeseburger	1	111
Beef	3oz (85 gm)	88
Turkey, ground	3oz (85 gm)	84
Shrimp	6 large	80
Ground beef, lean (15% fat)	3oz (85 gm)	77
Salmon	3oz (85 gm)	75
Turkey, dark meat	3oz (85 gm)	71
Chicken breast	3oz (85 gm)	49
Hot dog	1	44
Whole cow's milk	1 cup	34

Plant Sterols and Stanols

These are lipids found in plants that are **structurally similar** to cholesterol.

Sources: Grains, fruits, vegetables, nuts, seeds, and legumes.

Mechanism of Action:

- They **compete with cholesterol** for intestinal absorption.
- This **reduces** the amount of cholesterol that is absorbed.

Health Benefit: Consuming plant sterols/stanols can be an effective way to lower elevated blood cholesterol levels.

4

LIPIDS (FATS)

PART 2: Fat Digestion, Absorption, & Transport

Lipid Digestion, Absorption, & Transport

Lipid Digestion

Stomach: Minor digestion; an inactive lipase is activated by stomach acid.

Small Intestine (Primary Site):

- Fatty chyme triggers the release of hormones **CCK** and **Secretin**.
- **Secretin** → Liver produces bile; Pancreas releases bicarbonate.
- **CCK** → Gallbladder releases stored bile; Pancreas secretes **pancreatic lipase**.

Lipid Digestion, Absorption, & Transport

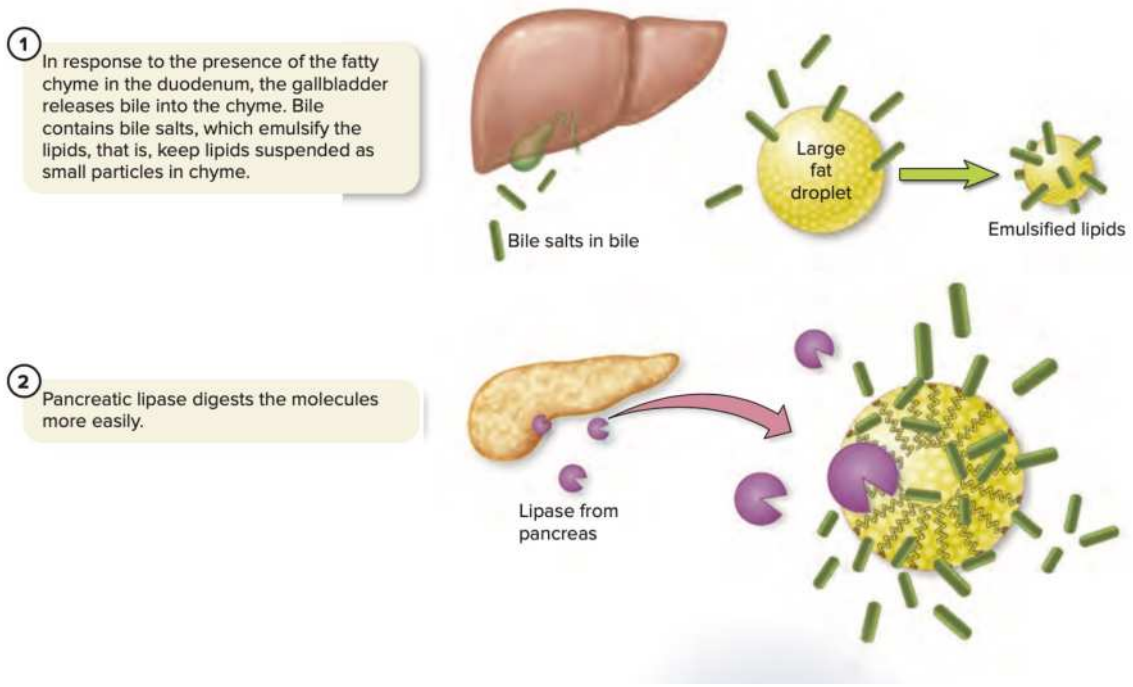
Role of Bile:

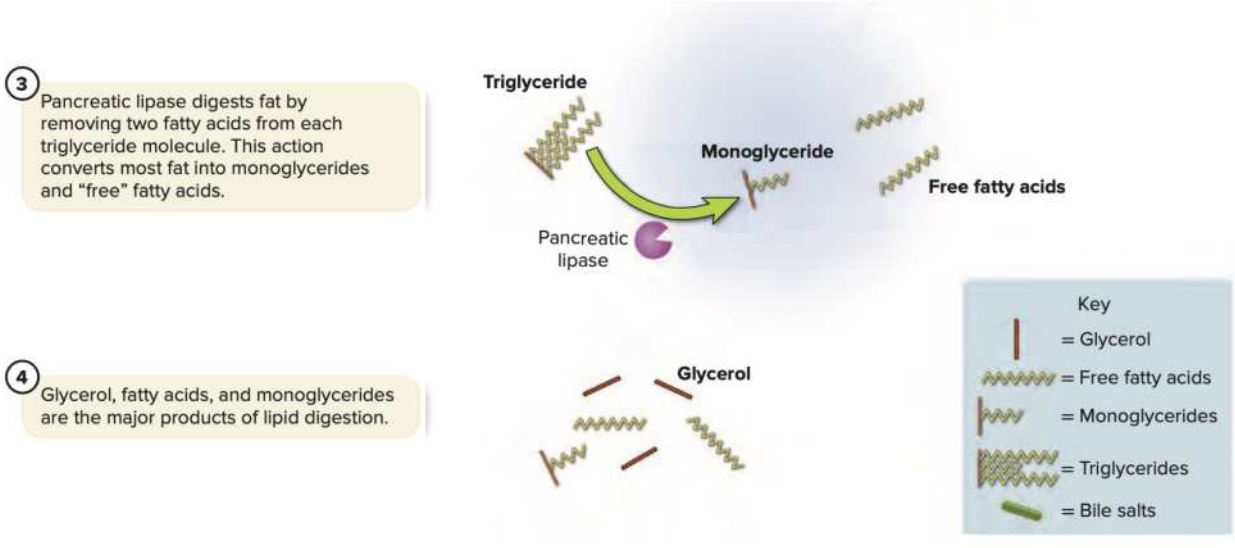
- **Emulsifies** lipids (breaks large fat droplets into small particles).
- Increases surface area for enzymes to attack.

Role of Pancreatic Lipase:

- Removes two fatty acids from each triglyceride.
- **Triglyceride → Monoglyceride + 2 Free Fatty Acids**.

Figure: Lipid Digestion





Lipid Absorption

Short- & Medium-Chain Fatty Acids: Diffuse directly into absorptive cells and enter the **bloodstream**.

Long-Chain Fatty Acids, Monoglycerides, & Cholesterol:

- These products cluster together with bile salts to form **micelles**.
- Micelles transport these lipids to the edge of the absorptive cells (enterocytes) for diffusion.

Inside the Absorptive Cell:

- Monoglycerides and fatty acids are **re-formed into triglycerides**.
- These reformed lipids are packaged into a **chylomicron**.

Lipid Transport

Chylomicrons (a type of lipoprotein) are coated with protein and phospholipids.

They are **too large** to enter blood capillaries.

Instead, they enter the **lacteal** (a lymph vessel) in the villus.

Lymphatic System → Thoracic Duct → Bloodstream (at left subclavian vein).

In the blood, **Lipoprotein Lipase (LPL)** breaks down triglycerides in the chylomicrons, releasing fatty acids for cells to use.

The liver clears the chylomicron remnants.

The Four Major Lipoproteins

Lipoproteins: Water-soluble structures that transport lipids through the bloodstream.

1. **Chylomicrons:**

   Formed in enterocytes (small intestine).

   Transport *dietary* fat from the GI tract to the body.

2. **VLDL (Very-Low-Density Lipoprotein):**

   Made in the **liver**.

   Carries triglycerides from the liver to other tissues.

The Four Major Lipoproteins

3. **LDL (Low-Density Lipoprotein):**

   Forms from VLDL after it unloads triglycerides.

   Carries **cholesterol** to tissues.

   Often called **"bad" cholesterol**.

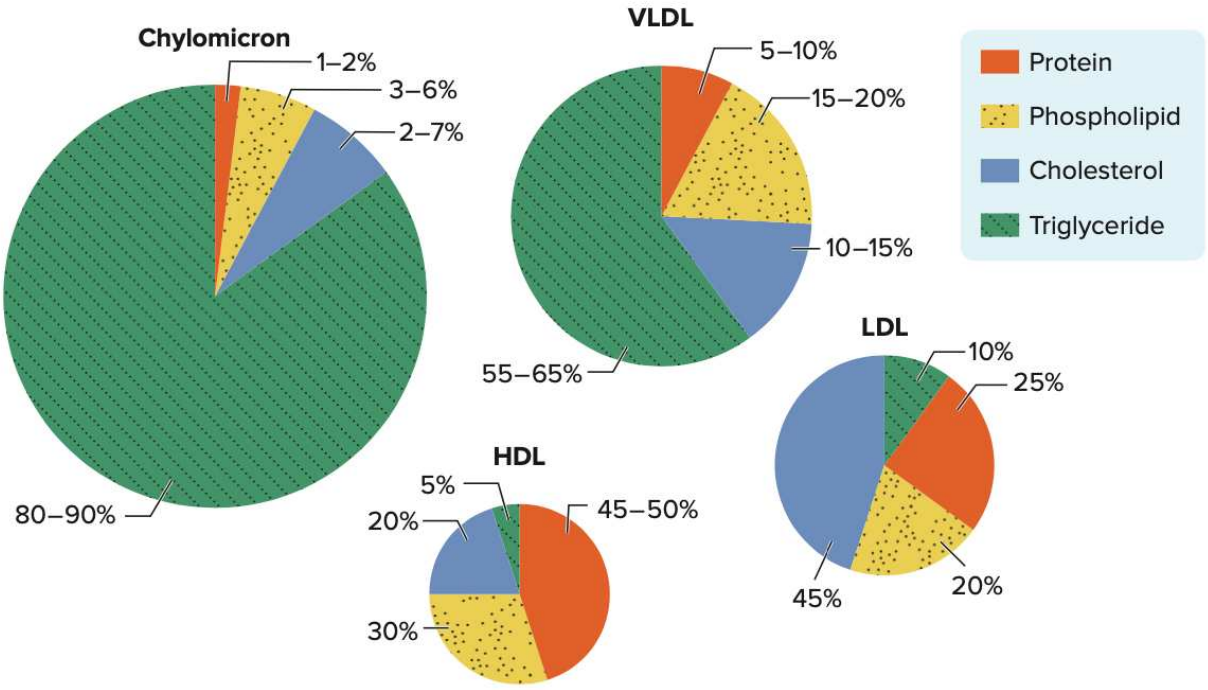
4. **HDL (High-Density Lipoprotein):**

   Protein shells released from the **liver**.

   Performs **reverse cholesterol transport**: picks up cholesterol from tissues and brings it back to the liver.

   Often called **"good" cholesterol**.

FIGURE Major lipoproteins.



Bile Recycling and Gallstones

Enterohepatic Circulation:

- The process of recycling bile salts.
- ~98% of bile salts are reabsorbed in the ileum and returned to the liver.
- **Soluble fiber** can bind to bile and prevent reabsorption, forcing the liver to use more cholesterol to make new bile.

Gallstones:

- Hard particles, primarily made of **cholesterol**, that form in the gallbladder.

Bile Recycling and Gallstones

Cholecystectomy (Gallbladder Removal):

- Bile drips directly from the liver into the small intestine.
- High-fat meals are poorly digested.
- Can lead to **steatorrhea** (presence of undigested fat/lipid in the stool).

Lipid Consumption Patterns

Average American Intake (2017-2020):

- Total Fat: **~37%** of total daily energy.
- Saturated Fat: **~12%** of total daily energy.

Acceptable Macronutrient Distribution Range (AMDR):

- Total Fat: **20% to 35%** of total calories.

2020-2025 Dietary Guidelines:

- Saturated Fat: **Less than 10%** of total calories.
- Trans Fat & Cholesterol: Keep intakes **as low as possible**.

Essential Fatty Acid AMDRs (Adults):

- Linoleic acid (n-6): 5% to 10% of total kcal.
- Alpha-linolenic acid (n-3): 0.6% to 1.2% of total kcal.

Table: Dietary Fat and Cholesterol Recommendations

Total fat	20–35% of total calories (45–75 g total fat for a 2000-kcal diet)
Saturated fat	Less than 10% of total calories (22 g saturated fat for a 2000-kcal diet)
Linoleic acid	5–10% of total calories (11–22 g linoleic acid for a 2000-kcal diet)
Alpha-linolenic acid	0.6–1.2% of total calories (1–3 g alpha-linolenic acid for a 2000-kcal diet)
Cholesterol	As little as possible while consuming a balanced and nutrient-dense diet
Trans fat	Keep intake as low as possible.

Lipids and Cardiovascular Disease (CVD)

CVD: Group of diseases affecting the heart and blood vessels (e.g., heart disease, stroke).

Most cases result from **Atherosclerosis**.

A long-term process where **plaque** (fatty buildup) forms inside arterial walls.

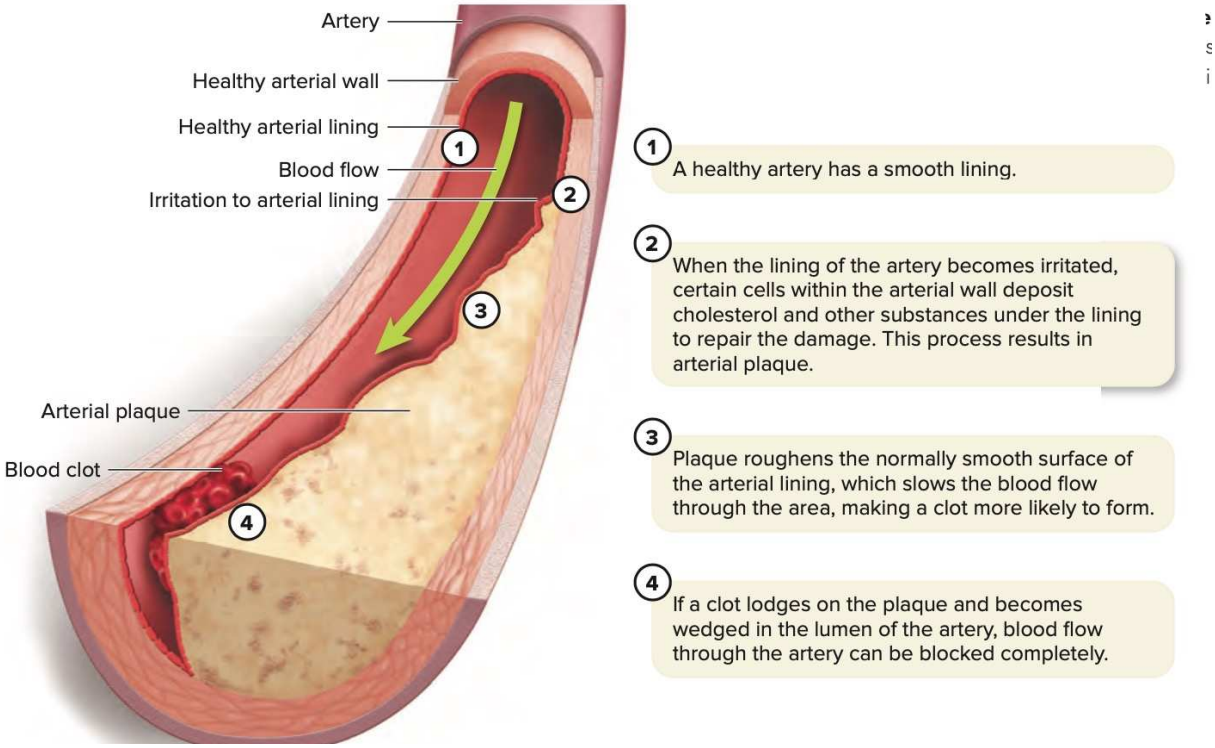
Atherosclerosis leads to **Arteriosclerosis**.

"Hardening of the arteries," or loss of arterial flexibility.

How Atherosclerosis Develops

1. **Injury:** The arterial lining becomes injured (e.g., from oxidation, infection, high blood pressure).
2. **Inflammation:** The immune system responds, causing inflammation.
3. **Plaque Formation:** White blood cells and other substances deposit cholesterol under the lining, contributing to **arterial plaque**.
4. **Narrowing:** Plaque narrows the artery's **lumen** (opening) and roughens the smooth surface.
5. **Clotting:** The roughened surface slows blood flow, making clots more likely to form.

FIGURE: Atherosclerosis



Consequences of Atherosclerosis

Thrombus: A fixed bunch of clots that stays in place and disrupts blood flow.

Myocardial Infarction (Heart Attack): Occurs when a thrombus completely blocks blood flow to a section of the heart muscle, causing it to die.

Stroke: Occurs when a clot blocks an artery in the brain, causing brain cells to die.

Embolus: A thrombus or piece of plaque that **breaks free** and travels through the bloodstream. It can lodge in a smaller artery, causing a heart attack or stroke.

Risk Factors: Hypertension & Homocysteine

Hypertension (High Blood Pressure):

- A major modifiable risk factor for CVD.
- Contributes to the initial injury of artery walls.
- Forces the heart to work harder to circulate blood through stiff arteries.

Homocysteine:

- A toxic by-product of protein metabolism.
- **Homocystinuria** (high levels in the blood) may injure arterial walls and increase CVD risk.
- Vitamins B-6, B-12, and folate help convert it to safer compounds.

TABLE Atherosclerosis: Major Risk Factors

Nonmodifiable Risk Factors
Family history of CVD (especially before 60 years of age) Advanced age Male sex Race/ethnic background
Modifiable Risk Factors
Diabetes mellitus Diet lacking balance and nutrient-dense options Elevated blood cholesterol (especially LDL cholesterol) Excess body fat Hypertension Physical inactivity Tobacco use or exposure to tobacco smoke Untreated sleep apnea

Lipoproteins and Atherosclerosis

HDL ("Good") Cholesterol:

Transports cholesterol **away from body cells** and back to the liver for disposal.

High HDL is protective.

LDL ("Bad") Cholesterol: Transports cholesterol **from the liver to cells** for their use.

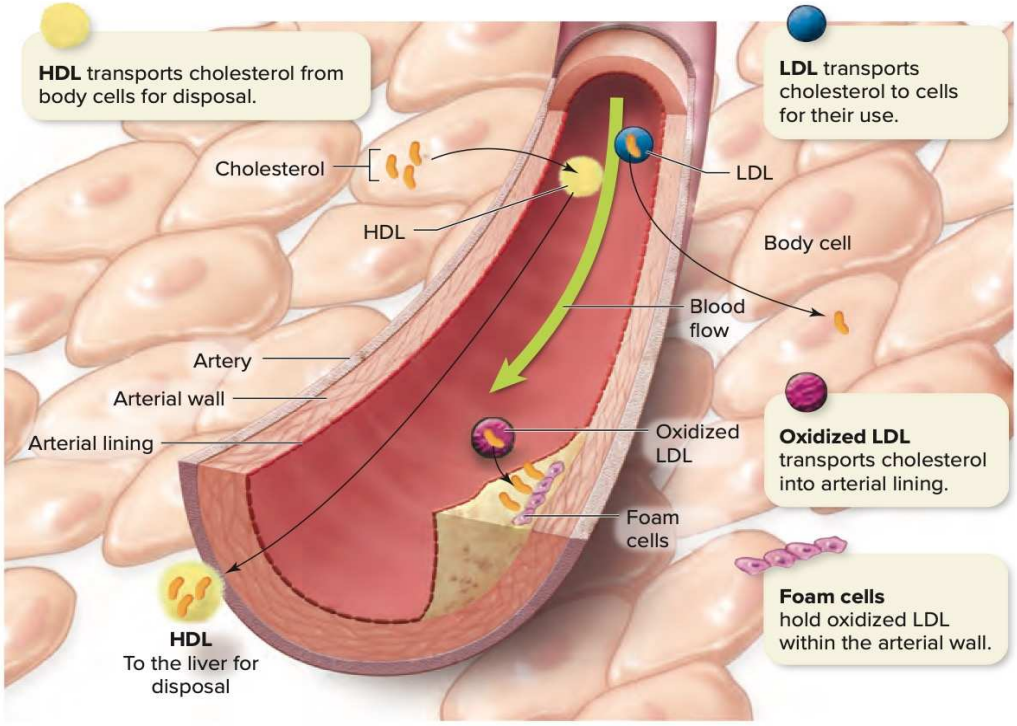
High LDL is a risk factor.

Oxidized LDL:

LDL that has been damaged by free radicals.

Oxidized LDL is taken up by immune cells, forming "foam cells" that become part of arterial plaque.

Lipoproteins and Atherosclerosis



Assessing Atherosclerosis Risk

Lipoprotein Profile: A series of blood tests to check: (Total Cholesterol, LDL Cholesterol, HDL Cholesterol, Triglycerides)

Total Cholesterol to HDL Ratio:

- A healthy ratio is **less than 3.5:1**.
- Example: 180 mg/dl Total / 30 mg/dl HDL = 6:1 ratio (This is a HIGH risk, even though total cholesterol is < 200).

High-Sensitivity C-Reactive Protein (hs-CRP):

- A protein produced by the liver in response to **inflammation**.
- Elevated CRP is a biomarker for atherosclerosis risk.

Table: Classification of Fasting Blood Lipid Levels

Total Cholesterol (mg/dl)	Classification
< 200	Desirable
200–239	Borderline high
≥ 240	High
LDL Cholesterol (mg/dl)	**Classification**
< 100	Optimal
100–129	Near optimal/above optimal
130–159	Borderline high
160–189	High
≥ 190	Very high
HDL Cholesterol (mg/dl)	**Classification**
< 40 (for males); < 50 (for females)	Low
≥ 40 (for males); ≥ 50 (for females)	Desirable
Triglycerides (mg/dl)	**Classification**
< 150	Normal
150–199	Borderline high
≥ 200	High

Reducing Risk: Dietary Changes

How Fats Affect Blood Lipids?

Saturated Fats (SFAs): Raise *both* LDL ("bad") and HDL ("good") cholesterol levels.

Trans Fats: Also raise blood cholesterol levels.

Monounsaturated Fats (MUFAs): Generally **lower LDL** cholesterol *without* reducing HDL.

Polyunsaturated Fats (PUFAs):

May **reduce both** total cholesterol and LDL cholesterol.

May also reduce beneficial HDL in some people.

Recommendation: Replace foods rich in saturated fat and trans fat with foods rich in monounsaturated and polyunsaturated fats.

Reducing Risk: Omega-6 vs. Omega-3

Omega-6 (Linoleic Acid): Found in many vegetable oils (corn, soybean).

Used to make eicosanoids that **increase inflammation** and **blood clotting**.

(Role in heart disease is controversial) .

Omega-3 (EPA & DHA): Found in fatty fish, flaxseed, walnuts.

Reduces the risk of developing heart disease.

Actions tend to be anti-inflammatory and reduce blood clotting.

Recommendation: Consume at least **8 ounces of seafood** (especially fatty fish like salmon, tuna, trout) per week.

Supplements: Doses higher than 3 g/day may interfere with blood clotting.

Omega-3 Benefits and Sources

Possible Health Benefits of Omega-3 Fatty Acids

May Reduce Risk of:

- Heart disease
- Arthritis
- Certain cancers
- Alzheimer's disease
- Depression

May Improve:

- Immune function
- Cognitive development in infants

TABLE Rich Food Sources of Omega-3 Fats

Fish/Shellfish (DHA and EPA)	Oils	Nuts and Seeds	Other
Herring, salmon, tuna, sardines, catfish, mackerel, trout, halibut, shrimp	Flaxseed, walnut, canola, soybean	Walnuts, flaxseeds	Algae

Reducing Risk: Other Dietary Modifications

Dietary Cholesterol:

- Has *less* effect on blood cholesterol than saturated fat does.
- Why? The liver synthesizes *less* cholesterol when dietary intake is high.

Increase Soluble Fiber: Reduces LDL cholesterol (by binding to bile, preventing its reabsorption).

For High Triglycerides (>150 mg/dl):

- Reduce intakes of **refined carbohydrates** (candy, pastries, soft drinks).
- Reduce **alcohol** intake.

Food Selection & Preparation:

- Limit processed/cured meats.
- Trim visible fat from meat.
- Use moist cooking methods (roasting, steaming) or stir-frying instead of deep frying.
- Drain fat from browned ground beef.

Reducing Risk: Lifestyle & Medications

Lifestyle Factors:

Weight Loss: Losing excess body fat (especially abdominal) is associated with better blood lipid levels.

Physical Activity: Regular (150 min/week) moderate-intensity activity can:

Reduce LDL and triglycerides.

Raise beneficial HDL levels.

Reducing Risk: Lifestyle & Medications

Medications (If lifestyle changes are not enough):

Statins: A class of drugs that interfere with the **liver's** metabolism of cholesterol, effectively reducing LDL and triglycerides.

Zetia®: A drug that inhibits the **intestinal absorption** of cholesterol

Summary

Lipids (Triglycerides, Phospholipids, Sterols) are essential for energy, structure, and hormone production.

Fatty Acids are Saturated (SFA), Monounsaturated (MUFA), or Polyunsaturated (PUFA, incl. Omega-3 & -6).

Triglycerides (Glycerol + 3 FAs) are 95% of fat in food/body.

Phospholipids act as emulsifiers and form cell membranes.

Cholesterol (a sterol) is vital for membranes and hormones; found only in animal foods.

Digestion requires bile (emulsifier) and pancreatic lipase.

Transport requires lipoproteins (Chylomicrons, VLDL, LDL, HDL).

CVD is linked to high LDL ("bad") and low HDL ("good").

Risk can be reduced by replacing SFAs/trans fats with MUFAs/PUFAs, increasing soluble fiber, exercising, and managing weight.



PRINCIPLE OF NUTRITION

Dr. Marwan Jalambo
Assistant Professor (Nutrition)

5

PART 1: Proteins: Amino Acids

Chapter 5 Overview

1. **Introducing Proteins**
2. **Amino Acids**
3. **Protein Synthesis and Structure**
4. **Protein in Foods**

Introducing Proteins

Definition: Proteins are large, complex organic molecules made of amino acids.

Chemical Elements:

- **Carbon (C)**
- **Hydrogen (H)**
- **Oxygen (O)**
- **Nitrogen (N)**

All living organisms contain proteins

Major Functions of Protein (Part 1)

Build New Cells: Used to build new cells and functional components.

Structural Components:

- **Collagen:** Provides structure for cartilage, ligaments, and bone.
- **Keratin:** Found in hair, nails, and skin.

Immune System: Form antibodies to tag invaders for destruction.

Movement: Contractile proteins in muscles.

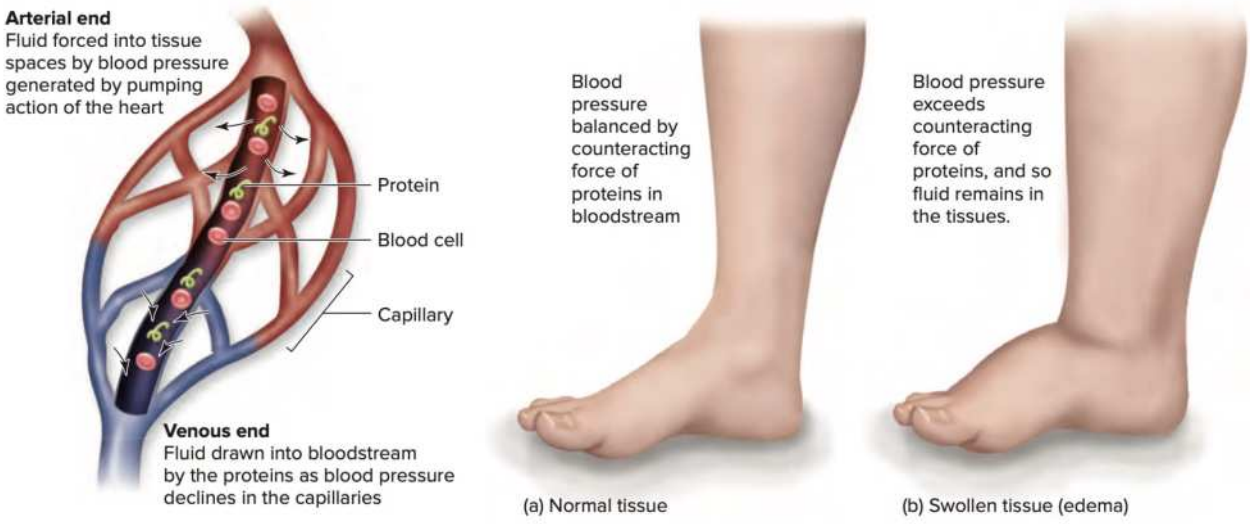
Blood Clotting: Necessary to stop blood loss.

Major Functions of Protein (Part 2)

Fluid Balance:

- Blood proteins, like albumin, help maintain the proper distribution of fluids.
- They counteract blood pressure by attracting fluid back into the bloodstream.
- During starvation, low blood protein causes fluid to leak into tissues, resulting in **edema** (swelling).

FIGURE: Normal fluid balance and edema



Major Functions of Protein (Part 3)

Acid-Base Balance:

- Proteins act as **buffers** to keep blood and tissue fluids within a proper pH range (7.35–7.45).
- They can bind to excess hydrogen ions (H+) to neutralize acidity.

Other Functions:

- **Enzymes:** Speed up chemical reactions.
- **Hormones:** Such as certain regulatory hormones.
- **Transporters:** Carry substances in the blood.
- **Energy Source:** Used as a last resort.

Amino Acids: The Building Blocks

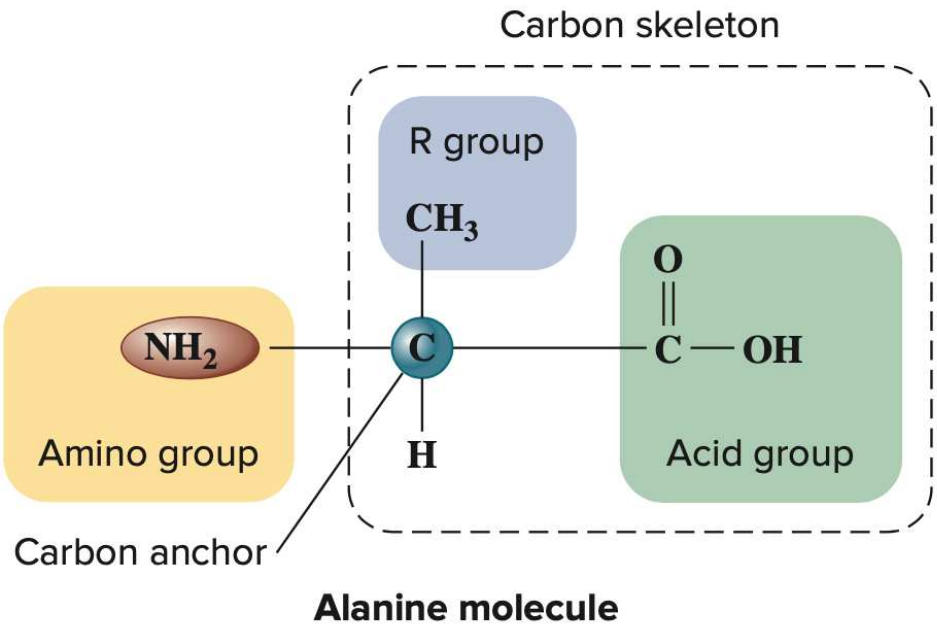
Proteins are comprised of smaller units called **amino acids**.

The human body uses 20 different amino acids to make proteins.

Basic Chemical Structure:

1. **Carbon**
2. **Amino Group** (contains nitrogen)
3. **Carboxylic Acid Group**
4. **R Group** (side chain) - This part is unique and determines the amino acid's properties.

FIGURE: Amino acid: Basic chemical structure



Alanine molecule

Classifying Amino Acids

Essential Amino Acids (EAAs):

- There are **9** EAAs.
- The body **cannot** make them or cannot make enough.
- They **must** be supplied by the diet.

Nonessential Amino Acids (NEAAs):

- There are **11** NEAAs.
- The body **can** synthesize (make) these.

Conditionally Essential Amino Acids:

- Normally nonessential, but become essential under certain conditions, such as illness, injury, or genetic disorders.
- Example: Glycine for injury recovery, Tyrosine for PKU.

TABLE: Essential and Nonessential Amino Acids

Essential		Nonessential	
Histidine	Threonine	Alanine	Cysteine*
Isoleucine	Tryptophan	Aspartic acid	Glutamine*
Leucine	Valine	Asparagine	Glycine*
Lysine		Glutamic acid	Proline*
Methionine		Serine	Tyrosine*
Phenylalanine		Arginine*	

*Under certain conditions, this amino acid can become essential.

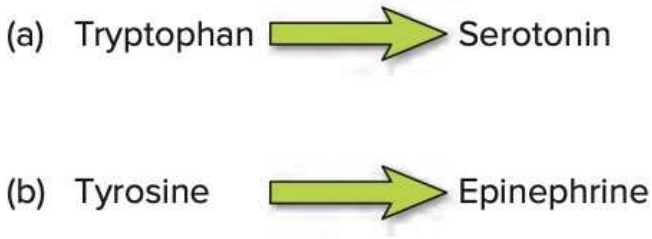
Amino Acid Derivatives

The body uses amino acids to form important nitrogen-containing compounds that are *not* proteins.

Examples: Creatine, Melanin (pigment). Amines (compounds with amino groups)

Tryptophan is a precursor for **Serotonin** (mood regulator).

Tyrosine is a precursor for **Epinephrine** (hormone).



Protein Synthesis

The instructions for making proteins are coded in **DNA (deoxyribonucleic acid)**, located in genes.

Cells assemble the 20 amino acids in specific sequences based on the DNA code.

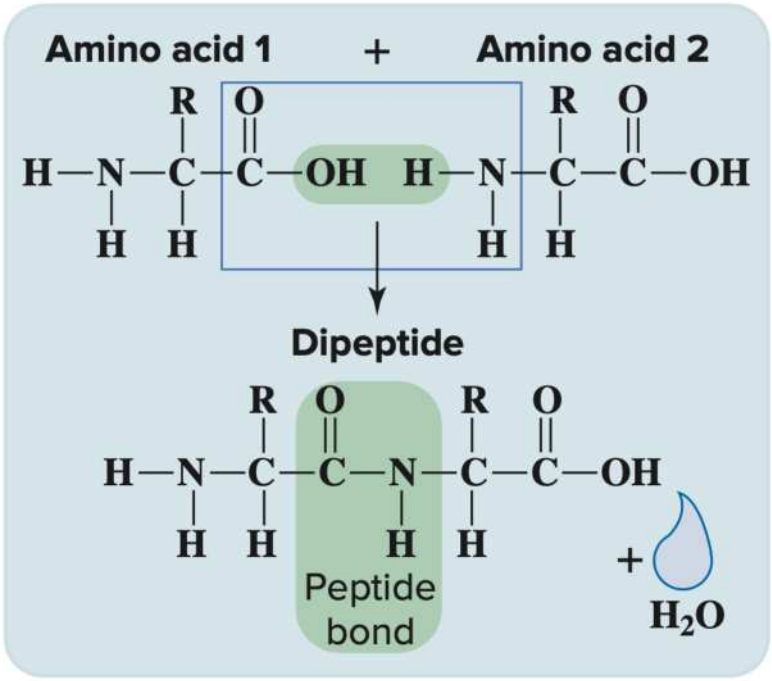
Amino acids are linked by **Peptide Bonds**.

Dipeptide: 2 amino acids.

Polypeptide: 50 or more amino acids.

If an essential amino acid is missing from the "pool," protein synthesis stops.

FIGURE: Peptide bond



The 4 Levels of Protein Structure

A protein's final shape is critical to its function.

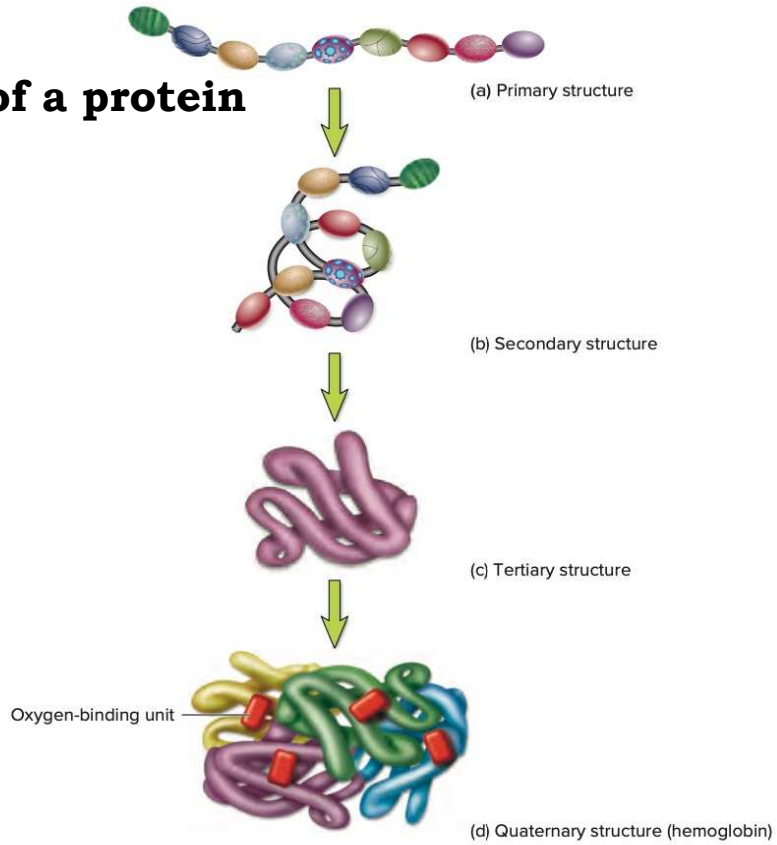
1. Primary Structure: The linear chain of amino acids linked by peptide bonds.

2. Secondary Structure: The coiling of the polypeptide chain.

3. Tertiary Structure: The final 3-dimensional, folded form of the protein.

4. Quaternary Structure: When two or more polypeptide chains associate (e.g., Hemoglobin).

FIGURE Shaping of a protein



Synthesis Errors & Denaturation

Synthesis Errors: A faulty DNA code (genetic mutation) can cause the wrong amino acid to be inserted.

Example: Sickle Cell Anemia

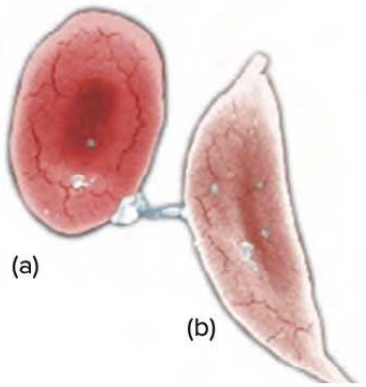
A genetic disorder where an incorrect amino acid is inserted into hemoglobin chains.

This causes the hemoglobin structure to be abnormal, impairing oxygen transport and causing red blood cells to become "sickle" shaped.

Protein Denaturation: Altering a protein's natural shape and function by unfolding it.

Causes: Heat (cooking an egg), acid (lemon juice in milk, stomach acid), alcohol, or physical agitation (whipping). This change is permanent.

FIGURE: Sickle cell anaemia.
This microscopic view of red blood cells shows
(a) normal, disk-shaped cells and
(b) (b) a sickle cell that contains abnormal haemoglobin.



Protein in Foods & Protein Quality

Protein Content:

Animal foods (meat, fish) generally provide more protein per serving than plant foods (vegetables).

Good plant sources include seeds, tree nuts, and legumes (beans, peas, peanuts).

Protein in Foods & Protein Quality

Protein Quality:

High-Quality (Complete) Protein:

Contains all 9 essential amino acids (EAAs) in the proper proportions.

Well-digested and absorbed.

Sources: Animal foods (meat, fish, poultry, eggs, dairy).

Low-Quality (Incomplete) Protein:

Lacks or has inadequate amounts of one or more EAA.

Less digestible.

Sources: Most plant foods.

Plant-Based Exceptions (Complete): Soy protein and quinoa.

Limiting Amino Acids

Definition: The essential amino acid found in the lowest concentration (shortest supply) in an incomplete protein.

Effect: If a diet lacks a limiting amino acid, the body's ability to make its own proteins is impaired.

Common Limiting AAs in Plants:

Grains (wheat, rice, corn): Low in **Lysine**.

Legumes (beans, peas, lentils): Low in **Methionine**.

Nuts & Seeds: Low in **Lysine**.

Protein Complementation

Definition: The practice of combining certain plant foods to provide all essential amino acids.

By combining foods with different limiting AAs, the meal provides a complete protein profile.

It is not necessary to eat complementary proteins at *every* meal, but rather throughout the day.

Examples of Complementary Protein Dishes

Dish	Combination
Red beans and rice	Legumes + Grains
Peanut butter on bread	Legumes + Grains
Hummus (chickpeas) with pita bread	Legumes + Grains
Black beans and cornmeal tortilla	Legumes + Grains
Tofu (soy) with brown rice and cashews	Legumes + Grains + Nuts

Summary

- **Proteins** are made of 20 amino acids and are unique for containing **Nitrogen**.
- They are essential for structure (collagen), function (enzymes, antibodies), and regulation (fluid balance, pH).
- **9 Essential AAs** must come from the diet; **11 Nonessential AAs** can be made by the body.
- **High-quality (complete) proteins** (animal foods, soy, quinoa) have all 9 EAAs; **incomplete proteins** (most plants) lack one or more.
- **Protein complementation** (e.g., beans and rice) combines plant proteins to get all 9 EAAs.



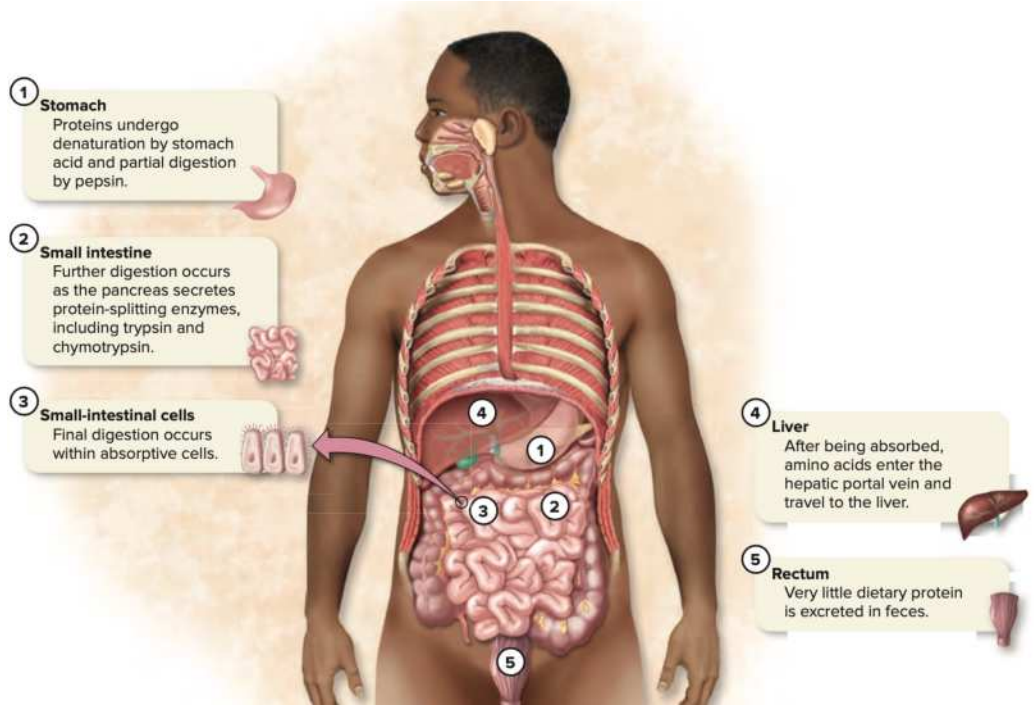
5

PART 2: Proteins: digestion and absorption

Chapter 5 Overview

1. Protein Digestion, Absorption, and Transport
2. Protein Metabolism
3. Meeting Protein Needs
4. Protein Undernutrition
5. Food Hypersensitivity Disorders
6. Plant-Based Diets
7. Nutritional Genomics

Summary of protein digestion and absorption



Protein Absorption and Transport

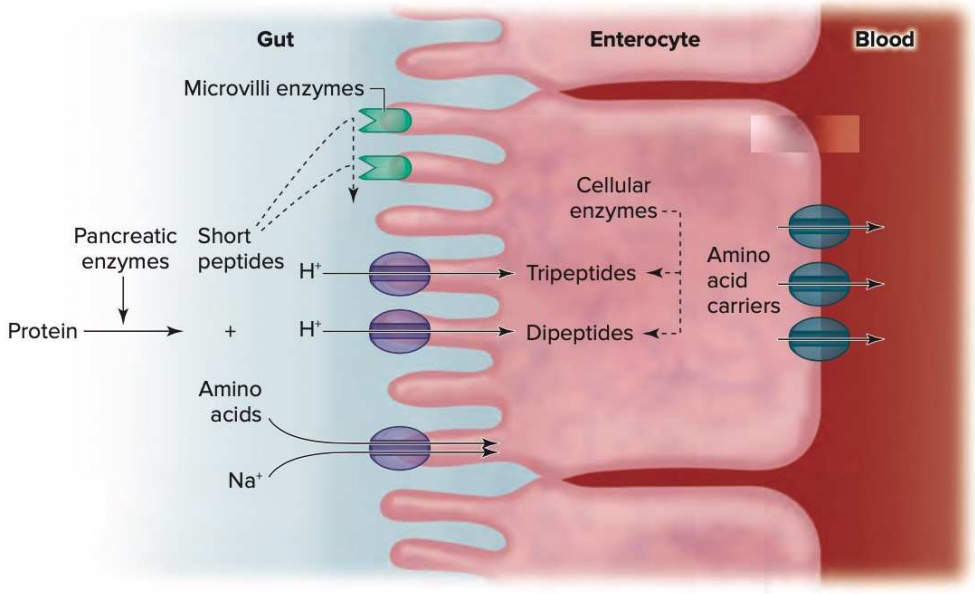
1. Absorption: Short peptides and amino acids are transported into the absorptive cells (enterocytes).

2. Inside the Cell: Dipeptides and tripeptides are broken down into individual amino acids.

3. Transport: Amino acids exit the cell and enter the capillary (bloodstream). They travel via the hepatic portal vein to the **Liver** .

4. In the Liver: The liver keeps some amino acids for its own needs and releases the rest into the general circulation. (Very little dietary protein is excreted in feces).

FIGURE Peptide and amino acid absorption and transport



Protein Metabolism: Turnover & Pool

Protein Turnover:

- The continuous cellular process of breaking down old or unneeded proteins into their component amino acids and recycling them to make new proteins.

Amino Acid Pool:

- The readily available supply of amino acids in the body's cells that can be used for protein synthesis.
- **Endogenous Source:** ~2/3 of the pool comes from protein turnover.
- **Exogenous Source:** ~1/3 of the pool comes from dietary (food) protein.

Protein Metabolism: Transamination vs. Deamination

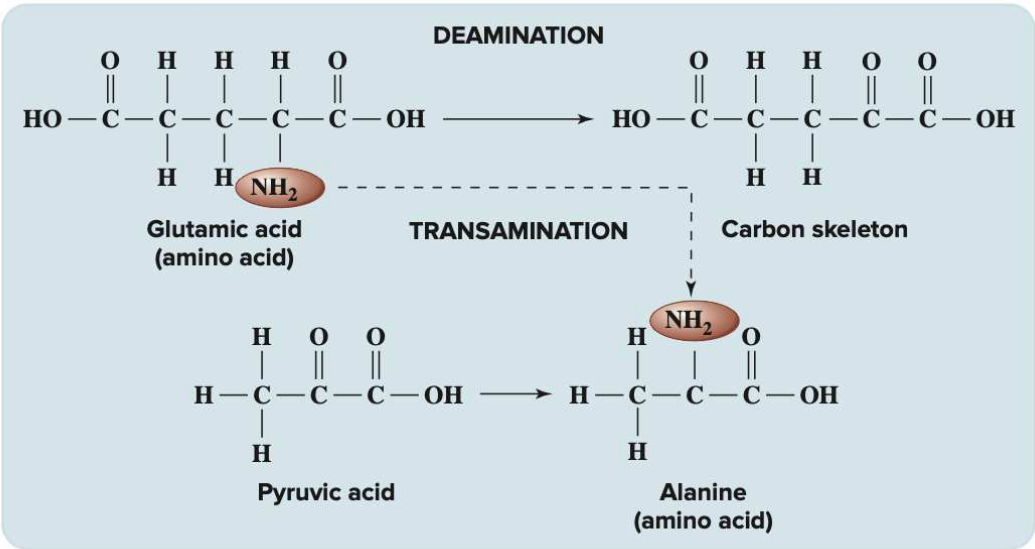
Transamination:

- The transfer of a nitrogen-containing group (amino group) from one unneeded amino acid to a carbon skeleton.
- **Purpose:** To create a new *nonessential* amino acid.

Deamination:

- The removal of the nitrogen-containing group (amino group) from an unneeded amino acid.
- **Result:** Leaves a **carbon skeleton** (used for energy or converted to glucose/fat) and the nitrogen group.

FIGURE: Deamination and transamination



Handling Excess Nitrogen (Ammonia & Urea)

The body **does not store** excess amino acids.

Step 1: Unneeded amino acids undergo **deamination**.

Step 2: The removed amino group (NH2) forms **Ammonia (NH3)** in the liver. Ammonia is toxic.

Step 3: The **Liver** converts toxic ammonia into **Urea** (a less toxic waste product).

Step 4: Urea is released into the bloodstream.

Step 5: The **Kidneys** filter urea from the blood and excrete it in **urine**.

Clinical Markers: BUN & UUN

Blood Urea Nitrogen(BUN): Blood test used to assess **kidney function**.

High BUN (>20 mg/dl): May indicate kidney disease (kidneys aren't filtering urea), high protein intake, congestive heart failure, or GI bleeding.

Low BUN (<6 mg/dl): May indicate liver failure, malnutrition, or overhydration.

Urine Urea Nitrogen(UUN): Urine test used as marker of **protein intake**.

High UUN (>20 g): May indicate high protein intake or excessive breakdown of body protein.

Low UUN (<12 g): May indicate inadequate protein intake, malnutrition, or a kidney problem.

Nitrogen Balance

Definition: Balancing nitrogen intake (from dietary protein) with nitrogen losses (in urine, feces, skin, hair, nails).

Nitrogen Equilibrium (Intake = Loss):

Occurs in a healthy adult meeting their needs.

Positive Nitrogen Balance (Intake > Loss):

The body is retaining more nitrogen than it loses; building tissues.

Occurs during: Growth, pregnancy, recovery from illness/injury, resistance exercise .

Negative Nitrogen Balance (Intake < Loss):

The body is losing more nitrogen than it retains; breaking down tissues.

Occurs during: Starvation, serious illness/injury, bed rest, fever.

Meeting Protein Needs: RDA & AMDR

RDA (Recommended Dietary Allowance):

- For healthy adults: **0.8 g/kg of body weight**.
- *Calculation Example:* A 75 kg (165 lb.) adult needs:
  - 75 kg×0.8 g/kg=60 g of protein/day.
- Protein needs **increase** during pregnancy, breastfeeding, periods of growth, and recovery from serious illness or burns.

AMDR (Acceptable Macronutrient Distribution Range):

- For healthy adults: **10% to 35%** of total daily energy (calories) should come from protein.

High-Protein Diets

Possible Advantages	Possible Disadvantages
Weight Loss Increased satiety, reduced hunger Spontaneous calorie reduction Maintenance of lean muscle mass	**Kidney Stress** Increased nitrogen burden on kidneys Changes in kidney function
Blood Sugar Regulation Improved blood glucose control Increased insulin sensitivity	**Cardiovascular Disease Risk** Potentially high saturated fat intake
Cardiovascular Effects Reduced LDL cholesterol Reduced blood triglycerides Reduced blood pressure	**Cancer Risk** Increased risk of some types of cancer, especially colorectal cancer
Bone Health Increased intestinal calcium absorption Increased bone growth-promoting hormones	**Osteoporosis Risk** Increased urinary excretion of calcium Increased risk of bone fragility, leading to fracture

Protein Undernutrition (PEM)

PEM (Protein-Energy Malnutrition): Occurs when the diet lacks sufficient **protein AND energy**.

Impact on Children: PEM is especially devastating for children.

Stunted growth

Weakness and irritability

Vulnerable to dehydration and infections

Can cause permanent brain damage and lower intelligence.

Protein Undernutrition (PEM)

Forms of PEM:

Marasmus (Starvation):

Results from a severe lack of energy *and* nutrients.

Signs: Extreme weight loss, loss of body fat, "skin and bones" appearance (wasting).

Kwashiorkor:

Results from consuming adequate energy but *insufficient high-quality protein*.

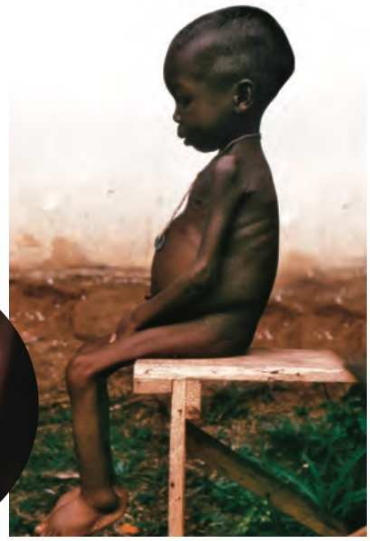
Often occurs when a toddler is weaned.

Key Sign: **Edema** (swelling) in the belly, legs, and cheeks.

Other signs: Stunted growth, sparse/brittle hair, skin patches.

Marasmic Kwashiorkor

Definition: A form of PEM where a child with kwashiorkor (edema) also begins to lack sufficient energy, leading to wasting.



Characteristics: The child exhibits signs of both conditions simultaneously:



Edema (swollen tissues)

Wasting (loss of muscle and fat).

Food Hypersensitivity: Allergy vs. Intolerance

Food Allergy:

An **IMMUNE RESPONSE** where immune cells mistake a food *protein* (allergen) for a harmful organism and mount an attack.

Signs: Hives, swelling, difficulty breathing, vomiting, diarrhea .

Anaphylaxis: A severe, life-threatening allergic reaction that can cause a drop in blood pressure (anaphylactic shock).

Top Allergens: Cow's milk, eggs, peanuts, tree nuts, soy, wheat, fish, shellfish, and sesame .

Food Intolerance:

An unpleasant physical reaction to food components.

It **does not** trigger an immune response.

Examples: Lactose intolerance, MSG sensitivity, caffeine, FODMAPs.

Celiac Disease

Definition: **Autoimmune disorder** (it is *not* a simple allergy or intolerance).

Mechanism:

Person consumes **Gluten** (a protein found in wheat, and barley).

The immune system reacts by producing antibodies against gluten.

These antibodies also attack the person's own intestinal tissue, damaging the **villi** (the small, finger-like projections that absorb nutrients).

Result: **Malabsorption** of nutrients, leading to undernutrition, abdominal pain, diarrhea, and poor growth.

Treatment: A lifelong, strict **gluten-free diet**.

Non-Celiac Gluten Sensitivity (NCGS)

Definition: Intolerance to gluten-containing foods in people who do *not* have celiac disease or a wheat allergy.

Symptoms:

IBS-like symptoms (abdominal pain, bloating, diarrhea).

Symptoms outside the gut (headache, joint/muscle pain, fatigue, "mental fog").

Diagnosis:

First, rule out celiac disease (blood tests, biopsy) and wheat allergy.

Then, use an elimination diet followed by a food provocation test (FPT) to see if symptoms return when gluten is reintroduced.

Treatment: Strictly avoiding gluten-containing foods

Phenylketonuria (PKU)

Definition: A rare **genetic metabolic disorder**.
Mechanism: Caused by a genetic mutation where cells cannot produce the active enzyme **phenylalanine hydroxylase**.

This enzyme normally converts the *essential* amino acid **Phenylalanine** into the *nonessential* amino acid **Tyrosine**.

Result: Phenylalanine and its toxic by-products build up in the body, damaging nerve cells and causing intellectual disability if untreated.

Treatment: A lifelong diet low in phenylalanine.
Avoid high-protein foods: nuts, dairy, eggs, meat, and other animal foods.

Tyrosine becomes conditionally essential for individuals with PKU.

TABLE: Common Types of Plant-Based Diets

Type of Plant-Based Diet	Animal Foods Included
Pescavegetarian diet	Cow's milk, other dairy products, eggs, fish, and other shellfish
Semivegetarian diet	Cow's milk, other dairy products, eggs, fish, other shellfish, and chicken
Lactoovovegetarian diet	Cow's milk, other dairy products, and eggs
Lactovegetarian diet	Cow's milk and other dairy products
Ovovegetarian diet	Eggs
Vegan diet	None

TABLE: Nutritional Considerations for Plant-Based Diets

Possible Advantages	Possible Disadvantages
High Vitamin C, vitamin E, and folate Phytonutrients Fiber Magnesium and potassium	**Low** Vitamin B-12, vitamin D, and riboflavin Zinc, iron, and calcium Omega-3 fatty acids Certain essential amino acids Energy
Low Saturated fatty acids Sodium	

Plant-Based Diets: Special Life Stages

Children:

Have higher protein and energy needs per pound of body weight.

Plant foods are bulky and high in fiber, which can make children feel full before they have met their high energy/protein needs.

Requires careful planning to ensure nutritional adequacy.

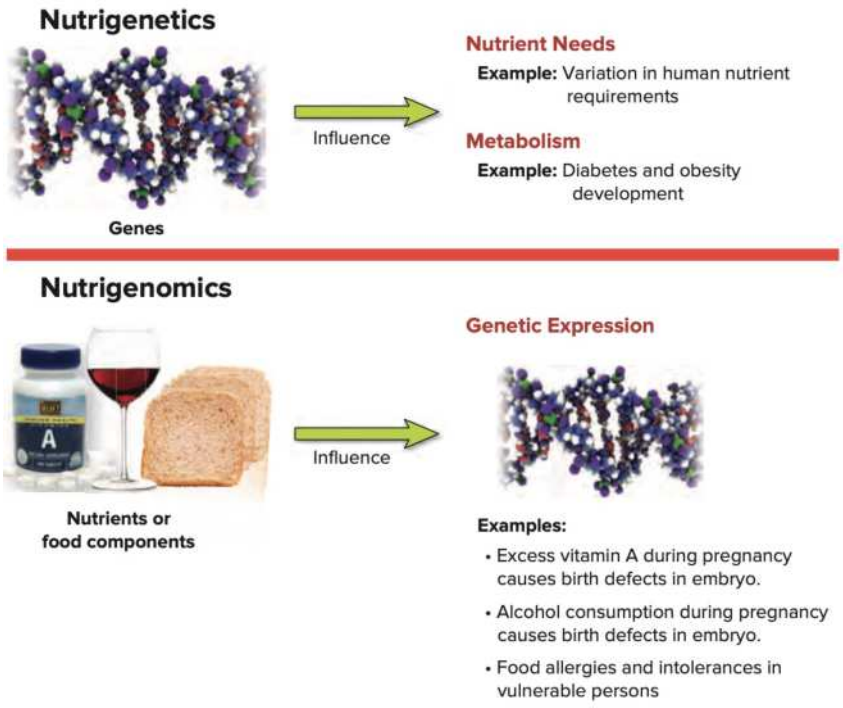
Pregnancy & Breastfeeding (Vegan):

Mothers should consult with health care providers about taking a **Vitamin B-12 supplement**.

Breast milk can be deficient in B-12 if the mother's diet lacks it.

B-12 deficiency in infants can lead to **severe developmental delays and neurological damage**.

Nutritional Genomics



Nutritional Genomics

The study of how diet affects genes and how genes affect responses to diet.

Nutrigenetics:

- The study of how inherited *genetic variations* (your genes) influence the body's *response to specific nutrients*.
- *(Genes ➔ Food Response)*
- Example: Explains why a high-fat diet lowers cholesterol in one person but not another.

Nutrigenomics:

- The study of how *nutrients* (food) affect the *expression of a person's genome* (genes).
- *(Food ➔ Gene Expression)*
- Example: Alcohol consumption during pregnancy causes birth defects.

Personalized Nutrition: Using this science to tailor dietary recommendations to an individual.

Summary

- Digestion (using **pepsin**, **trypsin**, **chymotrypsin**) breaks proteins into amino acids for absorption.
- Excess nitrogen is deaminated, forms **ammonia** (toxic), is converted to **urea** (in the liver), and is excreted by the **kidneys**.
- **PEM** (Protein-Energy Malnutrition) includes **Marasmus** (wasting) and **Kwashiorkor** (edema).
- **Celiac disease** is an autoimmune reaction to gluten; **food allergies** are an immune reaction to a protein; **intolerances** are non-immune reactions.
- **Plant-based diets** must be well-planned to avoid deficiencies, especially in **Vitamin B-12**, Vitamin D, calcium, iron, and zinc.



PRINCIPLE OF NUTRITION

Dr. Marwan Jalambo
Assistant Professor (Nutrition)

6

Part 1: Metabolism Fueling the Body

What Is Energy?

Energy is the power derived from food to perform cellular work

The body needs energy for:

Physical activities (running, jumping)

Physiological activities (breathing, heart beating)

Maintenance and repair of tissues

Energy has many forms: chemical, mechanical, electrical, heat, solar, and nuclear

Energy cannot be created or destroyed, only transformed

The body transforms chemical energy stored in food into a form cells can use

Sources of Energy

Human cells obtain energy by releasing the chemical energy stored in macronutrients and alcohol

Energy in food is measured in **kilocalories** (**Calories**)

Energy can be utilized from:

1. Carbohydrates
2. Lipids
3. Proteins
4. Alcohol

No energy is obtained from vitamins, minerals, or water

Energy Metabolism

Energy Metabolism: The sum of all chemical pathways that break down molecules to release energy and use energy to build new molecules

Metabolic reactions are either **catabolic** or **anabolic**

Anabolism vs. Catabolism

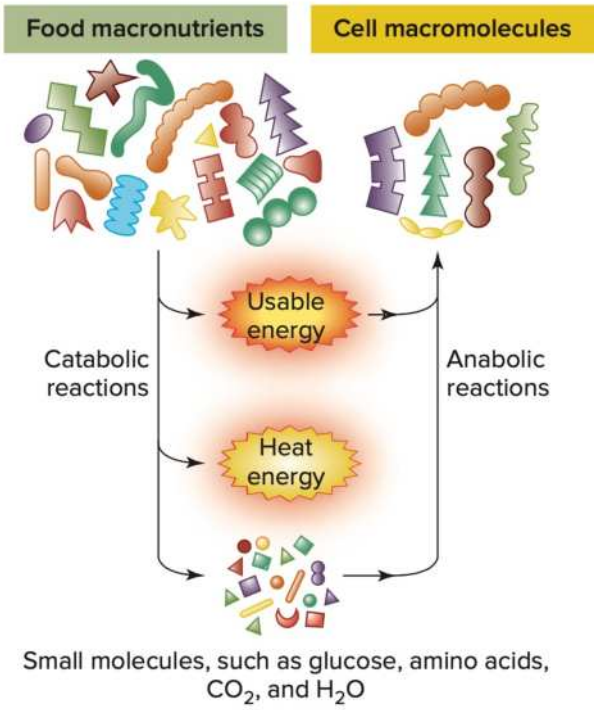
Catabolism:

- Metabolic pathways that break down larger molecules into smaller ones
- *Example:* Breakdown of glucose, fatty acids, and amino acids
- **Releases energy** some is captured as usable chemical energy, the rest becomes heat energy (maintains body temp)

Anabolism:

- Metabolic pathways that build larger molecules from smaller ones
- *Example:* Building glycogen from glucose, triglycerides from fatty acids, or proteins from amino acids
- **Requires energy input** (fueled by catabolic reactions)

Figure: Metabolism



Role of Enzymes and Coenzymes

Enzymes:

- Required for most chemical reactions in cells
- They *catalyze* (facilitate) reactions and remain unchanged

Coenzymes:

- Organic compounds that assist enzymes
- Enzymes generally cannot function without them
- Many **B vitamins** (thiamin, riboflavin, niacin, pantothenic acid, B-6) serve as coenzymes in energy metabolism

Table: B Vitamins and Their Coenzymes

B Vitamin	Coenzyme
Thiamin	Thiamin pyrophosphate (TPP)
Riboflavin	Flavin adenine dinucleotide (FAD, FADH2) Flavin mononucleotide (FMN)
Niacin	Nicotinamide adenine dinucleotide (NAD+, NADH) Nicotinamide adenine dinucleotide phosphate (NADP, NADPH)
Pantothenic acid	Coenzyme A (CoA)
Vitamin B-6	Pyridoxal phosphate (PLP)

Energy Shuttles: NAD+ and FAD

Energy is transferred in the form of electrons (e+), usually accompanied by hydrogen ions (H^+)

NAD+ (Nicotinamide adenine dinucleotide):

- Contains **Niacin**
- Picks up 2 e− and 2 H^+ → becomes **NADH** + H^+

FAD (Flavin adenine dinucleotide):

- Contains **Riboflavin**
- Picks up 2 e− and 2 H^+ → becomes **FADH2**

Adenosine Triphosphate (ATP)

ATP (Adenosine Triphosphate) is the "energy currency" of the cell

It is a high-energy phosphate molecule

Comprised of adenosine bound to three phosphate groups

When the bond between the last two phosphates is broken, **energy is released** to "pay" for cellular work

When ATP loses its last phosphate, it becomes **ADP (Adenosine Diphosphate)**

The ATP Cycle

Phosphorylation: An anabolic reaction that attaches a phosphate (P) group back onto ADP to re-form ATP

This process **requires energy input** (from catabolism of food)

ATP cycling is the primary way cells "earn" and "spend" energy

Mitochondria

The "powerhouse" of the cell

Mitochondria are organelles that synthesize *most* of the ATP cells need

They catabolize macronutrients and transfer the energy to ATP

The initial breakdown of glucose (glycolysis) occurs in the **cytoplasm**

Aerobic vs. Anaerobic Metabolism

1. Aerobic Metabolism:

Requires oxygen

Allows mitochondria to make large amounts of ATP

Used for sustained activity.

2. Anaerobic Metabolism:

Does not require oxygen

Used when oxygen supply is inadequate

Produces a much smaller amount of ATP

Can only be maintained for short periods

Phase 1: Glycolysis

The first phase of glucose catabolism ("glucose breakdown")

An **anaerobic** pathway that occurs in the **cytoplasm.**

Process: 1,6-carbon glucose molecule is broken down into 2 ,3-carbon **pyruvate** molecules.

Energy Yield:

- Uses 2 ATP to start
- Forms 4 ATP
- **Net gain of 2 ATP**
- Also forms 2 NADH

Fate of Pyruvate (Aerobic)

- When oxygen is available, pyruvate molecules enter the **mitochondria**
- Pyruvate is converted into a two-carbon compound: **Acetyl Coenzyme A (Acetyl CoA)**
- This reaction is **irreversible**—acetyl CoA cannot be used to re-form pyruvate
- This is the "point of no return" for glucose catabolism

Fate of Pyruvate (Anaerobic)

- Under anaerobic conditions (no oxygen), pyruvate is converted to **lactic acid**
- Lactic acid releases H+ to form **lactate**
- This happens in vigorously contracting muscles

The Cori Cycle

- Muscle cells lack the enzyme to make glucose from lactate, so they release lactate into the bloodstream
- **Liver cells** remove lactate from the blood
- The liver recycles the lactate back into **glucose** (via the Cori cycle)
- The new glucose is released into the bloodstream for cells to use

Note: This cycle is inefficient; it requires 6 ATP to make glucose from lactate

Phase 2: Citric Acid Cycle

Also known as the **TCA Cycle** or **Krebs Cycle**

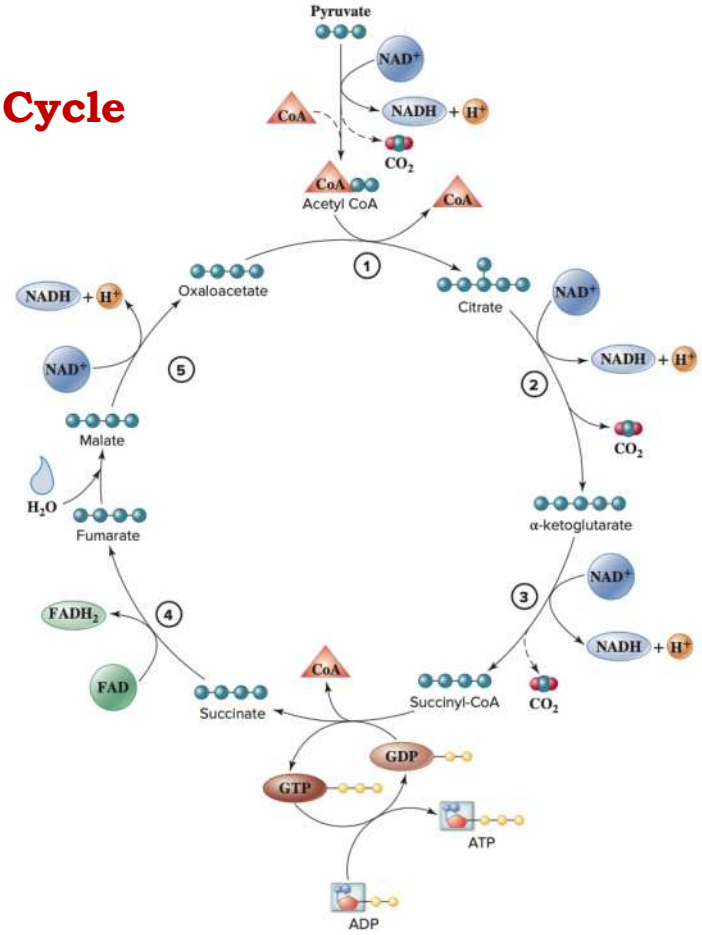
This is the second phase of glucose catabolism

Process:

1. Acetyl CoA enters the cycle and binds to oxaloacetate (4-carbon) to form citrate (6-carbon)
2. A complex series of reactions converts citrate into CO2, H2O, and intermediate compounds
3. Oxaloacetate is re-formed at the end, allowing the cycle to repeat

Main Products: The most important products are the coenzymes **NADH** and **FADH2** which carry high-energy electrons

Figure : Citric Acid Cycle



Phase 3: Electron Transport Chain (ETC)

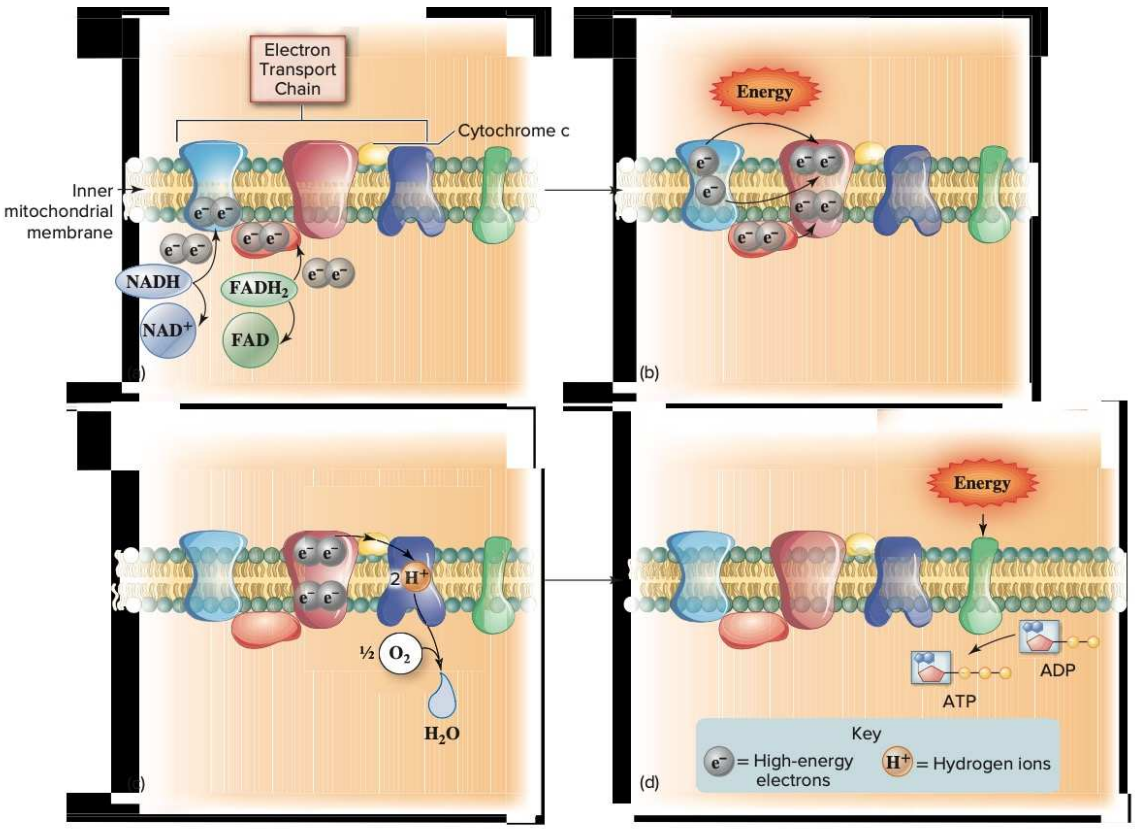
The final phase of aerobic metabolism

Occurs in the **inner membrane of the mitochondria**

Process:

1. NADH and FADH2 (from glycolysis and TCA cycle) carry hydrogen ions and high-energy electrons to the ETC
2. Electrons are passed down a "chain" of enzymes (cytochromes)
3. As electrons are passed, energy is released and used to pump H+
4. This energy is captured to attach Pi to ADP, forming **ATP**
5. At the end, **oxygen** accepts the electrons and H+ to form **water (H2O)**

FIGURE Electron transport chain



Total ATP Yield from Glucose

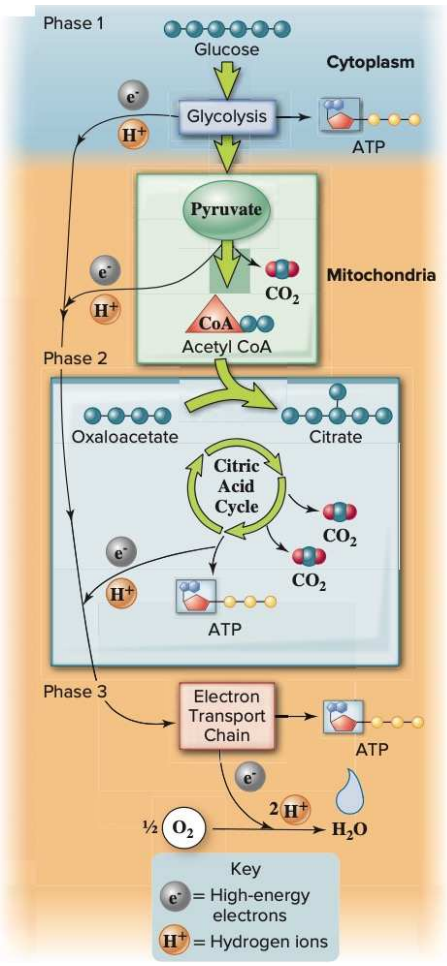
Yields vary because shuttles drop electrons at different points:

- 1 **NADH** → **2.5 ATP**
- 1 **FADH2** → **1.5 ATP**

The complete aerobic catabolism of **one glucose molecule** yields approximately **30** to **32** **molecules of ATP**

Summary: Carbohydrate Catabolism

This figure provides a complete overview of the 3 phases of glucose catabolism.




Thank you!

PRINCIPLE OF NUTRITION

Dr. Marwan Jalambo
Assistant Professor (Nutrition)

6

Part 2: Catabolism

Catabolism of Fructose and Galactose

Fructose:

- Catabolized **only in the liver**
- Feeds into the glycolytic pathway
- One product can be easily converted to glycerol, the backbone of triglycerides
- This makes fructose "uniquely fat-promoting"

Galactose:

- Converted into glucose (mainly in the liver)
- Follows the same catabolic pathway as glucose

Glycogenolysis

Glycogenolysis: The breakdown of stored glycogen into glucose molecules

In the Liver:

- Occurs when blood sugar levels fall
- The liver releases glucose into the bloodstream to supply all body cells

In Muscle Tissue:

- Releases a glucose intermediate that **cannot leave the muscle cell**
- The glucose must be catabolized for energy *within that muscle*

Provides a stockpile of quick energy for that muscle

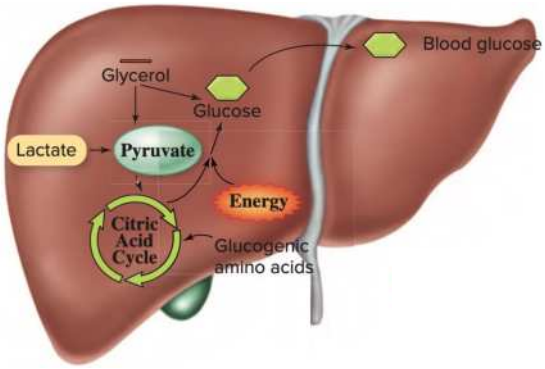
Gluconeogenesis

Gluconeogenesis: The synthesis of glucose from *noncarbohydrate* precursors

Occurs when glucose is not available.

Precursors include:

- Glycerol
- Lactate
- Pyruvate
- Many amino acids (glucogenic)



Primarily occurs in the **liver**

Obtaining Energy from Fat

Triglycerides are the most energy-dense macronutrient group

The body stores more energy as fat (triglycerides) than as glycogen.

Macronutrient	Fuel Content (kcal)	Approximate % of Total Energy Storage
Triglycerides (Fat)	**140,000**	**78** %
Protein	38,000	21 %
Carbohydrate	2000	1 %

Step 1: Lipolysis

Lipolysis: Breakdown of stored fat

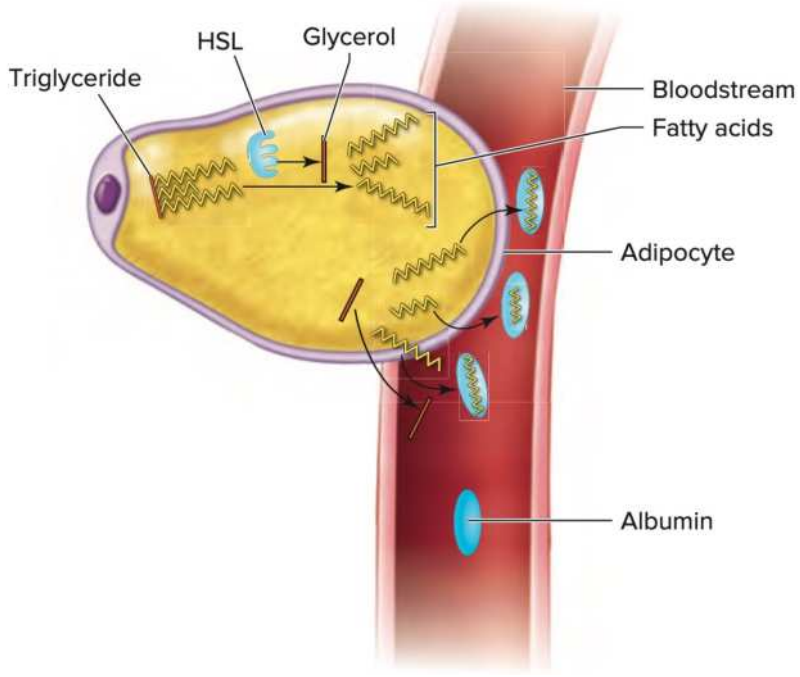
Occurs when insulin levels are low

The enzyme **Hormone Sensitive Lipase (HSL)** in fat cells (adipocytes) removes 3 fatty acids from the glycerol backbone.

1. **Glycerol** travels to the liver (can be used for gluconeogenesis).

2. **Fatty acids** bind to albumin (a protein) in the bloodstream for transport to other cells (like muscle) for energy.

FIGURE: HSL facilitates lipolysis in adipocytes (fat cell)



Step 2: Beta-Oxidation

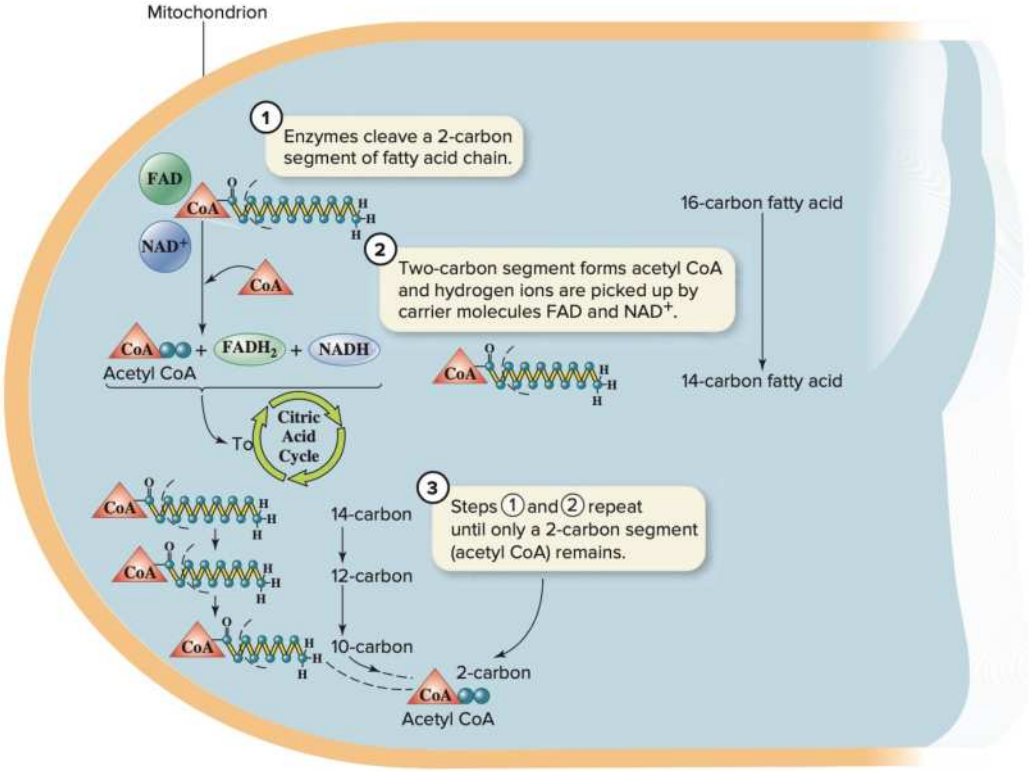
The catabolism of a fatty acid

Process:

1. **Activation:** The fatty acid is "activated" in the cytoplasm by binding to Coenzyme A (requires 2 ATP)
2. **Transport:** A molecule called **carnitine** shuttles the activated fatty acid into the mitochondria
3. **Cleavage:** Beta-oxidation cleaves the fatty acid chain into two-carbon segments, forming **acetyl CoA**

Each cleavage reaction also produces **1 NADH** and **1 FADH2**

FIGURE: Beta-oxidation.



ATP Yield from Fat

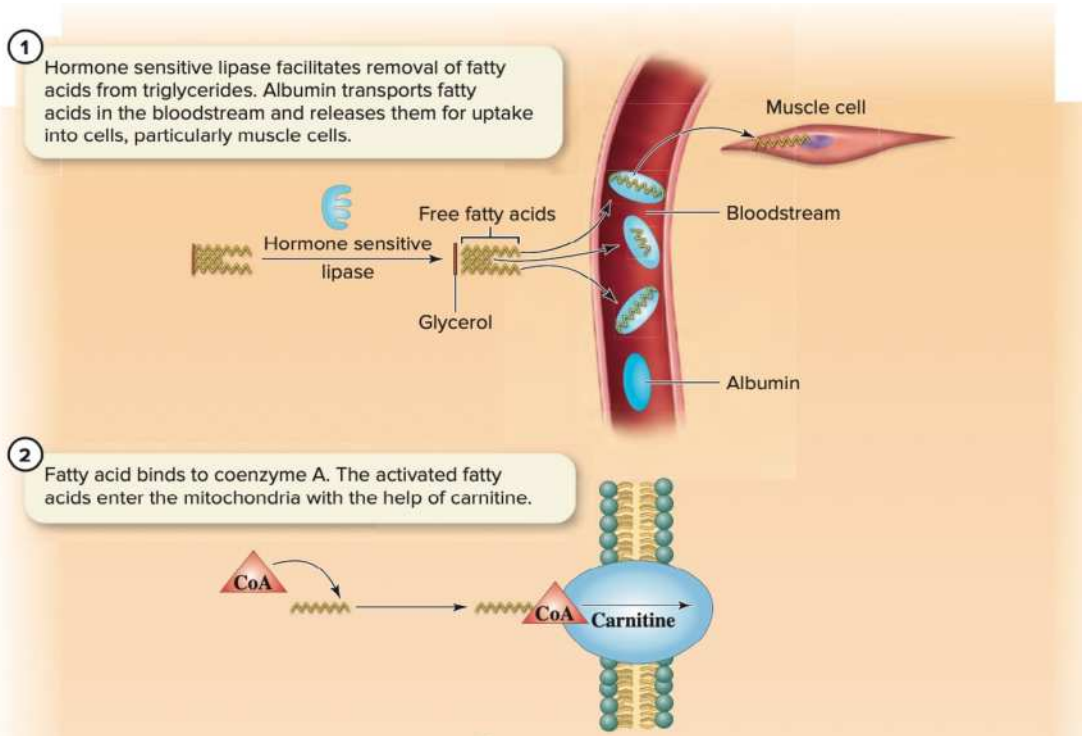
The acetyl CoA molecules from beta-oxidation enter the **Citric Acid Cycle**

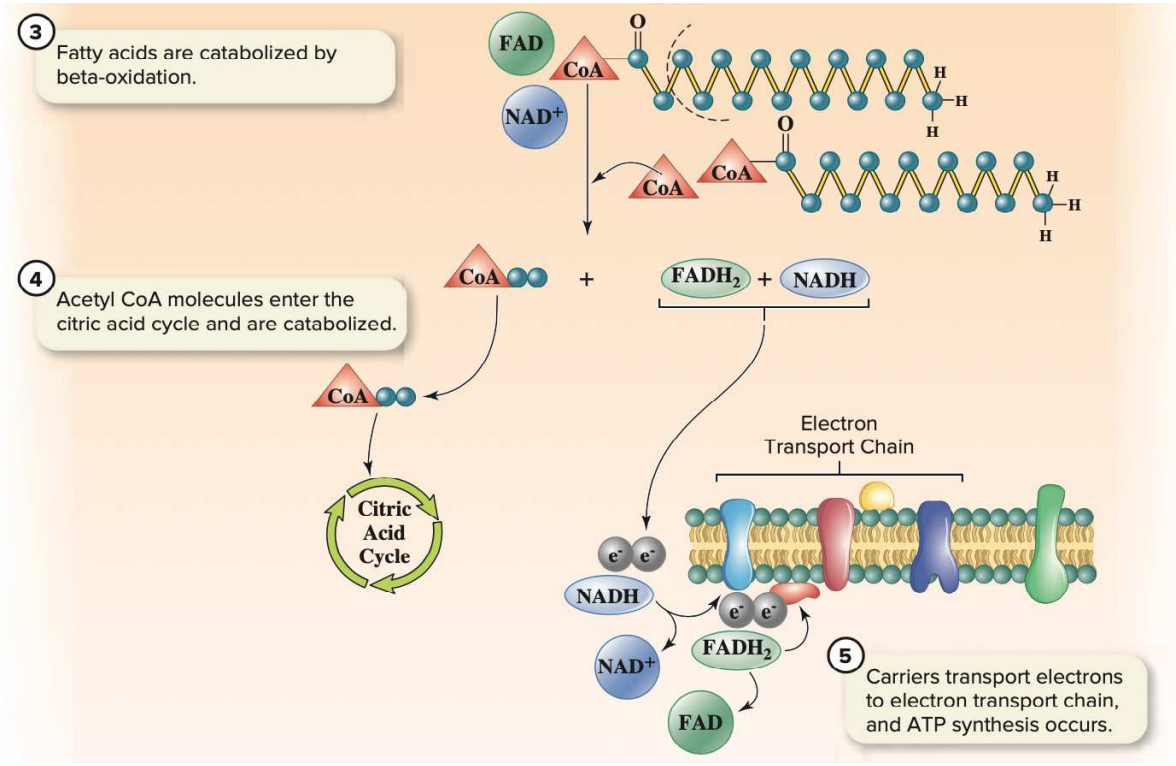
The NADH and FADH2 (from beta-oxidation *and* the TCA cycle) go to the **Electron Transport Chain**

Result: The complete oxidation of one 16-carbon palmitic acid molecule yields a **net of 106 ATP**

This is far more than the ~30-32 **ATP** from one glucose molecule

FIGURE: Summary of fat catabolism





Ketogenesis

Ketogenesis: Ketone body formation

When does it happen?

- During starvation or very low-carbohydrate intake, glucose is unavailable
- Fat catabolism increases, producing large amounts of acetyl CoA

Process: The liver converts excess acetyl CoA into **ketone bodies** (acetoacetate, beta-hydroxybutyrate, and acetone)

- Ketone bodies are released into the blood and can be used as fuel by most cells, including the brain (after adaptation)

Ketogenic Diet & Ketoacidosis

Ketogenic Diet:

- A high-fat, moderate-low protein, and very-low-carbohydrate diet
- Used therapeutically for epilepsy, obesity, and type 2 diabetes

Ketoacidosis:

- A **life-threatening** condition seen in *poorly controlled diabetes* (especially Type 1)
- A total lack of insulin causes extremely rapid, uncontrolled ketogenesis
- Ketone bodies accumulate so fast they lower the blood's pH
- Requires immediate medical care

Obtaining Energy from Protein

- Cells use amino acids primarily to synthesize essential proteins (enzymes, hormones, structural molecules)
- Catabolizing amino acids for ATP is a lower priority, used when other fuels are limited

Preparing Amino Acids for Catabolism

Step 1: Remove Nitrogen

- The nitrogen group must be removed via **deamination** or **transamination**
- This requires the coenzyme **PLP** (made from Vitamin B-6)

Step 2: Detoxify Ammonia

- Deamination produces **ammonia** (NH_3), which is toxic
- The **liver** converts ammonia into **urea**
- The **kidneys** filter urea from the blood and excrete it in urine

Glucogenic vs. Ketogenic Amino Acids

The remaining "carbon skeleton" can enter energy pathways

1. **Glucogenic Amino Acids:**
   - Carbon skeletons are broken down into **pyruvate** or intermediates of the **citric acid cycle**
   - These *can* be used to make glucose (via gluconeogenesis)
2. **Ketogenic Amino Acids:**
   - Carbon skeletons are broken down into **acetyl CoA**
   - These *cannot* be used to make glucose
   - Can be converted into ketone bodies

Energy Storage

When cells have adequate ATP, excess fuel is stored

Primary storage sites:

- **Glycogen:** Stored in liver and muscle cells
- **Triglycerides:** Stored in adipose tissue (fat cells)

- The body prefers to use stored fat and glucose; body proteins are used as a last resort

Storing Glucose

When cells have enough ATP, excess glucose is stored

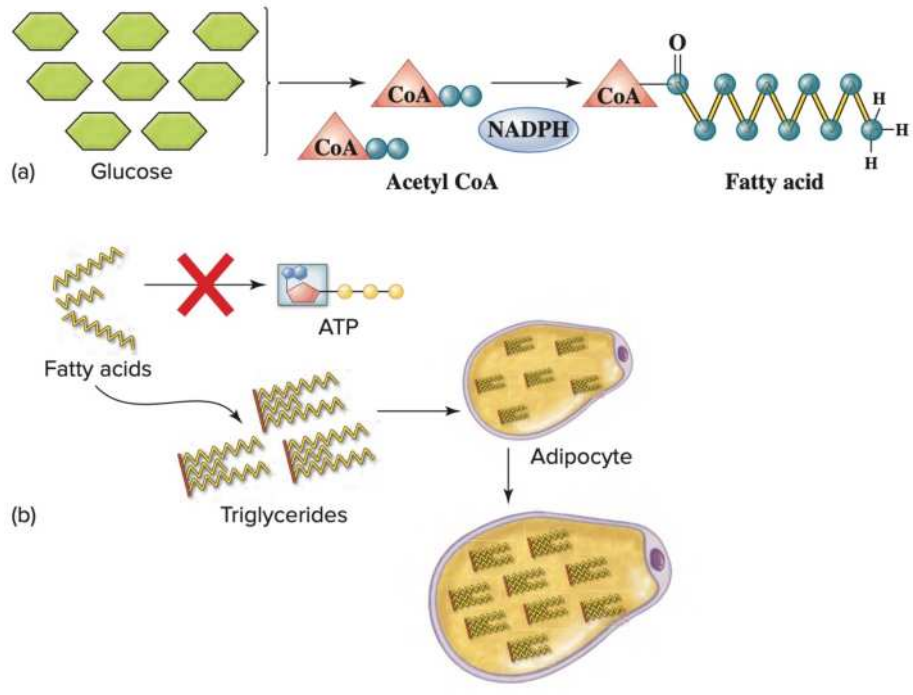
- **Pathway 1: Glycogenesis**
  - Glucose molecules are linked together to form its storage form, **glycogen**
  - Occurs in liver and muscle cells
  - Glycogen storage sites are **limited** and can fill up quickly

Storing Excess Glucose as Fat

Pathway 2: Lipogenesis

- Occurs when glycogen stores are full
- **Lipogenesis:** The production of fatty acids from acetyl CoA
- **Process:** Excess glucose → glycolysis → pyruvate → **acetyl CoA → fatty acid synthesis**
- Occurs in the cytoplasm (not mitochondria)
- The newly made fatty acids are combined with glycerol to form triglycerides for storage in adipocytes
- This is why high carbohydrate consumption can lead to body fat accumulation

FIGURE: Excess glucose and fat storage.



Storing Excess Protein

The body does **not** store excess amino acids

If not needed for protein synthesis, they are deaminated (nitrogen removed)

The carbon skeletons are then:

- Oxidized for energy
- Converted to glucose (and stored as glycogen)
- Converted to acetyl CoA (and stored as fat)

Hormonal Responses

Hormones are "chemical messengers" that regulate metabolic activities

They direct whether the body is in a **fed (anabolic) state** or a **fasted (catabolic) state**

- **Key metabolic hormones:**
  - Insulin
  - Glucagon
  - Cortisol
  - Epinephrine
  - Thyroid hormone

Insulin: The Anabolic Hormone

The primary anabolic hormone

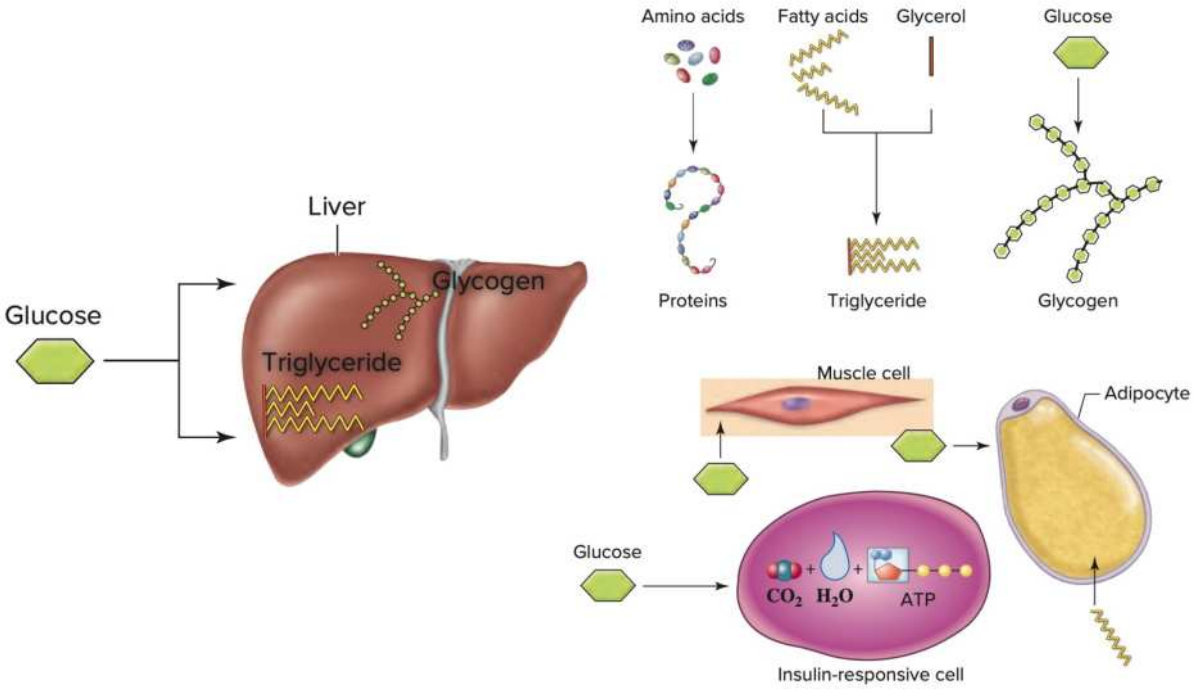
Secreted by: Beta cells in the pancreas

Trigger: A rise in blood glucose (after a meal)

Primary Actions (Promotes Storage):

- Helps glucose enter muscle and fat cells
- Increases **glycogenesis** (glycogen synthesis)
- Increases **triglyceride synthesis** (fat storage)
- Increases **protein synthesis**
- Slows down glycogen breakdown (glycogenolysis)

Figure: Actions of Insulin



The Catabolic Hormones

Instruct cells to use catabolic pathways (breakdown)

Triggered by: Fasting, low blood sugar, stress, and exercise

Overall Goal: Increase fuel mobilization from storage

- **Hormones:**
  - **Glucagon**
  - **Cortisol**
  - **Epinephrine**

Catabolic Hormones (Details)

Glucagon:

- Secreted by alpha cells of the pancreas when blood glucose is *low*
- Signals liver to increase **glycogenolysis** (glycogen breakdown) and **gluconeogenesis** (making new glucose)

Cortisol:

- "Stress hormone" from the adrenal glands
- Promotes **proteolysis** (protein catabolism) to supply amino acids for gluconeogenesis

Epinephrine (Adrenaline):

- From the adrenal glands; "fight-or-flight" response
- Stimulates **glycogenolysis** (in liver and muscle) and **lipolysis** (triglyceride breakdown) for quick energy

Thyroid Hormone

Produced by the thyroid gland

Function: Helps the body adapt by increasing or decreasing the overall **rate of metabolism**

Can be anabolic (promotes growth) or catabolic (increases glucose catabolism)

During starvation: Thyroid hormone production *decreases* to help conserve fuel stores

TABLE: Physiological Effects of Major Metabolic Hormones

Hormone	Overall Metabolic Effect	Effects in Liver	Effects in Adipose Tissue	Effects in Muscle
Insulin	Anabolic	Increases glycolysis and glycogenesis	Increases glucose uptake; increases lipogenesis	Increases glucose and amino acid uptake; increases glycogenesis and protein synthesis
Glucagon	Catabolic	Increases glycogenolysis and gluconeogenesis		
Cortisol	Catabolic	Increases glycogenolysis and gluconeogenesis	Increases lipolysis	Increases proteolysis
Epinephrine	Catabolic	Increases glycogenolysis and gluconeogenesis	Increases lipolysis	
Thyroid Hormone	Regulates metabolic rate	Increases glycogenolysis, gluco-neogenesis, and lipolysis	Increases lipolysis and lipogenesis	Increases glycogenolysis

Metabolic Responses to an Overnight Fast

Blood glucose levels fall → Pancreas releases **glucagon**

Priority 1: Glucagon stimulates liver **glycogenolysis** (glycogen breakdown) to maintain blood glucose.

Liver glycogen becomes depleted after ~12-24 hours.

Priority 2: Glucagon stimulates liver **gluconeogenesis** (making new glucose) from glucogenic amino acids.

During this time, liver and muscle tissue switch to using mainly **fatty acids** for fuel

Metabolic Responses to Starvation

An extended fast involves more drastic adaptations:

Adipose Tissue: Increases **lipolysis**, releasing large amounts of fatty acids into the blood

- **Liver:**
  - Increases **gluconeogenesis** (from amino acids) to fuel red blood cells and the nervous system
  - Increases **ketogenesis**, converting fatty acids into ketone bodies
- **Brain:** Adapts to use **ketone bodies** for energy
- **Muscle Tissue:** Begins **proteolysis** (breaking down muscle protein) to supply amino acids to the liver for gluconeogenesis

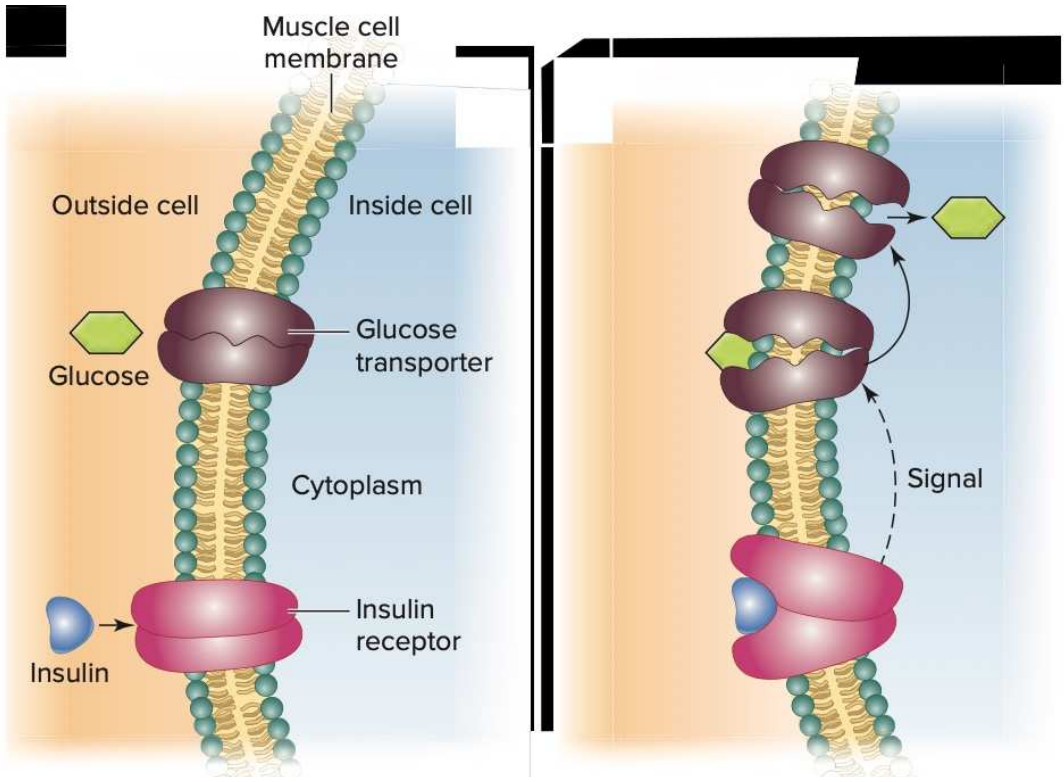
Obesity and Insulin Resistance

In obesity, fat and muscle cells can become **resistant to insulin**

The insulin receptors are no longer sensitive to insulin's signal

Result:

- Glucose and fatty acids are not taken up from the blood effectively
- Blood glucose levels stay abnormally high (hyperglycemia)
- This is a major risk factor for type 2 diabetes



Nonalcoholic Fatty Liver Disease (NAFLD)

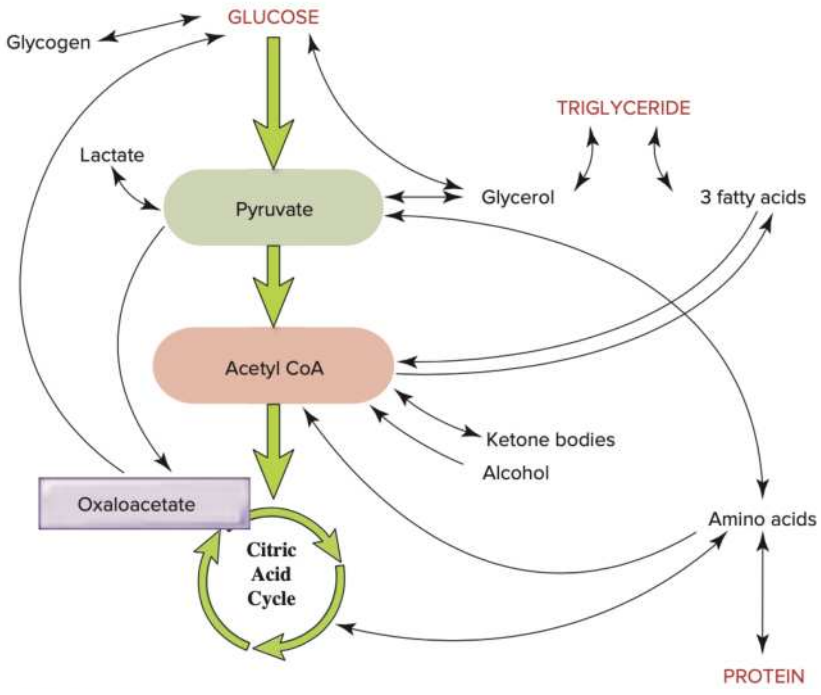
A buildup of fat in the liver not caused by alcohol

Causes:

1. **Insulin Resistance:** Insulin-resistant cells don't take up glucose/fat, so blood levels rise
2. **High Fructose Consumption:** Fructose is metabolized in the liver and rapidly promotes lipogenesis (fat synthesis)

- The liver's influx of glucose and fatty acids stimulates triglyceride synthesis, and this excess fat accumulates in liver cells

Summary of Energy Metabolism




Thank you!

PRINCIPLE OF NUTRITION

Dr. Marwan Jalambo
Assistant Professor (Nutrition)

7

PART 1: Vitamins Overview

Introducing Vitamins

1912: Polish chemist Casimir Funk coined the term **"vitamine"**

vita = necessary for life

amine = a type of nitrogen-containing substance

He discovered a substance (now known as thiamin) in rice bran that cured the disease beriberi.

The term was later changed to **"vitamin"** when scientists found not all are amines.

Have All Vitamins Been Discovered?

It is **unlikely** that any vitamins still need to be discovered.

- **Evidence:**
  1. Babies grow and thrive on **infant formulas** containing known essential nutrients.
  2. People with serious illnesses can be kept alive for years on **liquid synthetic feedings** that contain all known nutrients.
- If a vitamin remained undiscovered, these formulas would not be able to support life.

Vitamins: The Basics

Definition of a Vitamin: A vitamin is a complex organic compound that regulates metabolic processes and meets these criteria:

1. The body **cannot synthesize** it or make enough to maintain good health (it is an essential nutrient).
2. The compound **naturally occurs** in commonly eaten foods.
3. A **deficiency disorder** occurs when the substance is missing or not properly metabolized.
4. Good health is **restored** if the deficiency is treated by supplying the missing substance.

Vitamins vs. Macronutrients

Energy: Vitamins are **not** metabolized for energy, unlike carbohydrates, fats, and proteins.

Amounts: Foods contain much **smaller amounts** of vitamins than macronutrients.

Requirements: The body requires vitamins in **milligram (mg)** or **microgram (µg)** amounts, but needs macronutrients in **gram (g)** amounts.

Classifying Vitamins

1. **Fat-Soluble Vitamins:**
   - Vitamins A, D, E, and K.
   - Found in lipid portions of food.
   - Digested and absorbed with fats.
2. **Water-Soluble Vitamins:**
   - B vitamins (Thiamin, Riboflavin, Niacin, B-6, Pantothenic acid, Folate, Biotin, B-12).
   - Vitamin C.
   - Dissolve in watery components of food and the body.

Table: Classifying Vitamins

Fat-Soluble Vitamins	Water-Soluble Vitamins	Vitamin-Like Substances
A (retinol)	Thiamin (B-1)	Carnitine
D (cholecalciferol, ergocalciferol)	Riboflavin (B-2)	Choline
E (alpha-tocopherol)	Niacin (B-3)	Inositol
K (phylloquinones, menaquinones)	B-6 (pyridoxine)	Lipoic acid
	B-12 (cobalamin)	Taurine
	Biotin	
	Pantothenic acid (B-5)	
	Folate (folic acid)	
	C (ascorbic acid)	

Fat-Soluble vs. Water-Soluble: Key Differences

Storage:

Fat-Soluble: The body stores extra fat-soluble vitamins, primarily in the **liver and adipose tissue**.

Water-Soluble: The body stores only **limited amounts** (except B-12).

Excretion:

Fat-Soluble: More difficult to eliminate; do not dissolve in urine.

Water-Soluble: Kidneys filter and **eliminate excesses in urine**.

Toxicity:

Fat-Soluble: Can accumulate and cause **toxicity**.

Water-Soluble: Generally **not as likely** to be toxic.

Roles of Vitamins

Vitamins regulate cell division, development, growth, and maintenance of tissues.

Some act as hormones (e.g., Vitamin D).

Many participate in chemical reactions that release energy from macronutrients (e.g., Niacin as NAD, Riboflavin as $FADH_2$).

What is an Antioxidant?

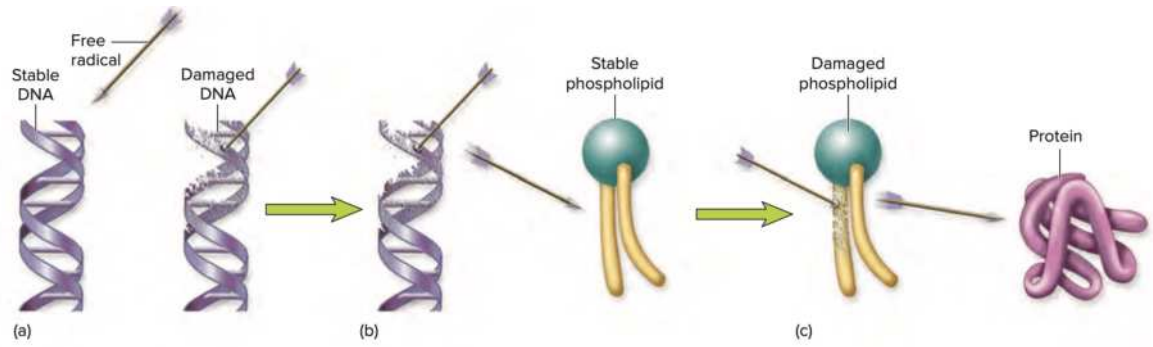
Oxidation: A chemical reaction where an atom or molecule **loses** one or more electrons.

Free Radical: The result of an oxidation reaction; a highly unstable substance with an **unpaired electron**.

Damage: Free radicals are highly reactive and "steal" electrons from stable molecules (proteins, fatty acids, DNA), causing damage.

Antioxidant: A substance that **protects cells by giving up electrons** to free radicals, neutralizing them. Vitamin E is an example.

Figure: Free Radical Activity



Bioavailability of Vitamins

Bioavailability: The amount of a vitamin that is actually available for use by the body.

Factors that affect vitamin bioavailability include:

- Life-stage (e.g., pregnancy) and nutrient status.
- Changes to normal GI transit time (e.g., diarrhea).
- Health conditions affecting fat absorption (e.g., cystic fibrosis).
- Food processing and preparation techniques.
- Source of the vitamin (natural vs. synthetic).

Vitamin Absorption

Primary site: Small intestine.

Absorption increases when the body's needs are higher (e.g., growth, pregnancy).

- **Fat-Soluble Vitamin Absorption:**
  - Requires processes similar to fat digestion.
  - **Bile** enhances absorption.
  - The presence of **dietary fat** stimulates bile release and improves absorption.
  - *Tip:* Adding salad dressing to vegetables enhances absorption of fat-soluble vitamins.

Preserving Vitamin Content

Storage:

- Eat fresh, perishable produce soon after purchase.
- Store at temperatures near freezing, in high humidity, and away from air.
- Store tomatoes, bananas, and garlic at room temperature.

Preparation:

- Wait to trim, peel, and cut fruits/vegetables until just before serving.
- Cook with skins on when possible.
- Cut into large pieces to reduce surface area exposure.

Preserving Vitamin Content (Cooking)

Water: Water-soluble vitamins (C, thiamin, folate) are easily destroyed or lost.

- Avoid high heat and exposure to alkaline substances (baking soda).
- Cook in small amounts of water and reuse the water (e.g., in soups).

Quick Cooking Methods:

- **Microwaving:** Cooks quickly without much water; conserves vitamins.
- **Steaming:** Conserves vitamin content.
- **Stir-frying:** Cooks quickly without water.

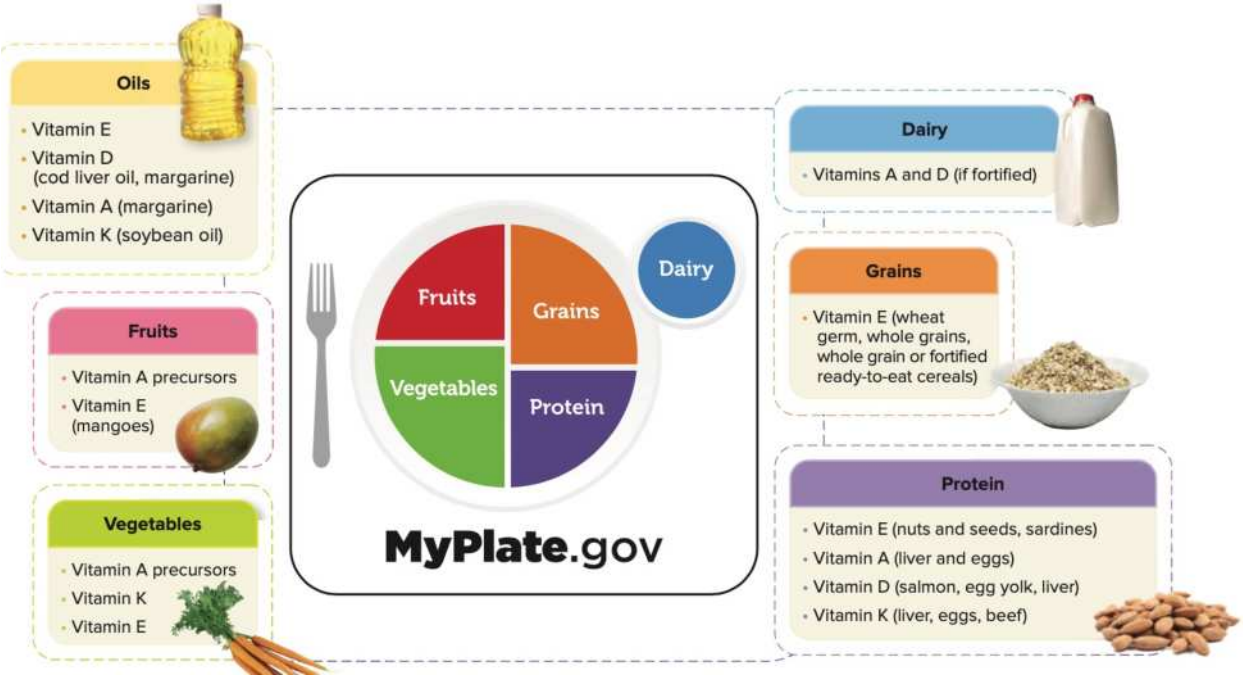
Fresh vs. Frozen or Canned

Canning: Heating process can destroy some vitamins but improves the bioavailability of others.

Frozen: Produce frozen immediately after harvesting and stored properly can be **just as nutritious** as fresh produce.

Cook frozen vegetables **without thawing** to conserve vitamin content.

Overview of Fat-Soluble Vitamins




Thank you!

PRINCIPLE OF NUTRITION

Dr. Marwan Jalambo
Assistant Professor (Nutrition)

7

PART 2: Fat-Soluble Vitamins

VITAMIN A

Vitamin A

Forms of Vitamin A:

1. **Retinoids (Preformed Vitamin A):**
   - Found in animal foods.
   - Includes retinol (most active form), retinal, and retinoic acid.
2. **Carotenoids (Provitamin A):**
   - Found in plants; many are yellow-orange pigments.
   - Beta-carotene, alpha-carotene, and beta-cryptoxanthin can be converted to retinol.

Functions of Vitamin A

Cell Production, Growth, and Development.

- **Epithelial Cells:**
  - Necessary for the production and maintenance of epithelial cells (skin, linings of digestive/respiratory tracts).
  - Helps cells secrete **mucus**, which forms a protective barrier.
- **Immune Function:**
  - Plays a role in the production and activity of white blood cells.
- **Bone Growth and Development:**
  - Participates in the bone-remodeling process.

Vitamin A and Vision

The **retina** contains **rods** and **cones**.

Rods: Enable adaptation to see in poorly lit environments.

Cones: Responsible for color vision; function in well-lit environments.

Both rods and cones need vitamin A (retinol) to function.

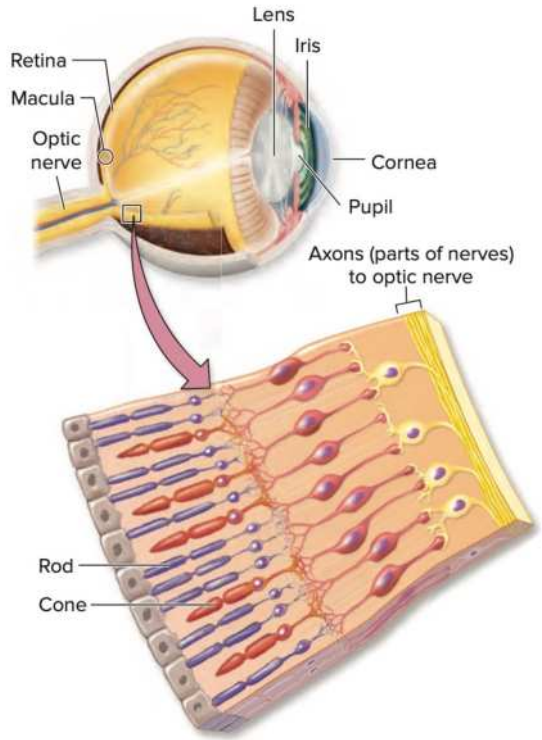
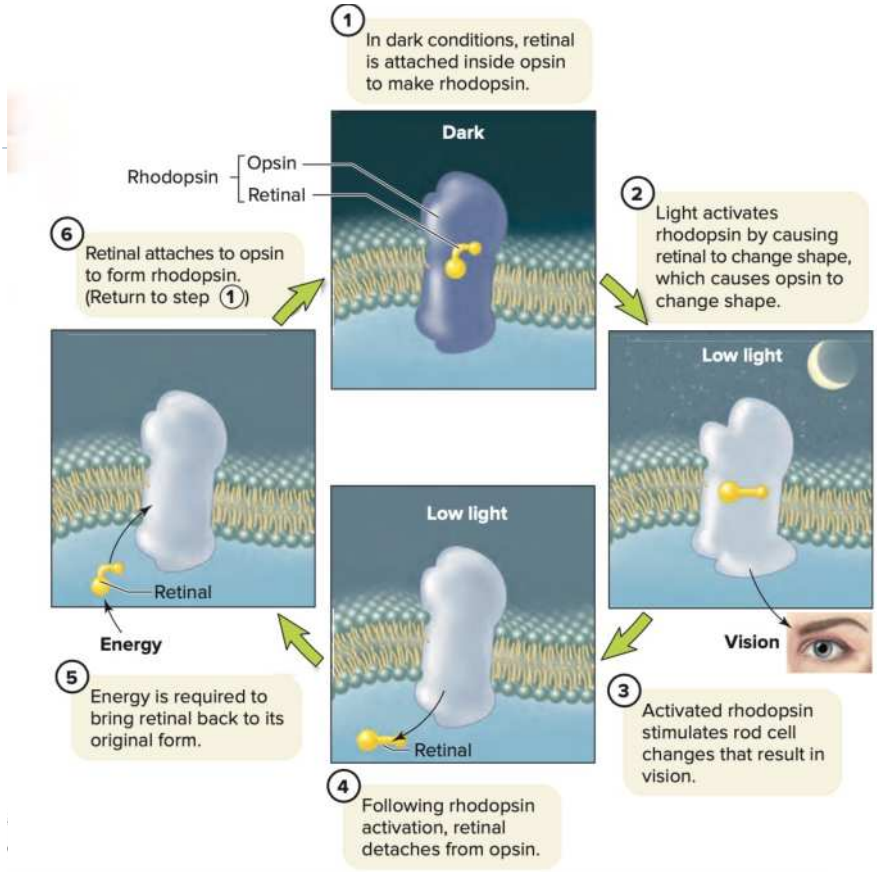


FIGURE: Vitamin A and the visual cycle. Rod cells in the retina need retinol, a form of vitamin A, to function properly.



The Visual Cycle

1. Rods convert retinol (from blood) to **retinal**.
2. In the dark, retinal binds to opsin to form **rhodopsin**.
3. Light strikes the rod, activating rhodopsin.
4. Retinal changes shape and splits from opsin.
5. This split sends a nervous impulse to the brain, resulting in vision.
6. Some retinal is recycled, but some is lost; it must be replaced by retinol from the blood.

Vitamin A Deficiency: Vision

Night Blindness:

- The inability to see in dim light.
- An early sign of vitamin A deficiency.
- Occurs when retinol is unavailable to replace retinal lost during the visual cycle.

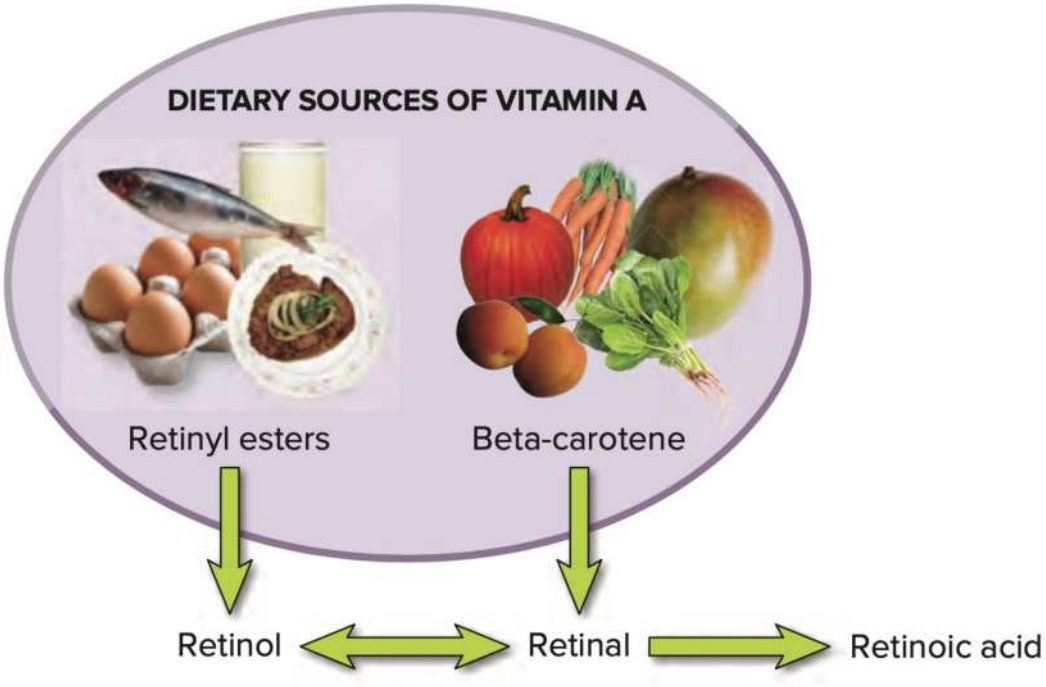
Xerophthalmia ("Dry Eye"):

- A chronic vitamin A deficiency.
- Epithelial cells stop producing mucus, and the cornea hardens (produces keratin) .
- Can easily be damaged by dirt and bacteria.
- Can lead to blindness if untreated.

Vitamin A Food Sources

- **Preformed Vitamin A (Retinyl Esters):**
  - Animal foods: Liver, butter, fish liver oils, eggs.
  - Fortified milk, yogurt, and cereals .
- **Provitamin A (Carotenoids):**
  - Carrots, spinach, leafy greens, pumpkin, sweet potatoes, broccoli, mangoes, cantaloupe.
- **Other Carotenoids (NOT Vitamin A):**
  - Lutein & Zeaxanthin (spinach, kale).
  - Lycopene (tomato products).

FIGURE Converting dietary sources into active forms of vitamin A



Vitamin A: Deficiency and Toxicity

- **Deficiency:**
  - In addition to vision problems, causes **excess keratin** production in skin (rough, bumpy).
  - Reduces immune system effectiveness.
- **Toxicity (Hypervitaminosis A):**
  - UL for adults = 3,000 µg/day.
  - Excessive intake (usually from supplements) can damage the liver.
  - **Teratogen:** Causes severe birth defects when taken in excess during pregnancy.

Carotenemia

Caused by eating too many beta-carotene-rich foods or supplements.

Characterized by **yellowing of the skin**.

Condition is **harmless**.

Skin color returns to normal when intake is reduced.



Vitamin A and Chronic Disease

- **CVD and Cancer:**
  - Diets rich in fruits and vegetables (high in carotenoids) are associated with lower risk.
  - However, research on **supplements** (Vitamin A or beta-carotene) has produced **mixed or no results** for preventing CVD or cancer .
- **Age-Related Macular Degeneration (AMD):**
  - A leading cause of impaired vision over age 55.
  - The macula contains lutein and zeaxanthin.
  - Supplements containing beta-carotene, vitamins C & E, and zinc **slowed the progression** in patients who already had advanced AMD.

Retinol Activity Equivalents

1 retinal activity equivalent (RAE) is equal to
1 μg retinol
12 μg beta-carotene
24 μg alpha-carotene or beta-cryptoxanthin

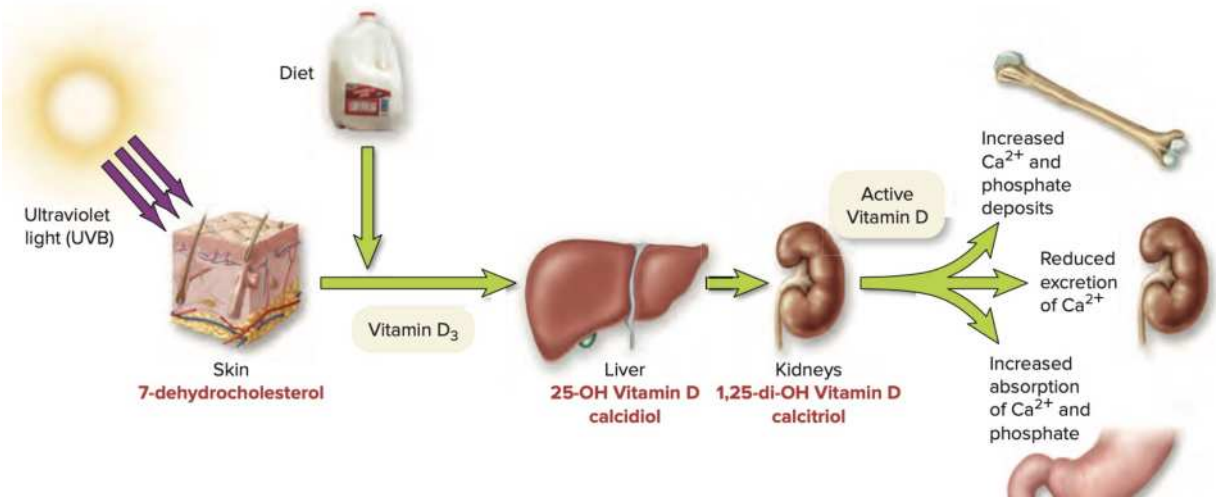
VITAMIN D

Vitamin D

- Also known as the **"sunshine vitamin"** because the body can make it with exposure to UV light.
- Considered both a vitamin and a hormone.
- Historically, sunlight or fish liver oil was used to prevent **rickets**, a condition causing soft, malformed bones in children.



Vitamin D Synthesis and Activation



Vitamin D Synthesis and Activation

1. **Skin:** UV light converts **7-dehydrocholesterol** (a cholesterol-derived substance) into Vitamin D_3.
2. **Liver:** Converts Vitamin D_3 into 25-hydroxyvitamin D (calcidiol).
    - *Blood levels of calcidiol are used to determine vitamin D status.*
3. **Kidneys:** Convert calcidiol into **1,25-dihydroxyvitamin D (calcitriol)**, the most biologically active form.

- *Requires both a healthy liver and healthy kidneys.*

Forms of Vitamin D

- **Vitamin D_2 (Ergocalciferol):**
    - Plant form of vitamin D.
- **Vitamin D_3 (Cholecalciferol):**
    - Animal form of vitamin D.
    - The form synthesized in human skin.

Functions of Vitamin D

- **Calcium and Phosphorus Regulation:**
  - Stimulates small intestine to **absorb calcium and phosphorus** from food.
  - *With vitamin D, calcium absorption increases from 10-15% to 30-80%.*
  - Stimulates bone cells to form calcium phosphate (mineral component of bone).
- **Works with Parathyroid Hormone (PTH):**
  - When blood calcium drops, PTH signals bones to release calcium and kidneys to increase vitamin D production.
- **Other Roles:**
  - Regulates neuromuscular and immune function; reduces inflammation.

FIGURE Maintaining normal blood calcium levels.

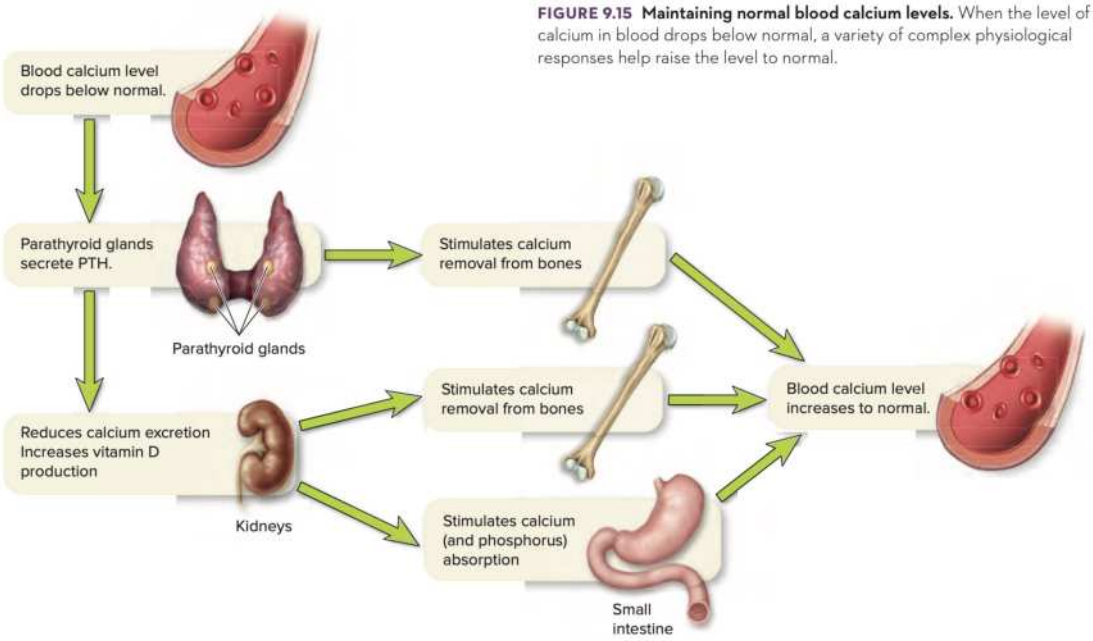


FIGURE 9.15 **Maintaining normal blood calcium levels.** When the level of calcium in blood drops below normal, a variety of complex physiological responses help raise the level to normal.

Vitamin D Food Sources

- **Few foods naturally contain vitamin D:**
  - Fish liver oils (e.g., cod liver oil).
  - Fatty fish (salmon, herring, catfish).
- **Fortified Foods:**
  - Cow's milk.
  - Ready-to-eat cereals.
  - Orange juice.
- **Other:**
  - Mushrooms grown under UV light.

Vitamin D and Sunlight

Sunlight exposure is crucial as food sources are limited.

- **Factors affecting synthesis:**
  - **Location:** results in poor synthesis during winter .
  - **Time of Day:** 10 a.m. to 3 p.m. is most intense.
  - **Skin Color:** Darker skin contains more **melanin**, which blocks UV radiation and *reduces* vitamin D synthesis.
  - **Other:** Clouds, shade, air pollution , and **sunscreen** limit synthesis.

Vitamin D Deficiency

Vitamin D is a **nutrient of public health concern** in the U.S..

Rickets (in Children):

At-risk: Breastfed infants (human milk is low in Vit D) , dark skin, minimal sun exposure.

Recommendation: Breastfed infants should get a 400 IU supplement per day.

Osteomalacia (in Adults):

Means "soft bones".

Bones contain normal collagen but low calcium, making them weak and easy to break.

At-risk: Adults confined indoors, those who cover skin, older adults (synthesis declines with age).

VITAMIN E

Vitamin E

History: Discovered as a component in vegetable oil necessary for reproduction in rats; named **tocopherol** (from Greek *toco* "childbirth").

Forms:

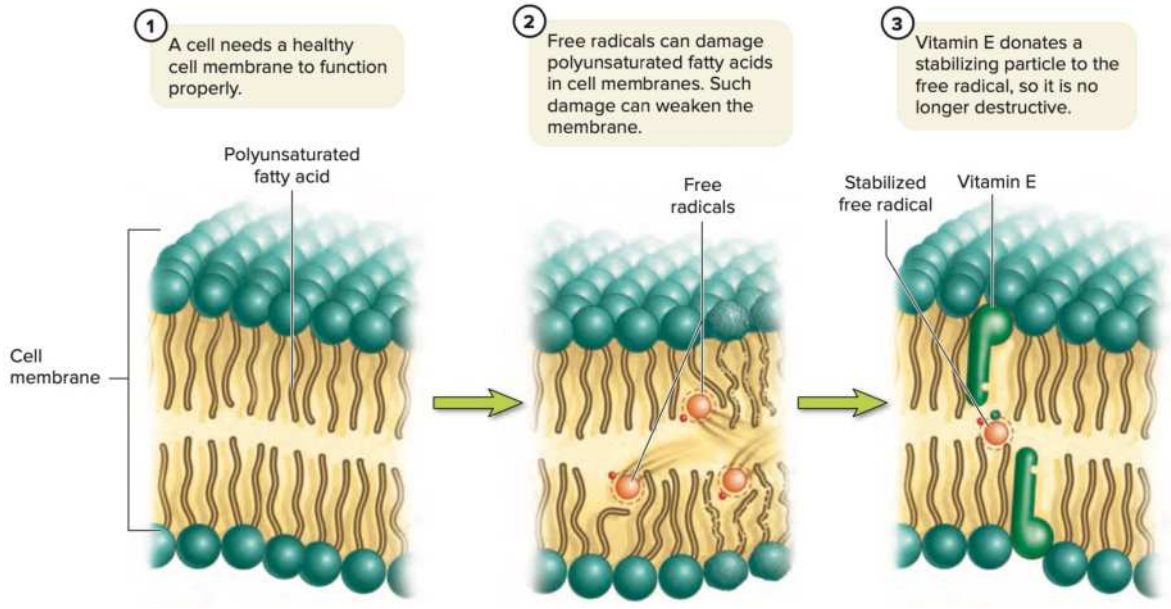
- Exists in eight forms (tocopherols).
- **Alpha-tocopherol** is the form used by the body and has the highest biological activity.
- Gamma-tocopherol is also in foods but has lower activity.

Function of Vitamin E

Potent Antioxidant.

- **Primary role:** Protects polyunsaturated fatty acids (PUFAs) in cell membranes from being damaged by free radicals.
- This protection may help prevent atherosclerosis, cancer, and premature aging.
- Also helps maintain nervous tissue and immune function.

FIGURE Vitamin E and the cell membrane.



Vitamin E Food Sources

Rich Sources:

- Plant oils (sunflower, safflower, canola, olive).
- Sunflower seeds.
- Almonds, almond milk.
- Peanuts.

Other Sources:

- Fish, whole grains, nuts, seeds.

Destruction:

- Lost during milling of grains (not restored by enrichment).
- Easily destroyed by oxygen, metals, light, and high heat (deep-frying).

Vitamin E: Deficiency and Toxicity

- **Deficiency:**
  - **Rare** in healthy people.
  - Occurs in people with diseases that interfere with fat absorption.
  - Can cause nerve damage, loss of neuromuscular control, and anemia (in preterm infants).
- **Toxicity:**
  - UL for adults = 1,000 mg/day.
  - High doses from **supplements** (not food) can be harmful.
  - **Interferes with Vitamin K's** role in blood clotting, leading to **uncontrolled bleeding (hemorrhage)**.

Vitamin E and Chronic Disease

Early epidemiological reports suggested vitamin E-rich diets lowered heart disease risk.

Current Status:

- Scientific enthusiasm for high-dose vitamin E supplements has **subsided**.
- Major trials have **failed to show** that supplements consistently reduce the risk of CVD or cancer.
- Use of supplements has been associated with an **increased risk of hemorrhagic stroke**.
- Results for cognitive decline (Alzheimer's) are conflicting.

VITAMIN K

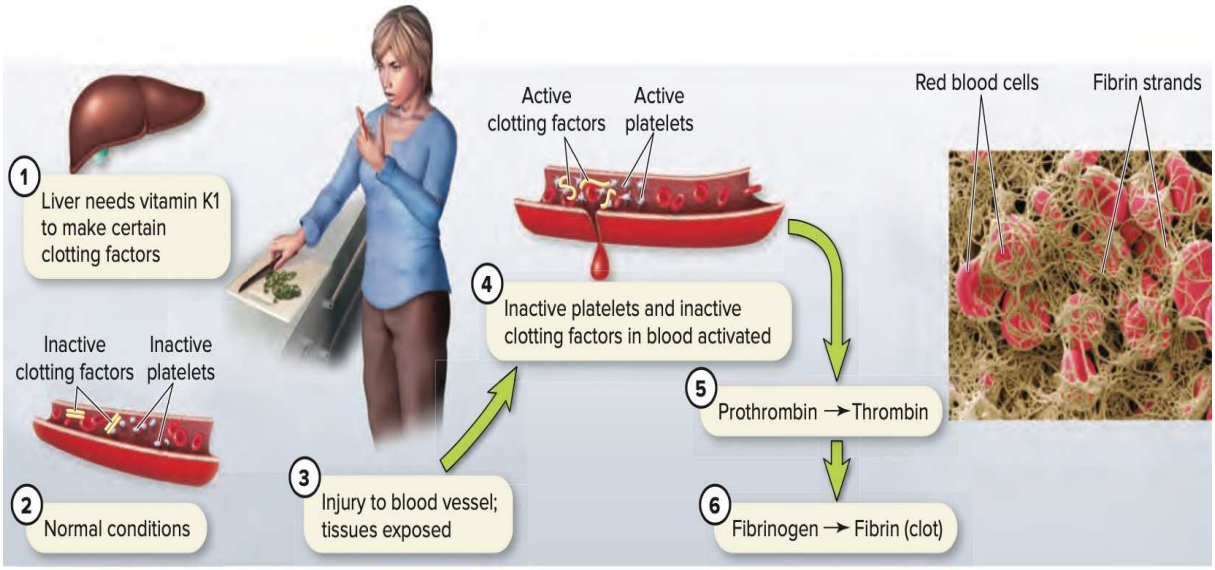
Vitamin K

- **History:** Discovered by Henrik Dam; a factor in alfalfa involved in blood clotting.
- Named "K" after *koagulation* (Danish for coagulation).
- **Forms:**
  - **K_1 (Phylloquinone):** From plants (green leafy veg).
  - **K_2 (Menaquinone):** In animal foods (eggs, butter, beef), fermented foods, and **synthesized by bacteria** in the intestinal tract.
  - **Menadione:** Synthetic form.

Functions of Vitamin K

- **Vitamin K_1 (Phylloquinone):**
  - **Blood Clotting:** The liver needs K_1 to produce four blood-clotting factors, including **prothrombin**.
  - Without K_1, blood does not clot effectively.
- **Vitamin K_2 (Menaquinone):**
  - **Bone Health:** Stimulates bone formation and inhibits bone breakdown.
  - **Calcium Regulation:** Prevents the deposition of calcium in soft tissues, such as blood vessels (prevents arterial calcification).

FIGURE Vitamin K1 and blood clotting.



Vitamin K and Warfarin

- **Warfarin** is a "blood-thinning" medication prescribed to people whose blood clots too easily.
- **Vitamin K interferes with warfarin's activity**.
- Patients on warfarin should **NOT** take vitamin K supplements.
- They should **maintain a consistent daily intake** of vitamin K from their diet (not eliminate it).

Vitamin K Food Sources

Vitamin K_1 (Phylloquinone):

- Green leafy vegetables (kale, spinach, turnip greens, salad greens).
- Broccoli, green beans.
- Soybean and canola oils.

Vitamin K_2 (Menaquinone):

- Egg yolks, butter, cheese, liver, beef.
- Fermented foods (sauerkraut, natto, kefir).

Vitamin K Deficiency

Adults:

- Deficiency is **rare**.
- At-risk: People on long-term antibiotics (which kill gut bacteria), or with fat malabsorption (like cystic fibrosis).
- Sign: Increased time for blood to clot.

Newborns:

- Babies are born with **low vitamin K stores** and an immature GI tract.
- At risk of serious, uncontrolled bleeding.
- **Recommendation:** All newborns receive a **single injection of vitamin K** immediately after birth.

Cystic Fibrosis (CF) and Fat-Soluble Vitamins

What is Cystic Fibrosis (CF)?

- An inherited disease caused by a **defective gene**.
- Leads to the **overproduction of thick and sticky mucus**.
- Negatively affects lungs, pancreas, liver, and intestines.

CF: Signs, Symptoms, and Diagnosis

- **Common Signs & Symptoms:**
  - Salty-tasting skin
  - Failure to gain weight normally in infancy
  - Abdominal pain, bloating, gassiness
  - Foul-smelling feces that float
  - Chronic cough with mucus accumulation

CF: Signs, Symptoms, and Diagnosis

Diagnosis:

- All newborns are screened.
- Positive blood test is followed by a **"sweat test"**.
- A high level of salt in sweat is a positive sign for CF.

CF and Vitamin Malabsorption

- **The Problem:** Thick mucus blocks ducts in the pancreas.
- **Result:** Pancreatic enzymes cannot reach the small intestine, leading to **impaired fat digestion and absorption**.
- **Consequence:** Because fat-soluble vitamins are absorbed *with* fat, individuals with CF are at high risk for deficiency.
  - **Vitamin D deficiency** is almost universal.
  - **Vitamin K deficiency** is common without supplements.
  - **Vitamin A** and **Vitamin E** are also concerns.
- **Treatment:** Includes supplementation of fat-soluble vitamins and use of pancreatic enzyme pills.

Vitamin	Major Functions in the Body	Adult RDA/AI (Adult RDA = Bold)	Major Dietary Sources	Major Deficiency Signs and Symptoms	Major Toxicity Signs and Symptoms
Vitamin A (preformed and provitamin A)	Normal vision and reproduction, cellular growth, immune sys- tem function	**700–900 µg RAE**	Preformed: liver, cow's milk, fortified cereals Provitamin: yellow- orange and dark green fruits and vegetables	Night blindness, xeroph- thalmia, poor growth, dry skin, reduced immune system functioning	**Adult Upper Intake Level (UL) = 3000 µg/day** Nausea and vomiting, headaches, bone pain and fractures, hair loss, liver damage, interference with vitamin K absorption
Vitamin D	Absorption of calcium and phosphorus, mainte- nance of normal blood calcium, calcification of bone, maintenance of immune function	**15–20 µg**	Vitamin D–fortified cow's milk, fortified cereals, fish liver oils, fatty fish	Rickets in children, osteomalacia in adults: soft bones, depressed growth, reduced immune system functioning	**Adult UL = 100 µg/day** Poor growth, calcium deposits in soft tissues

Vitamin	Major Functions in the Body	Adult RDA/AI (Adult RDA = Bold)	Major Dietary Sources	Major Deficiency Signs and Symptoms	Major Toxicity Signs and Symptoms
Vitamin E	Antioxidant	**15 mg** (alpha-tocopherol)	Vegetable oils and products made from these oils, certain fruits and vegetables, nuts and seeds, forti- fied cereals	Loss of muscular coordi- nation; retinal dam- age; hemolysis of red blood cells, resulting in anemia (premature infants)	**Adult UL = 1000 mg/day** Excessive bleeding as a result of interfering with vitamin K metabolism
Vitamin K	Production of active blood-clotting factors, prevention of soft tissue calcification	90–120 μg	Green leafy vegetables, canola and soybean oils, and products made these oils	Excessive bleeding	**Adult UL = undetermined** Unknown



PRINCIPLES OF NUTRITION

Dr. Marwan Jalambo
Assistant Professor (Nutrition)

8

PART 1: Water-Soluble Vitamins

Introducing Water-Soluble Vitamins

This chapter focuses on nine water-soluble vitamins:

1. **Vitamin C**
2. **The 8 B vitamins:**
    1. Thiamin (B1)
    2. Riboflavin (B-2)
    3. Niacin (B-3)
    4. Biotin
    5. Pantothenic acid (B-5)
    6. Pyridoxine (B-6)
    7. Folate (B-9)
    8. Cyanocobalamin (B-12)

Water-Soluble Vitamins: Basic Concepts

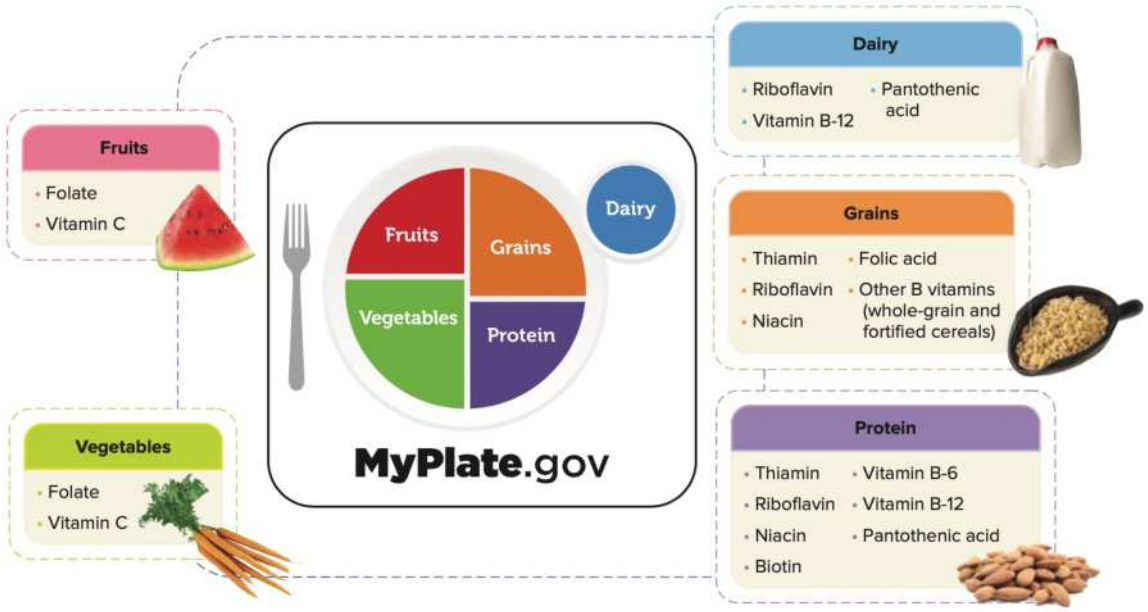
Solubility: They dissolve in the watery components of food and the body.

Storage: Compared to fat-soluble vitamins, excesses are **not stored in large amounts**.

Excretion: Excesses are filtered by the kidneys and eliminated in the urine.

Consumption: Because they aren't stored, they need to be consumed on a **regular basis** to prevent deficiency.

FIGURE: MyPlate: Water-soluble vitamins



Water-Soluble Vitamins as Coenzymes

Function: Most water-soluble vitamins function as components of specific **coenzymes**.

Role: Many chemical reactions involved in the metabolism of carbohydrates, fats, and amino acids involve coenzymes that contain B vitamins.

Synthesis:

1. Coenzyme forms in food are digested to release the free vitamin.
2. After absorption, cells combine the B vitamin with another compound to synthesize a coenzyme.
3. The coenzyme then activates an enzyme, allowing a reaction to occur.

FIGURE: Coenzyme synthesis

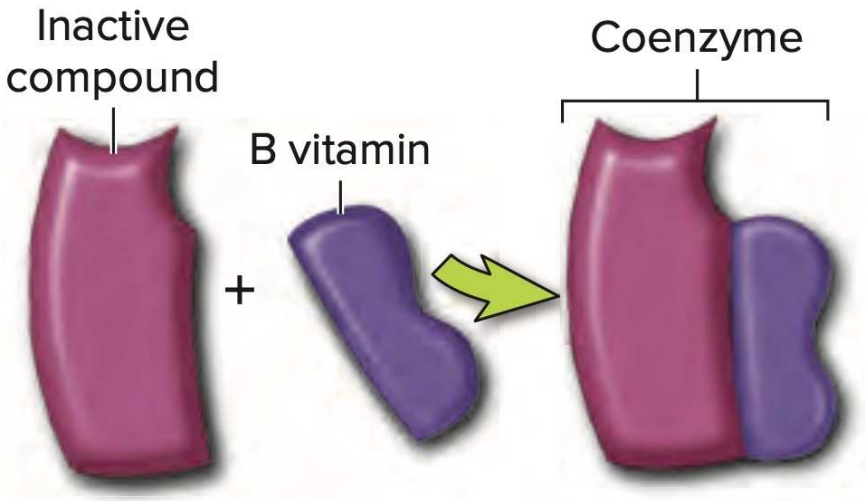
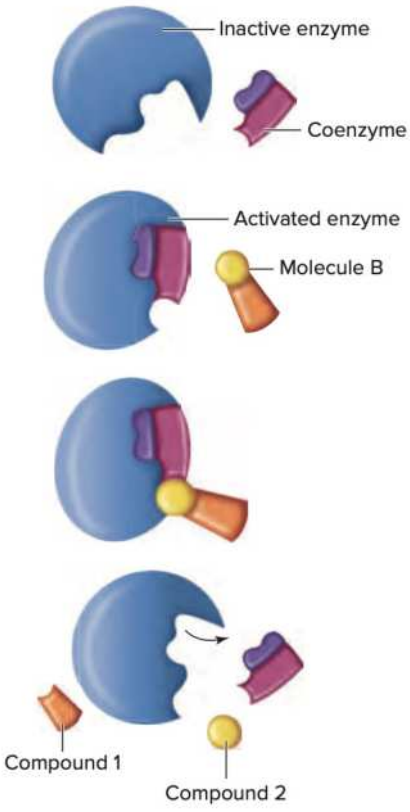


FIGURE: Coenzyme act



Vitamin B1 (Thiamin)

Thiamin (Vitamin B-1): Functions

Thiamin is part of the coenzyme **thiamin pyrophosphate (TPP)**.

TPP is necessary for:

1. Reactions that remove a CO2 molecule from a larger compound.
2. The breakdown of carbohydrates to release energy.
3. The metabolism of branched-chain amino acids (leucine, isoleucine, valine).
4. The synthesis of neurotransmitters (chemicals that allow nerve cells to communicate).

Thiamin: Food Sources

Good Sources: Whole-grain and enriched breads and cereals, legumes, and nuts.

Note: Overheating food can destroy thiamin.

Thiamin Content of Selected Foods

- White rice, enriched, dry, 1 cup: ~0.9 mg
- Lentils, raw, 1/2 cup: ~0.8 mg

- **RDA for adult female: 1.1 mg/day**
- **RDA for adult male: 1.2 mg/day**

Thiamin: Deficiency (Beriberi)

Disease: Beriberi.

Symptoms: People are very weak and have poor muscular coordination.

Affected Systems: Cardiovascular, digestive, and nervous systems.

A "Did You Know?" Case: An outbreak occurred on a ship in Thailand. The fishermen ate only polished rice (poor thiamin source) and **raw seafood**.

Thiaminase: Raw seafood contains thiaminase, an enzyme that **destroys thiamin**.

Thiamin: Deficiency (Wernicke-Korsakoff)

Condition: Wernicke-Korsakoff syndrome, a degenerative brain disorder.

Primary Cause: Occurs primarily in people with **alcohol use disorder**.

Mechanism: Alcohol reduces thiamin absorption and increases its excretion.

Signs & Symptoms:

Abnormal eye movements

Staggering gait

Distorted thought processes

Treatment: Eliminating alcohol and thiamin injections. Can be permanently disabling or fatal if untreated.

Thiamin: Toxicity

Toxicity from food or supplements is rare.

Reason: Excess thiamin is readily excreted in the urine.

Upper Level (UL): No UL has been established for thiamin.

Vitamin B2 (Riboflavin)

Riboflavin (Vitamin B-2): Functions & Sources

Coenzyme Forms:

Flavin mononucleotide (FMN)

Flavin adenine dinucleotide (FAD)

Functions: Play key roles in enzymatic reactions, including those for fatty acid and folate metabolism.

Sources:

Cow's milk, yogurt, and other dairy products.

Enriched cereals, almonds, and liver.

Mushrooms, broccoli, spinach.

Sensitivity: Riboflavin is stable to heat but **breaks down rapidly in light**. This is why milk is stored in opaque containers, not clear glass.

Riboflavin: Sources & Deficiency

Riboflavin Content of Selected Foods

- Beef liver, cooked, 3 oz: ~3.1 mg
- All-Bran cereal, 1/2 cup: ~2.7 mg
- Energy drink, 8 oz: ~1.7 mg
- Yogurt, plain, low-fat, 8 oz: ~0.5 mg

RDA for adult female: 1.1 mg/day

RDA for adult male: 1.3 mg/day



- **Deficiency:** Ariboflavinosis (rare).
- **Signs:** Fatigue , **glossitis** (swollen, sore, reddish-purple tongue) , and **cheilosis** (scaling/cracking at corners of mouth).

Riboflavin: Adequacy & Migraines

RDA: 1.1 mg/day (females), 1.3 mg/day (males).

Average U.S. Intake: Well above the RDA.

Toxicity: No side effects from large amounts; rapidly eliminated in urine.

UL: No UL has been established.

Vitamin B3 (Niacin)

Niacin (Vitamin B-3): Functions & Sources

Forms: Nicotinic acid and nicotinamide.

Coenzyme Forms:

Nicotinamide adenine dinucleotide (NAD)

Nicotinamide adenine dinucleotide phosphate (NADP)

Functions: Niacin coenzymes participate in at least 200 reactions, including pathways for releasing energy from macronutrients.

Sources: Enriched cereals, beef liver, tuna, salmon, poultry, pork, mushrooms.

Niacin: Synthesis from Tryptophan

The human body can synthesize niacin from the amino acid **tryptophan** (found in protein-rich foods).

Conversion Rate: About 60 mg of tryptophan yields about 1 mg of niacin.

Niacin Equivalents (NEs):

- Recommendations are given in NEs to account for both preformed niacin in food and niacin made from tryptophan.
- 1 NE = 1 mg of niacin.

Niacin: Deficiency (Pellagra)

Disease: Pellagra.

Risk Factors (in U.S.): Alcohol use disorder, eating disorders.

The "4 Ds of Pellagra":

1. **D**ermatitis
2. **D**iarrhea
3. **D**ementia
4. **D**eath



"Did You Know?" Corn:

1. Corn's niacin is tightly bound to a protein and resists digestion.

2. The traditional Mexican practice of soaking corn in **lime water** to make tortillas frees the niacin, enhancing its absorption and preventing pellagra.

Niacin: Toxicity & Clinical Use

UL (Adult): 35 mg/day (for nicotinic acid).

Clinical Use: Health care providers may prescribe **megadoses** of nicotinic acid to:

- Reduce elevated LDL cholesterol levels.
- Increase HDL cholesterol levels.

Toxicity Side Effects:

- Flushing of the skin (face and chest).
- GI tract upset (nausea, vomiting).
- Liver damage.

Pantothenic Acid (Vitamin B-5)

Pantothenic Acid (Vitamin B-5)

Coenzyme Form: Part of **Coenzyme A (CoA)**.

Functions:

- Helps release energy from macronutrients.
- Needed for fatty acid synthesis.

Name: "Pantothen" is Greek for "everywhere".

Sources: Widespread in foods. Rich sources include fortified cereals, liver, sunflower seeds, and mushrooms.

Deficiency: Very unlikely among healthy people.

AI (Adult): 5 mg/day.

UL: None established; no reports of toxicity.

Biotin

Biotin

Function: As a coenzyme, participates in chemical reactions that add CO2 to compounds.

Role: Promotes the synthesis of glucose and fatty acids.

Sources: Liver, eggs, peanuts, salmon, pork.

Deficiency: Severe deficiencies rarely occur.

- Why? Intestinal bacteria produce some biotin, and it's found in many foods.

"Did You Know?" - Raw Eggs:

- **Avidin**, a protein in *raw* egg whites, binds biotin and prevents its absorption.

- **Cooking eggs denatures avidin**, removing the risk

Pyridoxin (Vitamin B-6)

Vitamin B-6: Functions

Forms: A family of 3 compounds (pyridoxine, pyridoxal, pyridoxamine).

Primary Coenzyme: Pyridoxal phosphate (PLP).

Major Role: Facilitates enzymatic reactions involved in **amino acid metabolism**.

- Includes the conversion of tryptophan to niacin.
- Includes transamination reactions to form nonessential amino acids.

Homocysteine: PLP helps convert the toxic amino acid **homocysteine** to cysteine (a nonessential amino acid).

Vitamin B-6: More Functions

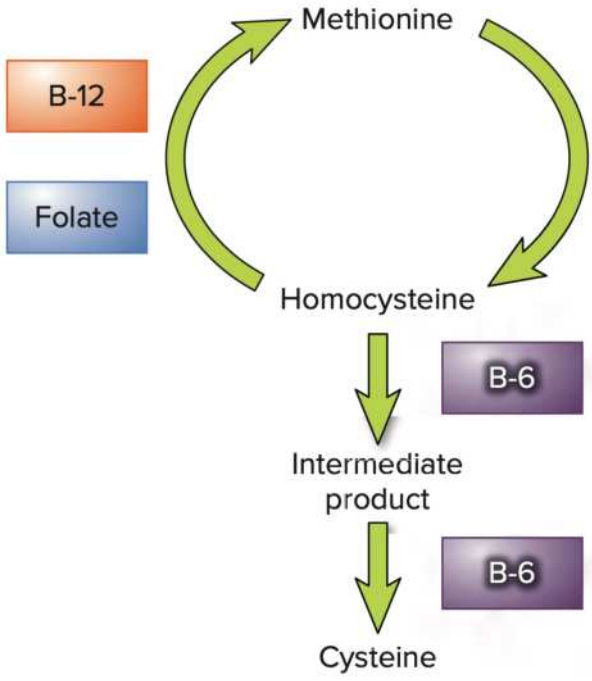
Heme Production:

- PLP participates in producing **heme**, the iron-containing portion of **hemoglobin**.
- Hemoglobin is the protein in red blood cells that transports oxygen.
- Deficiency can lead to a type of anemia.

Neurotransmitter Synthesis:

- Involved in making neurotransmitters like serotonin, dopamine, and histamine from their amino acid precursors.

Figure: Homocysteine Metabolism



Vitamin B-6: Sources & Deficiency

Sources: Pistachio nuts, walnuts, salmon, beef, potatoes, bananas, spinach.

Refining: Vitamin B-6 is lost when grains are refined and is **not added back** during enrichment (unlike thiamin, riboflavin, niacin, and folic acid).

Deficiency: Rare, but can result from alcohol use disorder.

Signs: Dermatitis, anemia, convulsions, depression, confusion

Vitamin B-6: Toxicity

UL (Adult): 100 mg/day.

Toxicity: Unlike most B vitamins, **megadoses of B-6 are toxic**.

Condition: Can cause severe sensory nerve damage, called **peripheral neuropathy**.

Symptoms: Walking difficulties and numbness of the hands and feet.

Note: The nerve damage resolves when megadoses are stopped.

Supplements: There is *not* enough scientific evidence to support using B-6 supplements for PMS or morning sickness.

Folate

Folate: Functions

Nomenclature:

- **Folate:** The group of related compounds naturally in food.
- **Folic Acid:** The synthetic form used in supplements and fortified foods.

Coenzyme Form: Tetrahydrofolate (THF).

Function: THF accepts and transfers single-carbon groups (like CH3).

Critical Roles:

1. **DNA Synthesis**
2. Amino Acid Metabolism

Because of its role in DNA synthesis, THF is needed for **cells to divide**.

Folate and Vitamin B-12: Working Together

The roles of folate and vitamin B-12 are interrelated.

Process (Recycling Methionine):

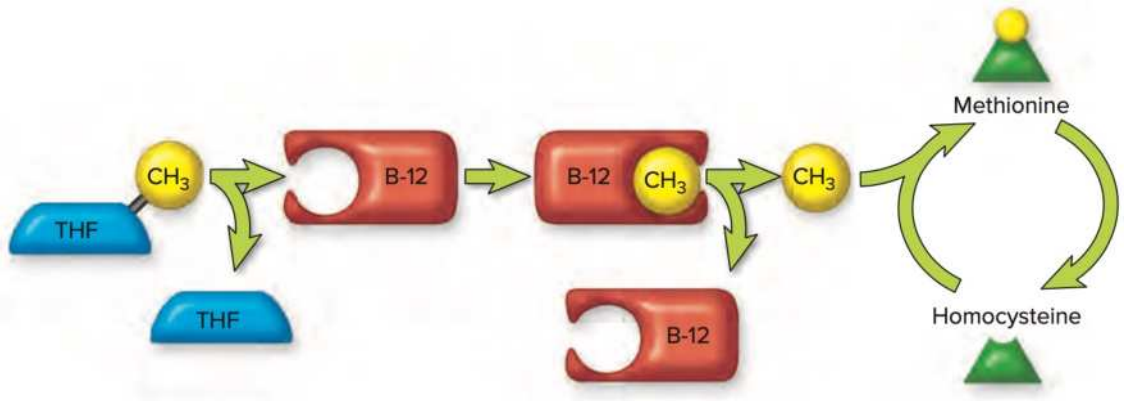
1. Folate (THF) transfers a methyl group (CH3) to vitamin B-12.
2. Vitamin B-12 then transfers that CH3 group to homocysteine.
3. This forms methionine.

The "Folate Trap": When vitamin B-12 is unavailable, folate cannot transfer its CH3 group. Folate gets "trapped" and cannot be used, leading to a **folate deficiency**, even if dietary intake is adequate.

Folate and Vitamin B-12: Working Together

The roles of folate and vitamin B-12 are interrelated.

Process (Recycling Methionine):



Folate: Sources

Natural Sources: Leafy vegetables ("foliage"), liver, legumes, asparagus, broccoli, and orange juice.

Fortified Sources: Enriched grain products (bread, flour, cereals) are among the **richest sources of folic acid** in the American diet.

Sensitivity: Folate is easily destroyed by heat, oxidation, and ultraviolet light. Food processing can destroy 50-90% of folate.

Tip: Eating fresh fruits and raw or lightly cooked vegetables retains the most folate.

Folate: Absorption & DFEs

Bioavailability: Natural folate and synthetic folic acid are absorbed differently.

- **Natural Folate (Polyglutamate):** Has a "string" of glutamates. Requires intestinal enzymes to cut off all but one before absorption.
- **Folic Acid (Monoglutamate):** Has only one glutamate. No digestion needed.

Result: Folic acid is **more bioavailable** than natural food folate.

Dietary Folate Equivalents (DFEs): Units that account for this difference.

Equation: DFE = µg natural food folate + (1.7 x µg folic acid)

Folate: Deficiency (Megaloblastic Anemia)

RDA (Adult): 400 µg (DFE)/day.

Risk: Increases during rapid growth (pregnancy, infancy, childhood).

Mechanism:

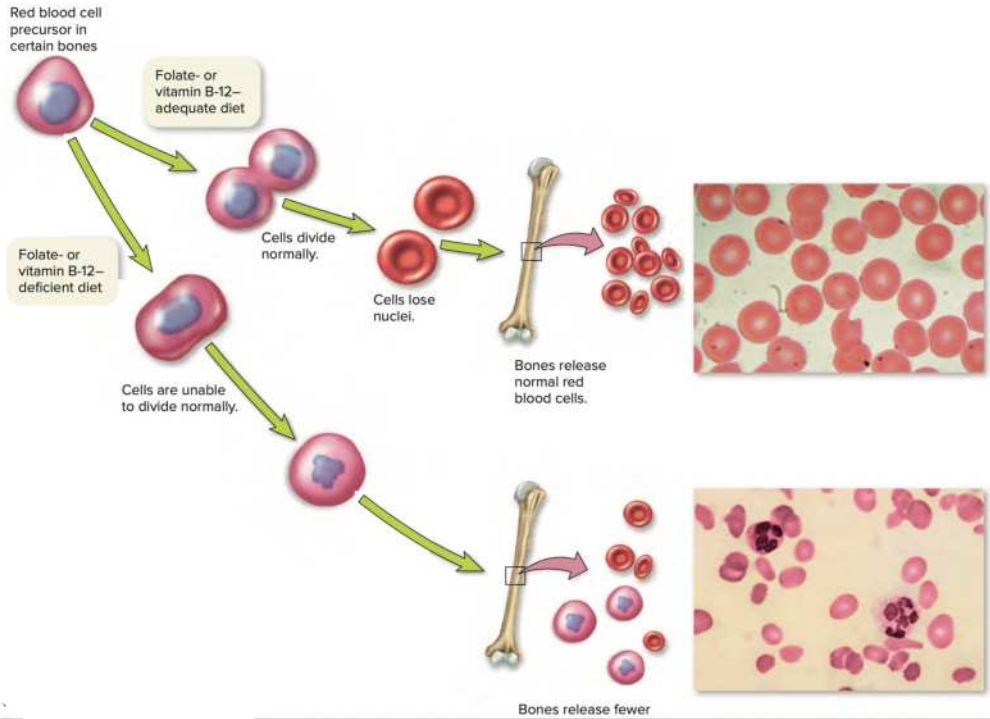
1. Folate deficiency affects rapidly dividing cells, especially **red blood cell (RBC) precursor cells** in bone marrow.
2. Without folate, these cells cannot make new DNA to divide.
3. The precursor cells enlarge but **cannot divide normally**.

Condition: Megaloblastic Anemia (or macrocytic anemia).

1. Characterized by large, immature RBCs (megaloblasts) that still have nuclei.
2. These abnormal cells don't carry oxygen properly.

Note: This *same* anemia is also caused by a **vitamin B-12 deficiency**.

Figure: Red Blood Cell Production



Folate: Deficiency (Neural Tube Defects)

Pregnancy: Folate needs increase to 600 µg (DFE)/day.

Why? Rapid DNA synthesis and cell division occur during embryonic development.

Neural Tube:

- Forms in the human embryo in the **first few weeks after conception** (often before a female knows she is pregnant).
- It develops into the brain and spinal cord.

Deficiency: Pregnant females with folate deficiency have a high risk of giving birth to infants with **Neural Tube Defects (NTDs)**.

NTDs: Spina Bifida & Anencephaly

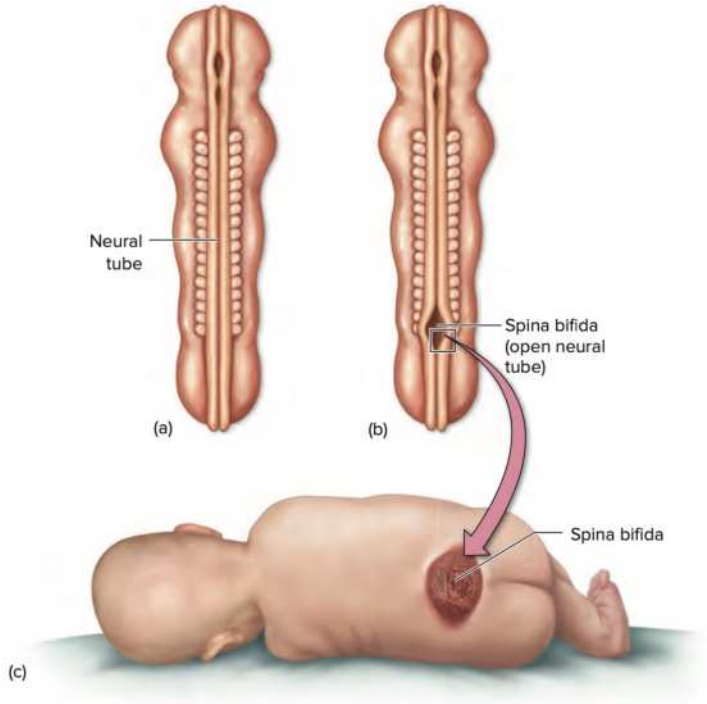
Spina Bifida: The embryo's spine does not form properly, and the bones fail to enclose the spinal cord. Can result in a sac bulging from the back and inability to walk.

Anencephaly: Much of the brain is malformed or missing; infants usually die shortly after birth.

Prevention:

- All females capable of becoming pregnant should consume **400 µg of *folic acid* daily**.
- FDA requires addition of folic acid to enriched flour and cereals.

NTDs: Spina Bifida



Folate: Toxicity & The B-12 Connection

UL (Adult): 1000 µg/day (for *synthetic* folic acid only).

Toxicity Concern: Excess folic acid can **"mask" a vitamin B-12 deficiency**.

How it Masks:

1. A B-12 deficiency causes megaloblastic anemia.
2. A folate deficiency *also* causes megaloblastic anemia.
3. Taking high-dose folic acid supplements will *cure* the anemia caused by a B-12 deficiency.
4. **BUT:** The folic acid does **NOT** prevent the **serious, irreversible nerve damage** that also accompanies B-12 deficiency.

Result: The anemia (an early warning sign) is gone, but the B-12 deficiency continues, allowing permanent nerve damage to progress undetected.

Cyanocobalamin (Vitamin B12)

Vitamin B-12 (Cobalamin): Functions

Coenzyme Roles:

1. Participates in cellular processes, including the metabolism of folate.
2. Needed for homocysteine metabolism (converts it to methionine, with folate).
3. Helps convert folate to its active coenzyme forms needed for DNA synthesis.

A Vital Function (No Folate Involved):

1. **Maintaining the myelin sheath** that wraps around and insulates nerve cells.
2. Myelin allows nerves to communicate effectively.

Deficiency Consequence: Without B-12, the myelin sheath deteriorates, which can lead to **paralysis and death**.

Vitamin B-12: Sources

Synthesis: Only bacteria, fungi, and algae can synthesize B-12.

Plants do NOT make vitamin B-12.

Natural Sources: People rely almost entirely on **animal foods**.

- Meat, cow's milk, dairy products, poultry, fish, shellfish, and eggs.
- Liver is one of the richest sources (B-12 is stored there).

Vegan Sources: Vegans must consume **fortified foods** (cereals, fortified almond/soy milk) or take supplements.

Vitamin B-12: Digestion & Absorption

Step 1 (Stomach): Hydrochloric acid (HCl) in gastric juice separates the B-12 from the animal protein it's bound to.

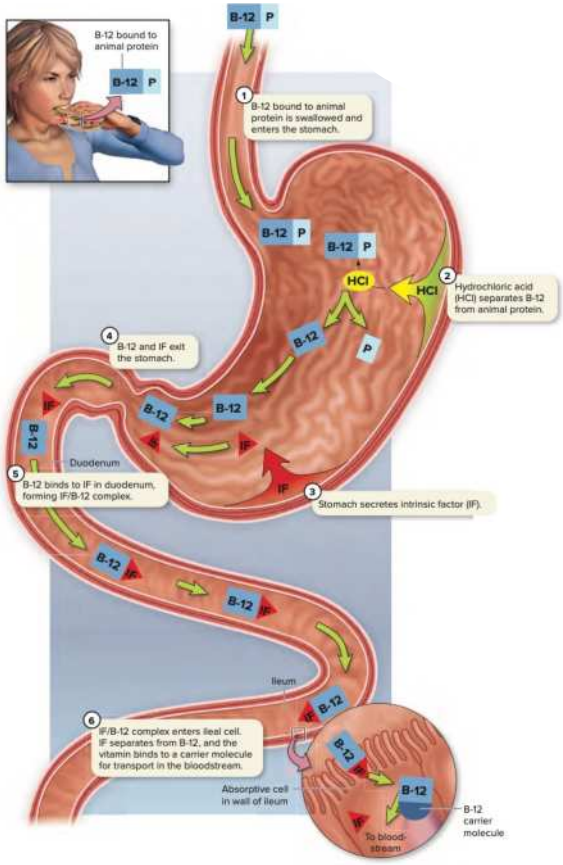
Step 2 (Stomach): Parietal cells in the stomach secrete a compound called **intrinsic factor (IF)**.

Step 3 (Small Intestine): The free B-12 binds to intrinsic factor (IF).

Step 4 (Ileum): The B-12/IF complex is absorbed in the ileum (the last part of the small intestine).

Note on Synthetic B-12: B-12 in fortified foods or supplements is *not* bound to protein, so it **does not need HCl** to be released. This makes it easier to absorb for people with low stomach acid.

FIGURE: Natural vitamin B-12 absorption.



Vitamin B-12: Storage & Adequacy

Storage: The liver stores about 50% of the body's B-12.

Reserves: A healthy liver has enough B-12 reserves to last **5 to 10 years**.

> This is why a new vegan may not show deficiency signs for many years.

RDA (Adult): 2.4 µg/day.

UL: No UL has been established.

Vitamin B-12: Deficiency Signs

Most cases of B-12 deficiency are **NOT** from inadequate intake, but from **problems with intestinal absorption**.

Signs & Symptoms:

- **Megaloblastic anemia** (same as folate deficiency)
- **Nerve damage** (unique to B-12 deficiency)
- Muscle weakness
- Smooth and shiny tongue
- Confusion, difficulty walking, and maintaining balance
- **Numbness and tingling** sensations, especially in hands and feet

Vitamin B-12: Absorption Issues

1. Food-Cobalamin Malabsorption:

- Caused by an inability to release B-12 from animal protein.
- Common in older adults because **HCl production in the stomach declines with age**.
- *These individuals CAN still absorb synthetic B-12* (from supplements/fortified foods) because it doesn't need HCl.

Vitamin B-12: Absorption Issues

2. **Pernicious Anemia:**
   - An **autoimmune disorder** that causes the stomach to **stop making intrinsic factor (IF)**.
   - Without IF, B-12 cannot be absorbed, regardless of intake.
   - This is a "deadly" condition if untreated.
   - **Treatment:** Must bypass absorption. Requires **monthly B-12 injections** or a nasal gel.

Ascorbic Acid (Vitamin C)

Vitamin C (Ascorbic Acid): Functions

Humans are one of the few species that **cannot synthesize** vitamin C; it is essential.

It serves as a nutrient **cofactor**, not a coenzyme.

Key Functions:

1. **Collagen Synthesis:**
   - Participates in reactions that form and maintain **collagen**.
   - Collagen is the fibrous protein that gives strength to connective tissues (bone, cartilage, tendons).
2. **Antioxidant Activity:**
   - Donates electrons to **free radicals** to neutralize them.
   - Helps **recycle vitamin E** (another antioxidant).

Vitamin C: Functions

3. **Immune Function:**
   - White blood cells (WBCs) have a high concentration of vitamin C.
   - WBCs generate free radicals to kill bacteria; vitamin C's antioxidant activity may protect the WBCs from self-destruction.
4. **Synthesis:**
   - Necessary for synthesizing bile, certain neurotransmitters, and hormones (like cortisol).
5. **Iron Absorption:**
   - **Enhances the absorption of nonheme iron** (from plants).
   - It donates an electron to convert Fe3+ (ferric iron) to Fe2+ (ferrous iron), which is the form the small intestine absorbs more readily.

Vitamin C: Sources

Best Sources: Plant foods.

- Peppers (red and green), citrus fruit, papaya, broccoli, cabbage, and berries.

Sensitivity: Vitamin C is very unstable. It is easily destroyed by:

- Heat
- Oxygen
- Light
- Alkaline conditions

Note: Storing C-rich foods in the refrigerator and eating them raw helps preserve the vitamin.

Vitamin C: Deficiency (Scurvy)

RDA (Adult): 75 mg/day (females), 90 mg/day (males).

Smokers: Need an **extra 35 mg/day** because smoke increases free radical formation in the lungs.

Disease: Scurvy.

Mechanism: Without vitamin C, the body forms weak connective tissue and cannot maintain existing collagen.

Vitamin C: Deficiency (Scurvy)

Signs & Symptoms:

- **Pinpoint bleeding** under the skin (petechiae).
- Skin bruises easily.
- **Gums become spongy and bleed**.
- Joints swell and ache.
- Teeth fall out.
- Fatal if untreated.

Vitamin C: Toxicity & The Common Cold

UL (Adult): 2,000 mg/day.

Toxicity:

- Exceeding the UL often causes **gastrointestinal upset and diarrhea**.
- Excess vitamin C is broken down to **oxalate**.
- Excess oxalate is excreted in urine and **raises the risk of kidney stones** in susceptible people.

The Common Cold:

- Routine supplementation (200 mg or more/day) does **NOT prevent** colds in the general population.
- It **MAY shorten the duration** of cold symptoms.
- It **MAY reduce the severity** of symptoms.

Vitamin	Major Functions in the Body	Adult RDA/AI (Adult RDA = Bold)	Major Dietary Sources	Major Deficiency Signs and Symptoms	Major Toxicity Signs and Symptoms
Thiamin	Part of TPP, coenzyme needed for carbohydrate metabolism and the metabolism of certain amino acids; may help produce neurotransmitters	**1.1–1.2 mg**	wheat germ, enriched breads and cereals, legumes, nuts	Beriberi and Wernicke-Korsakoff syndrome: weakness, abnormal nervous system functioning	None (Upper Level [UL] not determined)
Riboflavin	Part of FMN and FAD, coenzymes needed for carbohydrate, amino acid, and lipid metabolism	**1.1–1.3 mg**	Cow's milk, yogurt, and other dairy products; enriched breads and cereals	Inflammation of the mouth and tongue, eye disorders	None (UL not determined)

Vitamin	Major Functions in the Body	Adult RDA/AI (Adult RDA = Bold)	Major Dietary Sources	Major Deficiency Signs and Symptoms	Major Toxicity Signs and Symptoms
Niacin	Part of NAD and NADP, coenzymes needed for energy metabolism	**14–16 mg**	Enriched breads and cereals, beef, liver, tuna, salmon, poultry, pork, mushrooms	Pellagra Diarrhea Dermatitis Dementia Death	**Adult UL = 35 mg/day** Flushing of facial skin, itchy skin, nausea and vomiting, liver damage
Pantothenic acid	Part of coenzyme A that is needed for synthesizing fat and that helps release energy from carbohy- drates, fats, and protein	5mg	Beef and chicken liver, sunflower seeds, mushrooms, yogurt, fortified cereals	Rarely occurs	Unknown (UL not determined)

Vitamin	Major Functions in the Body	Adult RDA/AI (Adult RDA = Bold)	Major Dietary Sources	Major Deficiency Signs and Symptoms	Major Toxicity Signs and Symptoms
Biotin	Coenzyme needed for synthesizing glucose and fatty acids	30 µg	Eggs, peanuts, salmon, pork, liver, mushrooms, sunflower seeds	Rarely occurs: skin rash, hair loss, convulsions, and other neuro- logical disorders; developmental delays in infants	Unknown (UL not determined)

Vitamin	Major Functions in the Body	Adult RDA/AI (Adult RDA = Bold)	Major Dietary Sources	Major Deficiency Signs and Symptoms	Major Toxicity Signs and Sympto ms
Vitamin B-6	Part of PLP, coenzyme needed for amino acid metabolism, involved in neurotransmitter synthesis and hemoglobin synthesis	**1.3–1.7 mg**	Meat, fish, poultry, potatoes, bananas, spinach, sweet red peppers, broccoli	Dermatitis, anemia, depression, confu- sion, and neurological disorders such as convulsions	**Adult UL = 100 mg/day** Nerve destruct ion
Folate	Part of THFA, coenzyme needed for DNA syn- thesis and conversion of cysteine to methionine, preventing homocysteine accumulation	**400 µg**	Dark green, leafy vegetables; papayas; asparagus; broccoli; orange juice; enriched breads and cereals (folic acid)	Megaloblastic anemia, diarrhea, neural tube defects in embryos	**Adult UL = 1000 µg/ day** May stimulat e cancer cell growth

Vitamin	Major Functions in the Body	Adult RDA/AI (Adult RDA = Bold)	Major Dietary Sources	Major Deficiency Signs and Symptoms	Major Toxicity Signs and Symptoms
Vitamin B-12	Part of coenzymes needed for various cellular pro- cesses, including folate metabolism; maintenance of myelin sheaths	**2.4 µg**	Animal foods; fortified cereals; fortified almond, soy, or rice milks	Pernicious anemia: megaloblastic anemia and nerve damage resulting in paralysis and death	None (UL not determined)
Ascorbic acid (vitamin C)	Substance needed for connective tissue syn- thesis and maintenance, antioxidant, synthesis of neurotransmitters and certain hormones, immune system functioning	**75–90 mg** (people who do not smoke cigarett es)	Peppers, citrus fruits, cherries, broccoli, cabbage, berries	Scurvy: poor wound healing, pinpoint hemorrhages, bleeding gums, bruises, depression	**Adult UL = 2000 mg/ day** Diarrhea and GI tract discomfort



8

Part 2: Vitamin-Like Compounds

Vitamin-Like Compounds

Definition: Substances necessary to maintain normal metabolism.

Difference from Vitamins: They are **NOT considered essential nutrients** because the body can synthesize them.

- **Examples:**

  Choline, Carnitine, Inositol, Taurine, Lipoic acid

Need: The need for them generally increases during periods of rapid tissue growth (e.g., in preterm infants).

Choline

Functions:

- Part of the neurotransmitter **acetylcholine** (associated with attention, learning, memory, muscle control).
- Needed for the production of **phospholipids** in cell membranes.

Sources: Widely distributed. Rich sources include liver, eggs, beef.

AI (Adult): 425-550 mg/day.

Toxicity (UL = 3.5 g/day):

- Excessive intake can cause a **fishy body odor**.
- Low blood pressure.

Other Vitamin-Like Compounds

Carnitine:

Function: Transports long-chain fatty acids into mitochondria, where they are metabolized for energy.

Sources: Primarily meat and dairy foods.

Inositol:

Function: Found in cell membranes; plays important regulatory roles in cells.

Sources: Animal and plant foods.

Other Vitamin-Like Compounds

Taurine:

Function: Involved in vision, antioxidant activity in WBCs, and central nervous system function.

Sources: Animal foods only.

Lipoic Acid:

Function: Component of metabolic reactions that remove CO2 from compounds; also an antioxidant.

Sources: Meats, liver, yeast.

Table: Summary of Vitamin-Like Compounds

Substance	Adult AI	Major Functions	Major Dietary Sources	Major Toxicity Signs
Choline	425-550 mg	Neurotransmitter (acetylcholine) and phospholipid synthesis.	Widely distributed; human biosynthesis.	(UL=3500 mg) Fishy body odor, reduced blood pressure.
Carnitine	Not est.	Transports long-chain fatty acids for energy.	Meat, fish, poultry, milk; human biosynthesis.	(High doses) Nausea, vomiting, diarrhea.
Inositol	Not est.	Regulatory role in cells.	Fruits, beans, grains, nuts; human biosynthesis.	Unknown.
Taurine	Not est.	Vision, antioxidant, insulin action, cell growth.	Animal foods; human biosynthesis.	Unknown.
Lipoic acid	Not est.	Component of metabolic reactions; antioxidant.	Meats, liver, yeast; human biosynthesis.	Unknown.

Diet and Cancer: What is Cancer?

Cancer: A group of chronic diseases characterized by cells that have undergone damage (mutations) to certain genes.

Malignant Cells:

- They are "out of control".
- They divide repeatedly and frequently.
- They do not die easily.
- They do not perform their specialized functions.

Malignant Tumors: As a result of rapid growth, cancer cells often form masses.

Tumor Nutrition: Tumors stimulate the body to form blood vessels that divert nutrients *away* from healthy cells and *into* the tumor, causing body wasting.

Cancer Terminology

Metastasized: When malignant cells break away from the tumor, move to, and invade other parts of the body.

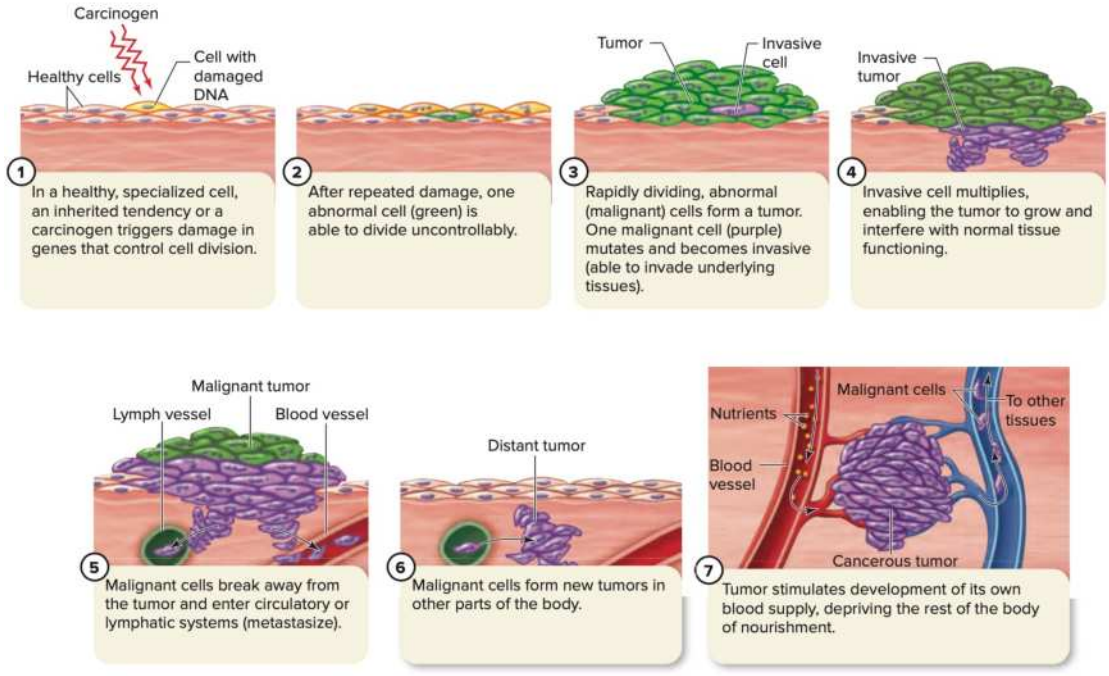
Benign Tumor:

- A noncancerous mass.
- The tumor's cells **do not** destroy nearby tissues or metastasize.
- Usually harmless, unless it grows large enough to interfere with healthy structures (like a blood vessel or brain tissue).

Carcinogen:

- An environmental factor that triggers cancer.
- Examples: Radiation, tobacco smoke, certain viruses, or chemicals.

FIGURE: Cancer development and progression



Risk Factors for Cancer

1. Aging (most cancers occur in people > 65)
2. Family history of cancer
3. **Using tobacco**
4. Exposure to radiation
5. Exposure to certain environmental substances
6. Certain viral and bacterial infections
7. Elevated levels of certain hormones
8. **Consuming alcohol and certain foods in excess**
9. **Being physically inactive**
10. **Obesity and excess body fat**

Diet and Cancer Development

Alcohol: A known carcinogen. Two or more drinks daily increases risk of mouth, throat, esophagus, liver, breast, colon, and rectum cancers.

Aflatoxin: A toxin from molds that grow on nuts or grains; increases risk of liver cancer.

Processed Meat: (e.g., "deli" meats). Sodium nitrate/nitrite added as preservatives can form **nitrosamines**, which are carcinogens.

High-Temp Cooking: Frying, grilling, or broiling meats at high temperatures may cause the formation of carcinogens called **heterocyclic amines**.

Limiting charred areas may reduce risk.

Diet and Cancer Prevention

Fruits and Vegetables: High intakes of fruits and vegetables **reduce the risk** of cancer, especially lung, oral, esophageal, stomach, and colon cancers.

Why? They are rich sources of:

- Vitamin C
- Carotenoids
- Other phytochemicals

Antioxidants: These compounds may have antioxidant activity, which protects DNA from free radical damage, preventing cancer-causing mutations.

Vitamins and Cancer Risk (The Research)

The relationship between specific vitamins and cancer risk is unclear and often **conflicting**.

Supplements: Research has generally **NOT** shown that taking dietary supplements with antioxidants (like C or E) reduces cancer risk. Some studies suggest they may *increase* risk.

Folate: Epidemiological studies are conflicting. Some show an inverse association, while others show no effect or even an *increased* risk of cancer.

Vitamin E: Unclear. One large study (SELECT) found that vitamin E (alpha-tocopherol) supplementation **increased prostate cancer risk**.

Vitamin C: Some evidence suggests it may be useful in *treating* cancer, but it may also *interfere* with certain cancer-fighting medications.

Reducing Your Cancer Risk

The best course of action is prevention. The American Cancer Society recommends:

- **Avoiding exposure to tobacco**.
- **Achieving and maintaining a healthy weight** and body composition.
- Adopting a **physically active lifestyle**.
- Eating a balanced diet that **limits intake of processed meats** and **emphasizes plant foods**, including fruits, vegetables, and whole grains.


Thank you!

PRINCIPLES OF NUTRITION

Dr. Marwan Jalambo
Assistant Professor (Nutrition)

9

Part 1: Water

Water: Learning Outcomes

1. Discuss the functions of water in the body .
2. Explain differences between intracellular and extracellular water .
3. Identify typical sources of water intake and loss .

Water and Body Composition

Water is a unique, essential nutrient that requires no digestion and is easily absorbed .

Survival is only a few days without water, versus weeks or months without other nutrients .

- **50% to 75%** of the body is water weight .
  - **Lean muscle tissue** is ~73% water .
  - **Fat tissue** is ~20% water .

An average young male's body (more lean tissue) is ~60% water .

A person's percentage of body water declines with age (from ~75% at birth to ~45% in older adults) .

TABLE: Functions of Water in the Body

Water
Is a solvent
Is a major component of blood, saliva, sweat, tears, mucus, and joint fluid
Removes wastes
Helps transport substances
Lubricates tissues
Regulates body temperature
Helps digest foods
Participates in many chemical reactions
Helps maintain proper blood pH

Water and Membrane Transport

Selectively Permeable Membrane: A barrier that allows certain substances to pass but not others .

Osmosis: The diffusion of a solvent (like water) through a selectively permeable membrane .

Water moves from a region with *less* dissolved material (dilute) to a region with *more* dissolved material .

This process stops when concentrations are equal on both sides

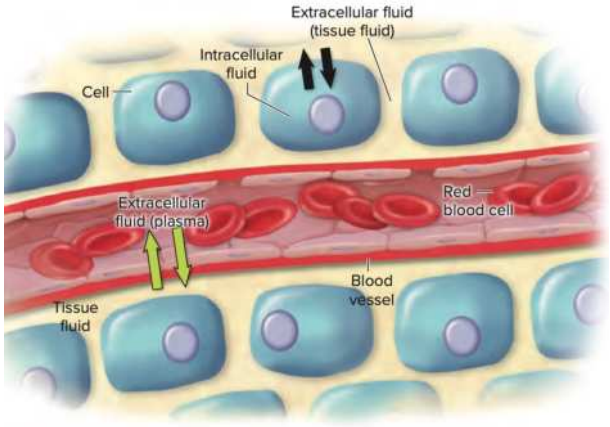
Water Distribution: Intracellular vs. Extracellular

Intracellular Water:

- Water *inside* the cells .
- Makes up about **two-thirds** of the body's water.

Extracellular Water:

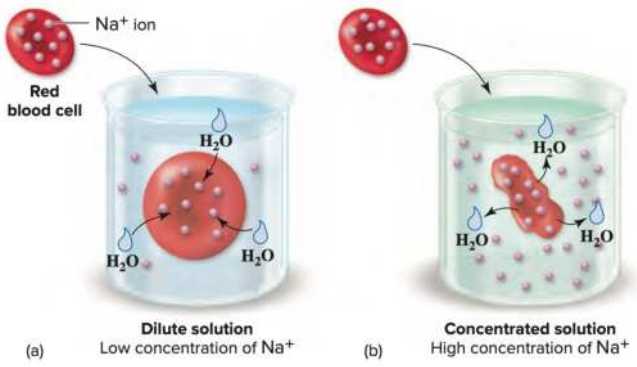
- Water *surrounding* the cells (tissue, or interstitial, fluid).
- Water in the fluid portion of blood (plasma).



Fluid Balance and Ions (Electrolytes)

Fluid balance is maintained by controlling concentrations of **ions (electrolytes)** .

- **Intracellular volume** depends on:
  - Potassium and phosphate ions .
- **Extracellular volume** depends on:
  - Sodium and chloride ions .
- Imbalances cause water to shift, making cells swell or shrink .



Sources of Water

Total Water Intake (AI):

Young Females: ~11 cups (2.7 L) .

Young Males: ~15.5 cups (3.7 L) .

Sources of Intake:

Beverages: Water, juice, milk, coffee, tea, etc. .

Foods: Fruits and vegetables are 60-95% water by weight .

Metabolic Water: Water formed by cells as a by-product of metabolism .

Inactive people: ~1 to 1.5 cups/day .

Active people: ~2 to 2.5 cups/day .

Water Balance and Hydration: Learning Outcomes

1. Discuss how the body maintains its water balance .
2. Describe dehydration and water intoxication .

- **Key Terms:**
  - **Hydration:** The body's water status .
- **Dehydration:** Body water depletion

Water Balance: Input vs. Output

A healthy person's average daily water input equals their average daily output .

Average balance is ~2.5 liters (2500 ml) per day .

Factors influencing balance: environmental conditions, fever, vomiting, diarrhea, exercise, and diet .

FIGURE: Daily water balance



Kidneys and Water Balance

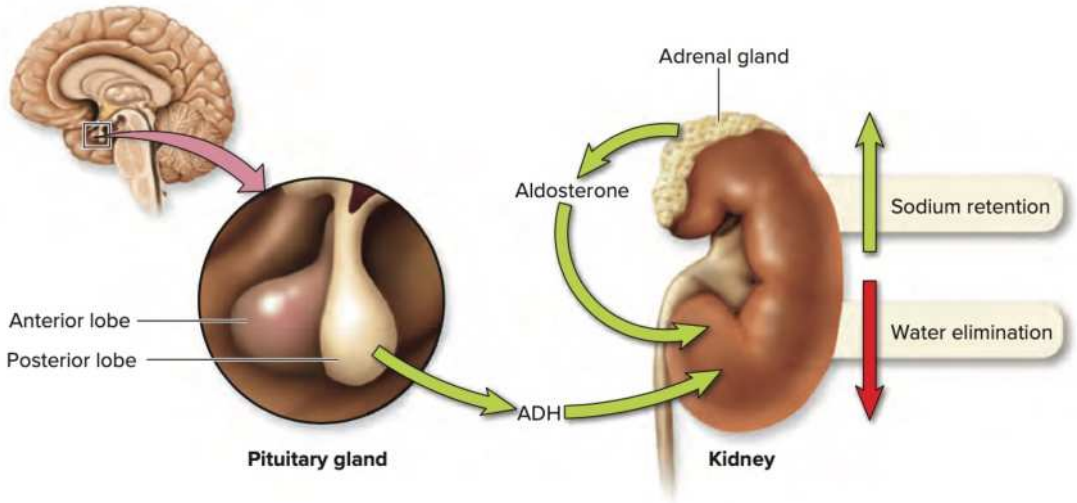
The **kidneys** are the major regulator of the body's water content and ion concentrations .

They maintain hydration by filtering excess ions (water follows) to become urine .

Healthy kidneys produce ~1 to 2 liter of urine per day .

Kidney stones are hard masses of minerals and waste products that form in the kidneys .

FIGURE Effects of antidiuretic hormone and aldosterone on kidneys

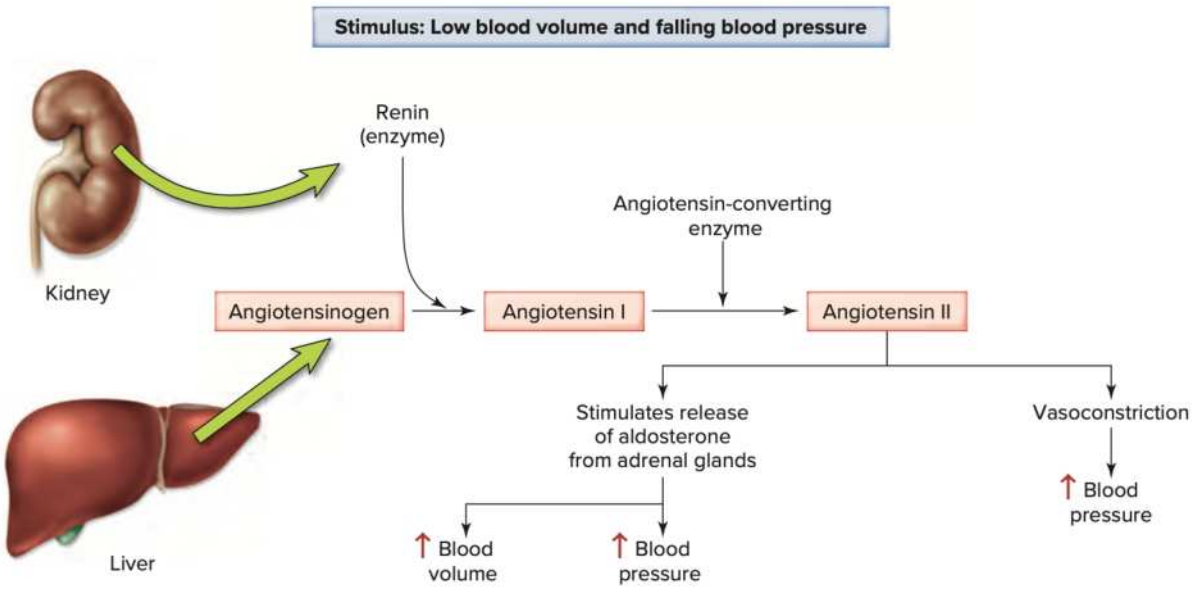


Hormones for Water Balance

In response to dehydration:

1. **Antidiuretic Hormone (ADH):**
   - Released from the posterior pituitary gland .
   - Stimulates kidneys to **conserve water** .
2. **Aldosterone:**
   - Secreted by the adrenal glands .
   - Signals kidneys to **reduce sodium elimination** (water follows sodium, so water is also conserved) .

Renin-Angiotensin System



Renin-Angiotensin System

This system regulates blood volume and blood pressure .

Process:

1. Kidneys secrete **Renin** (enzyme) in response to low blood volume/pressure .
2. Renin activates Angiotensinogen (from liver) to **Angiotensin I** .
3. Angiotensin-converting enzyme (ACE) (in lungs) converts it to **Angiotensin II** .

Angiotensin II Actions:

1. Stimulates release of Aldosterone .
2. Acts as a **vasoconstrictor** (constricts blood vessels) to increase blood pressure .

Dehydration

Assessing Hydration:

- **Urine Volume:** Adequately hydrated = sufficient urine. Limited intake = less urine .
- **Urine Color:** Light yellow = adequate. Dark = potential dehydration . (Note: Can be affected by B-vitamins, medications, etc.) .

Signs of Dehydration:

Thirst: Primary regulator . Less accurate in older adults .

Treatment:

- **Oral rehydration therapy** (solutions of water and electrolytes) .
- Serious cases require intravenous (IV) fluids .

Diuretics and Water Intoxication

Diuretic: A substance that increases urine production .

- **Caffeine:** A diuretic, but caffeinated beverages still contribute to water needs .
- **Alcohol:** Also a diuretic . It *inhibits ADH secretion*, causing kidneys to eliminate more urine and leading to dehydration .

Water Intoxication:

- Occurs when excessive water is consumed in a short time or kidneys can't filter water properly .
- Dilutes blood sodium, causing **hyponatremia** (low blood sodium) .
- Water moves into cells (including brain cells), causing swelling .
- Symptoms: Drowsiness, nausea, confusion, seizures, coma, and death .



9

Part 2: Major Minerals

Minerals: The Basics - Learning Outcomes

1. Explain the difference between major and trace minerals .
2. Classify mineral nutrients as major, trace, or possible essential nutrients .
3. Describe key roles of minerals in the human body .
4. List good food sources of minerals .

Mineral Classification

Major Minerals:

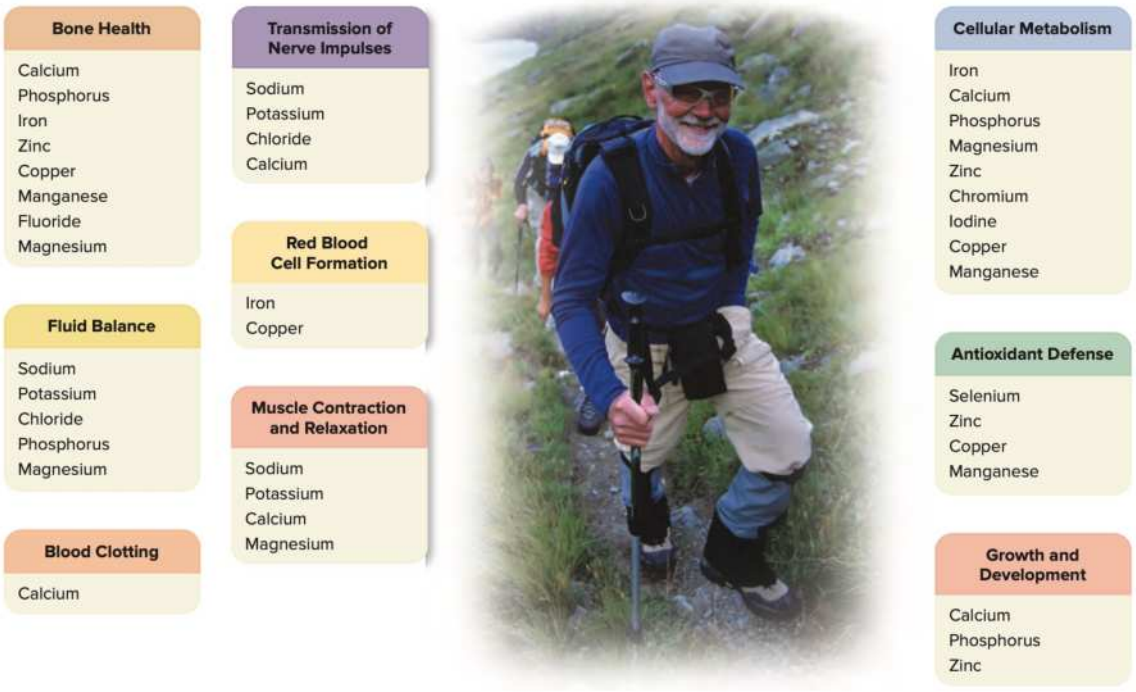
Required in amounts of **100 mg or more** per day .

Trace Minerals:

Required in amounts **less than 100 mg** per day .

Minerals are **indestructible** (unlike vitamins) but are water-soluble and can leach into cooking water .

FIGURE: Minerals and their functions



Functions of Minerals

Minerals have many roles, but do not provide energy .

Key Roles:

- **Structural Components:** e.g., Calcium and phosphorus in bones and teeth .
- **Inorganic Ions:**
  - e.g., Calcium ions (Ca^{2+}) for blood clotting .
  - e.g., Sodium ions (Na^{+}) for fluid balance .
- **Cofactors:** Ions (like Mg^{2+} or Cu^{2+}) that an enzyme requires to function .
- **Components of:** Enzymes, hormones, or other molecules (e.g., Iron in hemoglobin) .

Sources of Minerals & Bioavailability

Bioavailability: The body's ability to absorb and use minerals .

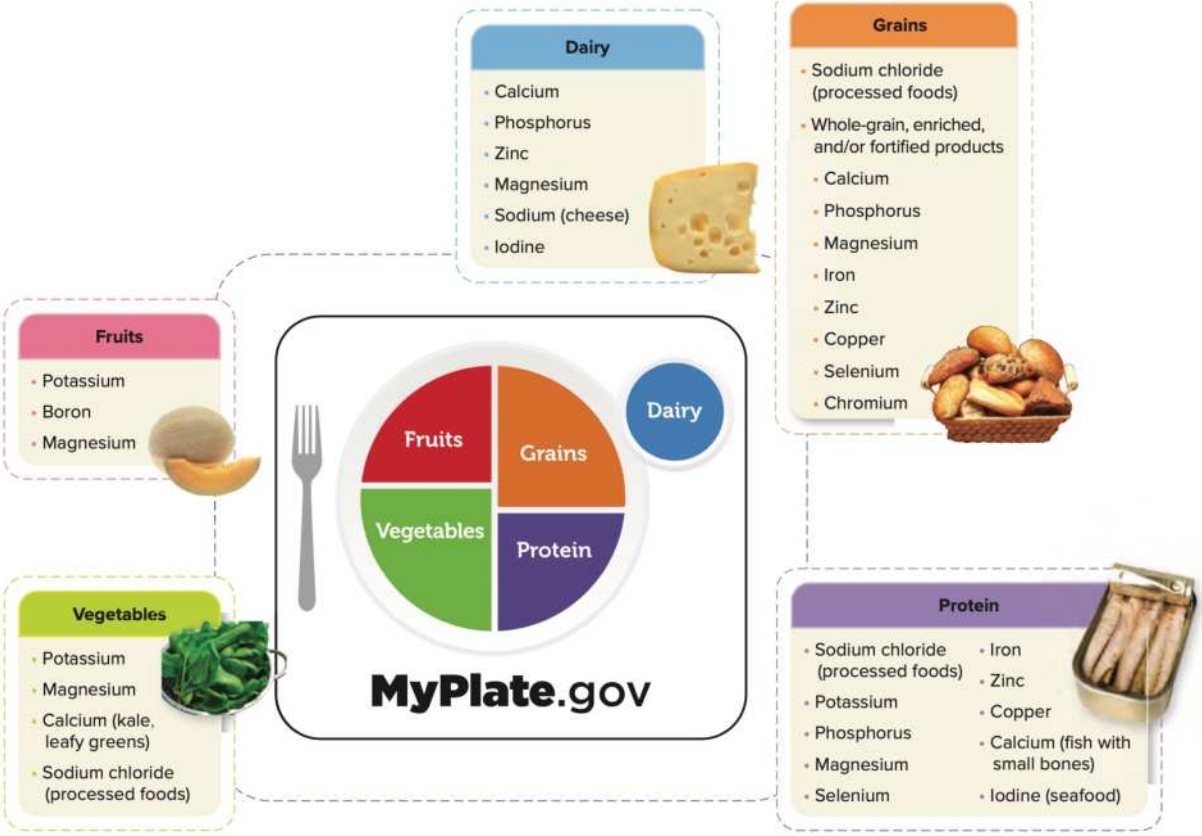
Factors that INCREASE bioavailability:

- Periods of growth (infancy, puberty), pregnancy, breastfeeding .

Factors that DECREASE bioavailability:

- **Phytic acid:** Found in whole grains, seeds, and beans .
- **Oxalic acid:** Found in spinach, collard greens, sweet potatoes .

Animal foods are often more reliable mineral sources (e.g., iron, calcium) .

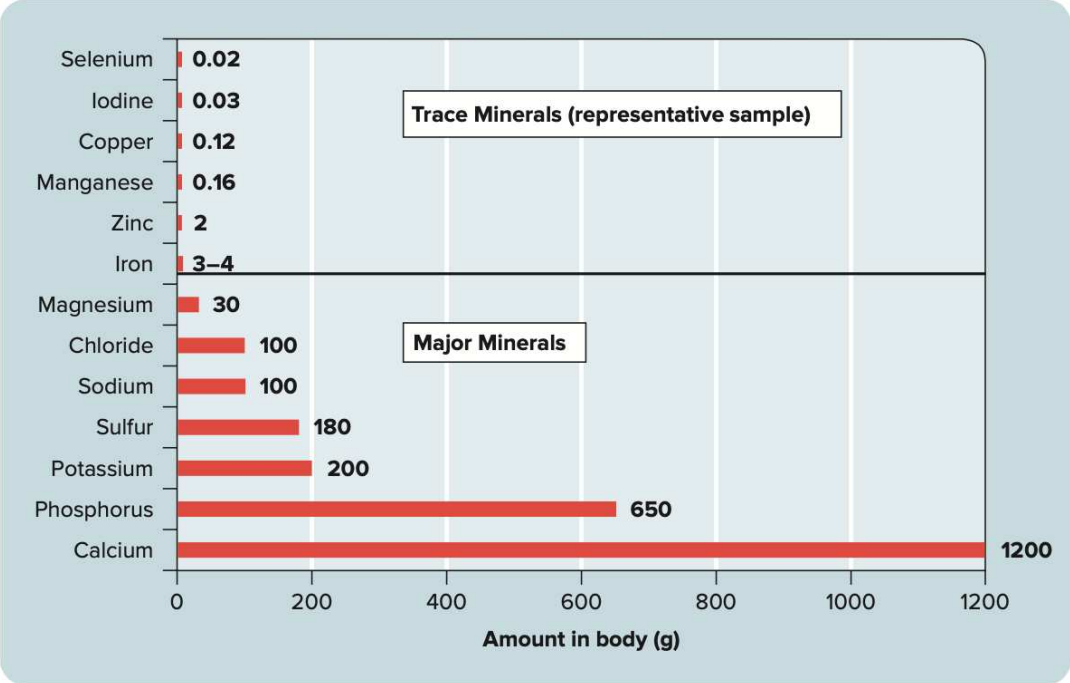


Mineral Storage in the Body

Major minerals are stored in larger quantities than trace minerals .

Calcium (1200 g) and **Phosphorus (650 g)** are the most abundant minerals in the body, mostly stored in the skeleton .

Mineral Storage in the Body



CALCIUM

Ca

Calcium: Learning Outcomes & Functions

Learning Outcomes:

1. Explain major functions of calcium .
2. Describe roles of hormones in blood calcium homeostasis.
3. List major food sources of calcium .
4. Discuss calcium deficiency and toxicity .
5. Explain the relationship between calcium and chronic diseases .

Calcium Functions:

1. Most abundant mineral in the body .
2. Formation of bones and teeth .
3. Muscle contraction .
4. Blood clot formation .
5. Nerve impulse transmission .
6. Cell metabolism .

Maintaining Blood Calcium Homeostasis

When Blood Calcium FALLS:

- **Parathyroid glands** secrete **Parathyroid Hormone (PTH)** .
- **PTH Actions:**
  1. Stimulates Vitamin D activity .
  2. Increases **osteoclast** activity (bone breakdown) to release calcium .
  3. Increases calcium retention in kidneys .
  4. Increases calcium absorption in small intestine .

Maintaining Blood Calcium Homeostasis

When Blood Calcium RISES:

- **Thyroid gland** secretes **Calcitonin** .
- **Calcitonin Actions:**
  1. Inhibits Vitamin D activity .
  2. Increases **osteoblast** activity (bone formation) .
  3. Decreases calcium retention in kidneys .
  4. Decreases calcium absorption in small intestine .

Bone Mineralization

Osteoclasts: Bone cells that break down bone tissue .

Osteoblasts: Bone cells that add (build) bone tissue where needed .

Hydroxyapatite: A crystalline structure of calcium and phosphorus that forms on bone tissue .

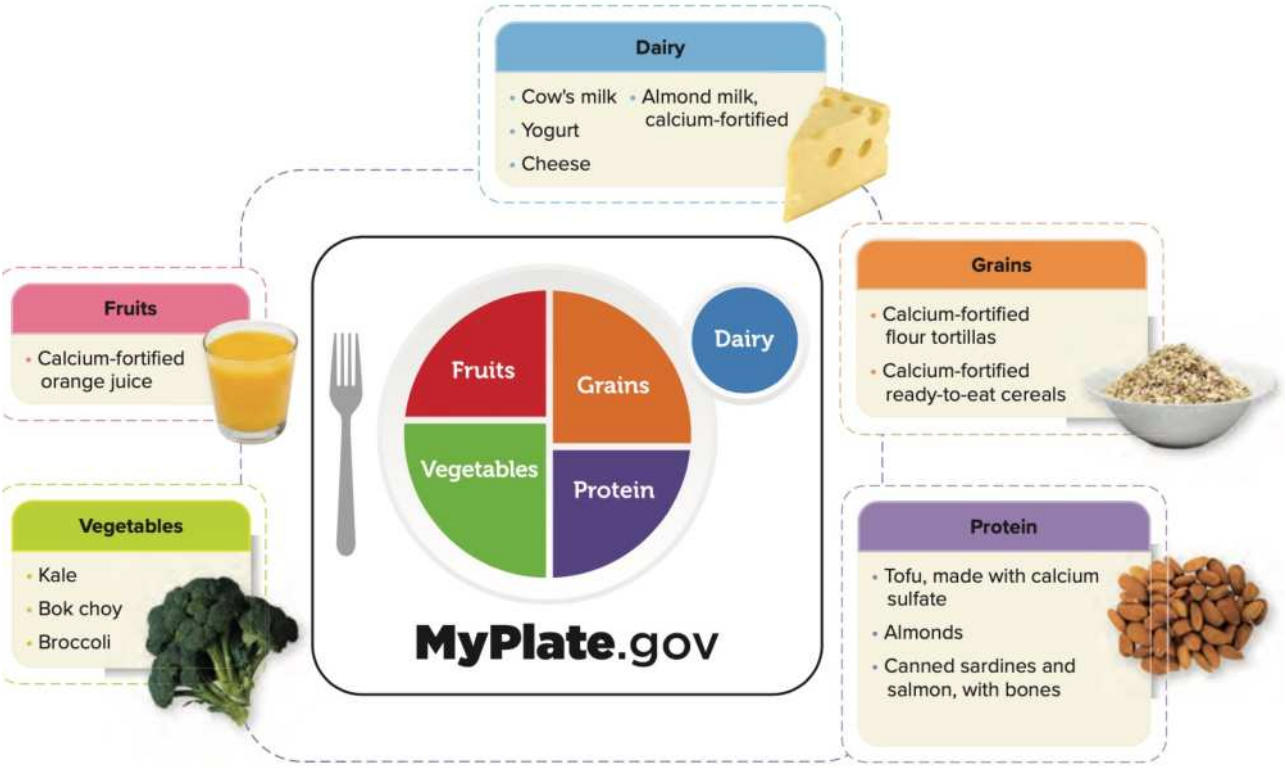
Bone Mineralization: The process by which bone tissue gains strength and rigidity

Sources of Calcium

Major Sources: Milk products (milk, yogurt, cheese), canned fish (with bones), leafy greens (kale, collard, bok choy), broccoli, calcium-fortified foods .

Bioavailability Varies:

- ~30% of calcium in cow's milk is bioavailable .
- ~5% of calcium in spinach is bioavailable (due to oxalic acid) .



Calcium Absorption & Adequacy

RDA (Adults): 1000–1200 mg/day .

RDA (Ages 9-18): 1300 mg/day (to allow for bone mass growth)

Adults absorb ~1/3 of calcium in food . (Higher in infants, children, and pregnant women) .

Enhance Absorption	Decrease Absorption
Vitamin D (calcitriol)	Vitamin D deficiency
Stomach acid	Reduced stomach acid
Lactose (infants only)	Phytic and oxalic acid
	High fiber intake
	High phosphorus intake
	Chronic diarrhea

Calcium Deficiency: Osteoporosis

Osteoporosis: A chronic disease characterized by low bone mass and reduced bone structure .

Osteopenia: A precursor condition; weak bones that are susceptible to fractures .

Bones become weak and susceptible to fractures (hip, spine, wrist) .

Peak Bone Mass: 85-90% is acquired by age 20 .

Bone loss begins in mid-adulthood and accelerates significantly in women after **menopause** due to declining estrogen .

Osteoporosis: Risk Factors

Nonmodifiable Risk Factors	Modifiable Risk Factors
Female	Having anorexia nervosa
Growing older	Low estrogen/testosterone
White or Asian ancestry	Inadequate calcium & vitamin D
Family history of osteoporosis	Use of steroids/anticonvulsants
Small, thin-boned body frame	Being physically inactive
	Smoking cigarettes
	Consuming excessive alcohol

Reducing Osteoporosis Risk & Calcium Toxicity

Risk Reduction:

Maximize peak bone mass with proper diet and regular exercise, especially from childhood through adolescence .

Perform **weight-bearing activities** .

Calcium Toxicity:

UL (Adults): 2000–2500 mg/day .

Caused by taking too many calcium-containing antacids or supplements .

Hypercalcemia: Higher-than-normal concentration of calcium in blood .

Symptoms: Kidney stones, bone pain, muscle weakness, fatigue, hypertension .

PHOSPHORUS

P

Phosphorus: Learning Outcomes

Learning Outcomes:

1. Explain the major functions of phosphorus .
2. List rich food sources of phosphorus .
3. Discuss phosphorus deficiency and toxicity ..

Phosphorus: Functions

1. Second most abundant mineral in the body .
2. ~85% is found with calcium in bones and teeth .
3. Essential for every cell .
4. Synthesize **ATP** (energy) .
5. Synthesize phospholipids, DNA, and RNA .
6. Regulate acid-base balance .
7. Activate enzymes and hormones .

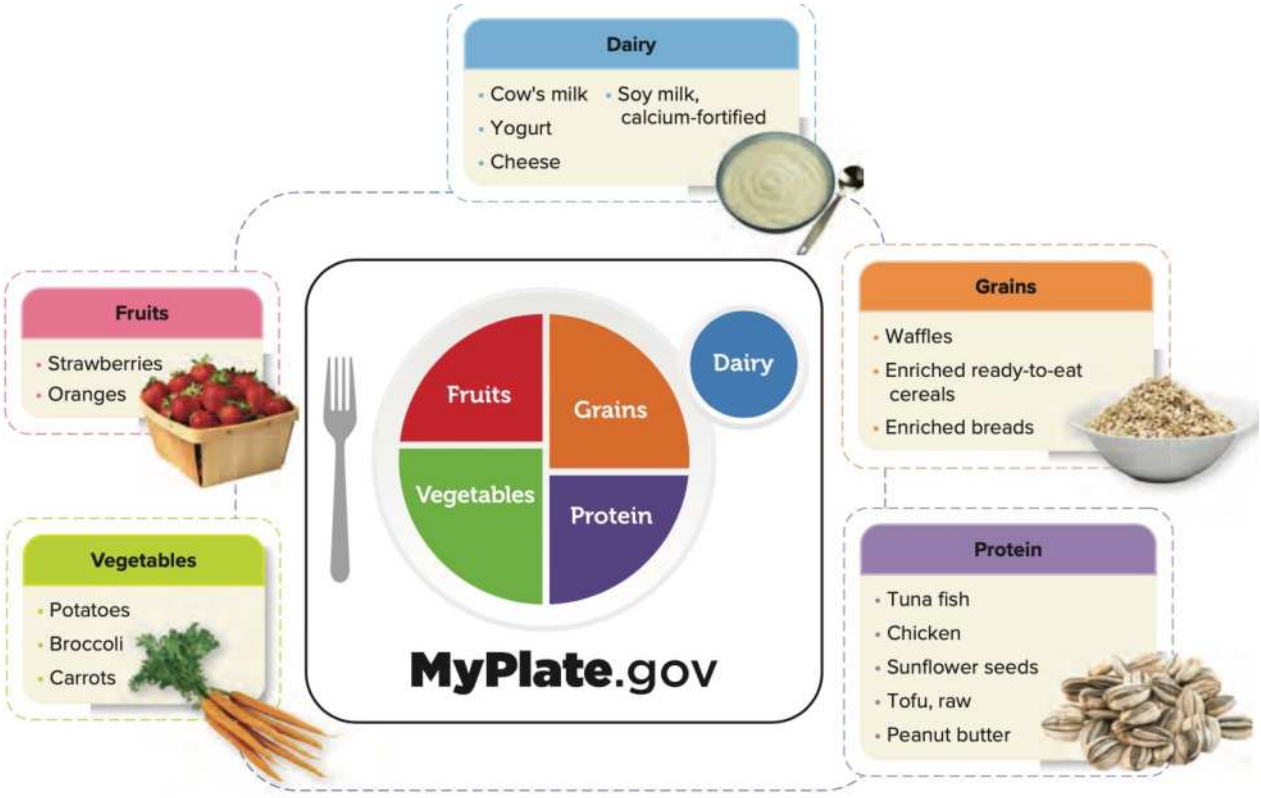
Sources of Phosphorus

Primary Sources: Dairy foods, meat, and cereal grains .

Fruits and vegetables are generally poor sources .

Bioavailability:

- **Higher** from animal foods (inorganic form) .
- **Lower** from plant foods (organic form, like **phytate**) .



Phosphorus: Adequacy, Deficiency, Toxicity

- **RDA (Adults):** 700 mg/day .
- **Deficiency:** Rare, as it's wide in the food supply .
- **Symptoms:** Bone loss, decreased mineralization, poor appetite, weakness, bone pain . Rickets in children .
- **Toxicity:**
  - **UL (Adults):** 4000 mg/day .
  - **Hyperphosphatemia:** Elevated blood phosphorus .
  - Can cause calcium-phosphate deposits in soft tissues (e.g., blood vessels), contributing to CVD .
  - High risk for individuals with **chronic kidney disease** .

SODIUM

Na^+

Sodium: Learning Outcomes

- **Learning Outcomes:**
  1. Explain the major functions of sodium .
  2. List rich food sources of sodium .
  3. Discuss causes and treatment of hyponatremia .
  4. Describe hypertension (risk factors, prevention, treatment) .
- **Primary Source:** Table salt (Sodium Chloride) .
  1. **1 tsp of table salt = 2325 mg of sodium** .

Sodium: Functions

1. Major role in maintaining normal **fluid balance** (as a primary *extracellular* ion) .
2. Transmission of nerve impulses .
3. Normal muscle function .

- Transport of substances (e.g., glucose) into cells .

Sources of Sodium

- Most sodium (75%+) comes from **processed foods** and **restaurant foods** .
- Fresh vegetables, raw meats, and grains are naturally low in sodium .
- Used as a flavor enhancer and preservative .
- **High-Sodium Foods:** Canned soups, sauces, gravies, hot dogs, deli meats, cheeses, pickled foods, and salted snacks .

Sodium: Adequacy, Deficiency, Toxicity

- **AI (Adults):** 1500 mg/day .

Chronic Disease Risk Reduction (CDRR) Guideline: Limit intake to **less than 2300 mg/day** .

- *Average U.S. adult intake is ~3500 mg/day .*

Deficiency (Hyponatremia):

- Uncommon .
- Can result from severe diarrhea/vomiting or prolonged endurance exercise .
- Symptoms: Headache, nausea, fatigue, muscle cramps, seizures, coma .

Toxicity:

- High-sodium diets are associated with increased risk of **hypertension** .
- **Sodium sensitive:** Individuals whose kidneys are less effective at eliminating excess sodium, causing water retention and increased blood pressure .

Sodium and Hypertension (High Blood Pressure)

- **Hypertension:** Persistently elevated blood pressure .
- A major public health problem; "silent killer" (often no symptoms) .
- Increases risk of CVD, heart disease, stroke, and kidney failure .
- **Blood Pressure Measurements:**
  - **Systolic:** Maximum pressure when ventricles *contract* (top number) .
  - **Diastolic:** Pressure when ventricles *relax* (bottom number)

Hypertension: Risk Factors & Treatment

Risk Factors:

Treatment:

1. **Dietary Modifications:**
   - **DASH Diet:** (Dietary Approaches to Stop Hypertension).
   - Low in sodium, fat, and cholesterol.
   - High in fruits, vegetables, and low-fat dairy.
2. **Lifestyle Changes:**
   - Losing excess body fat.
   - Increasing physical activity.
   - Managing stress.
3. **Medications**.

POTASSIUM

K^+

Potassium: Learning Outcomes

Learning Outcomes:

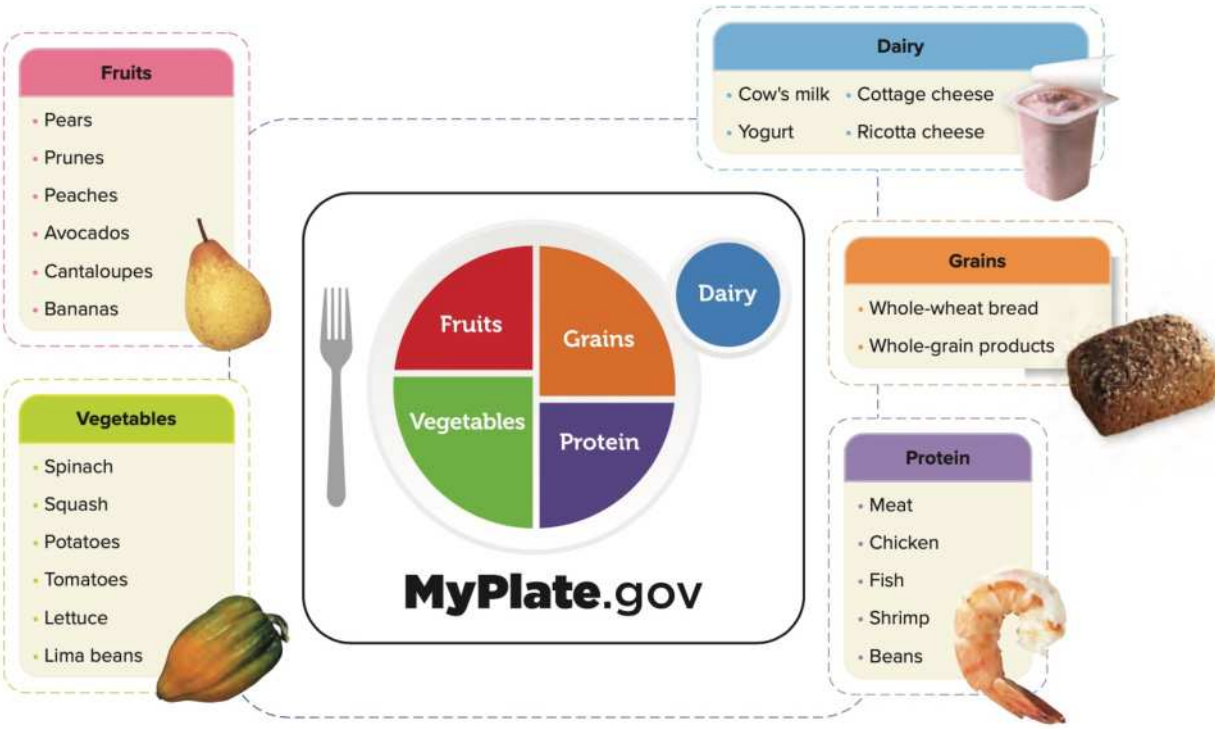
1. Explain the major functions of potassium .
2. List rich food sources of potassium .
3. Discuss potassium deficiency and toxicity .

Potassium: Functions

1. Primary positively charged ion in **intracellular fluid** (inside cells) .
2. Maintains proper fluid balance .
3. Transmitting nerve impulses .
4. Contracting muscles .
5. Associated with **lower blood pressure values** .

Sources of Potassium

- **Good Sources:** Fresh fruits and vegetables, cow's milk, whole grains, dried beans, and meats.
- **Poor Sources:** Most processed foods (which are high in sodium).



Potassium: Adequacy, Deficiency, Toxicity

- **AI (Adults):**
  - Females: 2600 mg/day .
  - Males: 3400 mg/day .
- Potassium is a **"nutrient of public health concern"** as intakes are often low .
- **Deficiency (Hypokalemia):**
  - Risk from excessive sweating, vomiting, diarrhea, or kidney diseases .
  - **Symptoms:** Muscle cramps, confusion, constipation, loss of appetite . Life-threatening .
- **Toxicity (Hyperkalemia):**
  - No UL from food consumption .
  - Caused by **taking potassium supplements** .
  - Can cause the heart to stop beating .

MAGNESIUM

Mg^{+2}

Magnesium: Learning Outcomes

Learning Outcomes:

1. Explain the major functions of magnesium .
2. List rich food sources of magnesium .
3. Discuss magnesium deficiency and toxicity .

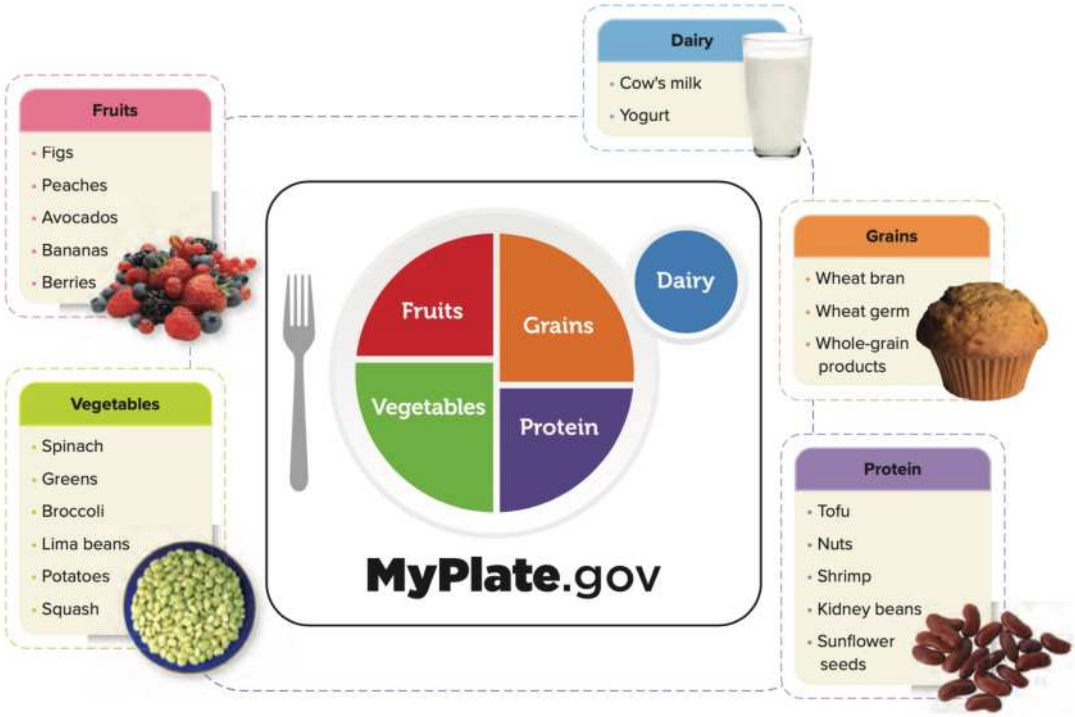
Magnesium: Functions

1. Participates in **more than 300 chemical reactions** .
2. Muscle contraction and relaxation .
3. Enzyme function and energy production .
4. DNA and protein synthesis .
5. Helps regulate blood pressure and blood glucose .
6. Maintains strong bones and a healthy immune system .

Sources of Magnesium

- Magnesium is a critical component of **chlorophyll** (green pigment in plants).
- **Richest Sources:** Spinach, green leafy vegetables, whole grains, beans, nuts, seeds, and chocolate.
- **Poor Sources:** Refined grains (magnesium is lost in processing).
- Also found in "hard" tap water.

Sources of Magnesium



Magnesium: Adequacy & Deficiency

- **RDA (Adults):** 310–420 mg/day.
- Many Americans consume *less* than the recommended amount.
- **Deficiency:**
  - Rare among healthy people.
  - **Risk Groups:** People with alcohol use disorder, poorly controlled diabetes, or using certain diuretics; older adults.
  - **Symptoms:** Irritability, weakness, loss of appetite, muscle twitching.
  - **Severe Symptoms:** Rapid heartbeat, inability to relax muscles, disorientation.

Magnesium: Toxicity

Hypermagnesemia: High blood magnesium.

Cause:

- Toxicity from *food* sources is rare.
- More likely to occur from ingesting excessive **laxatives, antacids, or dietary supplements** containing magnesium.

- **UL (Adults): 350 mg/day**
  - *This UL applies **only** to magnesium-containing medications and supplements, **not** food.*

Symptoms: Diarrhea.

Severe Toxicity: (Often in kidney failure) Weakness, slowed breathing, coma, and death.

CHLORIDE

Cl^-

Chloride: Learning Outcomes

Learning Outcomes:

1. Explain the major functions of chloride .
2. List rich food sources of chloride .
3. Discuss chloride deficiency and toxicity .

Chloride: Functions

1. The primary **anion** (negatively charged ion) in **extracellular fluid** .
2. Regulates fluid balance .
3. Provides electrolytic balance (with sodium) .
4. Production of **hydrochloric acid (HCl)** in the stomach .
5. Regulation of acid-base balance .
6. Transmission of nerve impulses .
7. Immune response .

Sources of Chloride

Primary Source: Table salt or sea salt (sodium chloride) .

Other Sources:

- Salt substitutes (potassium chloride) .
- Vegetables: Seaweed, rye, tomatoes, celery, and olives .

Chloride: Adequacy, Deficiency, Toxicity

1 tsp of table salt supplies ~3700 mg of chloride .

AI (Adults): 2300 mg/day .

Deficiency:

- Unlikely, due to high salt intake in most diets .
- Can occur from extreme vomiting, diarrhea, or sweating .
- **Symptoms:** Fatigue, loss of appetite .

Toxicity:

- **UL (Adults):** 3600 mg/day .
- **Cause:** Excessive table salt (chloride) intake or severe dehydration .
- **Consequence:** Can contribute to **hypertension** .

SULFUR

S

Sulfur: Learning Outcomes

Learning Outcomes:

1. Explain the major functions of sulfur .
2. Describe the relationship between sulfur and specific amino acids .

Sulfur: Functions

1. A component of organic compounds, including:
   - **Amino Acids:** Methionine and cysteine .
   - **Vitamins:** Biotin and thiamin .
2. Essential for the activity of many enzymes .
3. Part of antioxidant molecules (e.g., **glutathione**) .
4. Provides strength to hair, skin, and nails (via proteins keratin and collagen) .

Sulfur: Sources, Deficiency, Toxicity

Sources:

- **Protein-rich foods** .
- Meat, poultry, fish, eggs, cow's milk, cheese, legumes, and nuts provide ample sulfur .

Deficiency:

- Rare .
- Occurs only with severe protein deficiency .

Toxicity:

- Unlikely from dietary sources .

No AI, RDA, or UL is established for sulfur .


Thank you!

PRINCIPLE OF NUTRITION

Dr. Marwan Jalambo
Assistant Professor (Nutrition)

10

Trace Minerals

Defining Trace Minerals

Major Minerals: Required by the human body at ≥100 mg/day.

Trace Minerals: Essential mineral nutrients.

- Required by the body in small amounts (generally < 100 mg/day).
- Perform vital roles in the body.
- Obtaining adequate amounts from food can be difficult.

Table: Summary of Trace Minerals (Part 1)

Mineral	Adult RDA/AI	Major Functions	Major Dietary Sources	Major Deficiency Signs	Major Toxicity Signs
Iron (Fe)	Female: 18 mg Male: 8 mg	Component of hemoglobin & myoglobin Energy generation Immune function	Meat, animal foods, whole-grain & enriched breads/cereals	Fatigue Small, pale RBCs Low hemoglobin Poor immune function	UL: 45 mg/day • Intestinal upset • Organ damage • Death
Zinc (Zn)	Female: 8 mg Male: 11 mg	Component of numerous enzymes Immune system function	Seafood, meat, whole grains	• Skin rash • Diarrhea • Depressed taste/smell • Hair loss • Poor growth	UL: 40 mg/day • Intestinal upset Depressed immune function • Can reduce copper absorption
Copper (Cu)	0.9 mg	Promotes iron metabolism Component of antioxidant enzymes Connective tissue synthesis	Liver, cocoa, legumes, whole grains, shellfish	Anemia Reduced immune function • Poor growth	UL: 10 mg/day Vomiting Abnormal nervous system Liver damage

Table: Summary of Trace Minerals (Part 2)

Mineral	Adult RDA/AI (RDA)	Major Functions	Major Dietary Sources	Major Deficiency Signs	Major Toxicity Signs
Selenium (Se)	55 µg	Component of an antioxidant system	Meat, eggs, fish, seafood, whole grains	Muscle pain/weakness Form of heart disease	UL: 400 \mu g/day Nausea, vomiting • Hair loss Weakness Liver damage
Iodine (I)	150 µg	Component of thyroid hormones	Iodized salt, saltwater fish, dairy products	Goiter Congenital hypothyroidism	UL: 1100 \mu g/day Reduced thyroid gland function
Fluoride (F)	Female: 3 mg Male: 4 mg	Increases resistance of tooth enamel to cavities Stimulates bone formation	Fluoridated water, tea, seaweed	Increased risk of tooth decay (no true deficiency)	UL: 10 mg/day Stomach upset Staining of teeth Bone deterioration

Table: Summary of Trace Minerals (Part 3)

Mineral	Adult RDA/AI (RDA Bold)	Major Functions	Major Dietary Sources	Major Deficiency Signs	Major Toxicity Signs
Chromium (Cr)	Female: 20-25 µg Male: 30-35µg	Enhances insulin action	Egg yolks, whole grains, pork, nuts, mushrooms	Blood glucose remains elevated after meals	Unknown May interact with medications
Manganese (Mn)	Female: 1.8mg Male: 2.3 mg	Cofactor for enzymes (e.g., carbohydrate metabolism)	Nuts, oats, beans, tea	None in healthy humans	UL: 11 mg/day Abnormal nervous system function
Molybdenum (Mo)	45 µg	Component of certain coenzymes	Liver, peas, beans, cereal products, cow's milk	None in healthy humans	UL: 2000 \mu g/day Rarely occurs from diet, Supplements may cause joint pain


Thank you!